

Bayesian Time-Series Econometrics

Book 2 - algebraic derivations

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Fourth edition

Bayesian Time-Series Econometrics

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Cover illustration: Thomas Bayes (d. 1761) in Terence O'Donnell, *History of Life Insurance in Its Formative Years* (Chicago: American Conservation Co., 1936), p. 335.

To my wife, Mélanie.

To my sons, Tristan and Arnaud.

Contents

I	Bayesian statistics	1
3	Three applied examples	3
4	Further aspects of Bayesian priors and posteriors	7
5	Properties of Bayesian estimates	15
II	Simulation methods	17
6	The Gibbs sampling algorithm	19
7	The Metropolis-Hastings algorithm	21
8	Mathematical theory	25
III	Econometrics	29
9	The linear regression model	31
10	Applications with the linear regression model	37
IV	Vector autoregressions	51
11	Vector autoregressions	53
12	Further aspects of Bayesian vector autoregressions	67
14	Bayesian VAR: advanced applications	77
V	Vector autoregression extensions	81
15	Vector error correction	83
16	Vector autoregressive moving average	101
VI	Nowcasting	105
17	Mixed frequency vector autoregression	107
18	Dynamic factor model	109

19 MIDAS regression	113
References	121

PART I

Bayesian statistics

Three applied examples

derivations for equation (1.3.3)

$$f(y|p) = \prod_{i=1}^n p^{y_i} (1-p)^{1-y_i} = p^{\sum_{i=1}^n y_i} (1-p)^{\sum_{i=1}^n 1-y_i} = p^m (1-p)^{n-m} \quad (\text{a.1.3.1})$$

derivations for equation (1.3.5)

The derivative is given by:

$$\frac{d \log(f(y|p))}{dp} = \frac{m}{p} - \frac{n-m}{1-p} \quad (\text{a.1.3.2})$$

Set the value to 0 and solve for p :

$$\begin{aligned} \frac{m}{p} - \frac{n-m}{1-p} &= 0 \\ \Leftrightarrow \frac{m}{p} &= \frac{n-m}{1-p} \\ \Leftrightarrow m(1-p) &= p(n-m) \\ \Leftrightarrow m - mp &= np - mp \\ \Leftrightarrow m &= np \\ \Leftrightarrow p &= \frac{m}{n} \end{aligned} \quad (\text{a.1.3.3})$$

derivations for equation (1.3.11)

$$f(y|p) = \prod_{i=1}^n \frac{\lambda^{y_i} e^{-\lambda}}{y_i!} = \frac{\prod_{i=1}^n \lambda^{y_i} \prod_{i=1}^n e^{-\lambda}}{\prod_{i=1}^n y_i!} = \frac{\lambda^{\sum_{i=1}^n y_i} e^{-n\lambda}}{\prod_{i=1}^n y_i!} \quad (\text{a.1.3.4})$$

derivations for equation (1.3.13)

The derivative is given by:

$$\frac{d \log(f(y|\lambda))}{d\lambda} = \frac{\sum_{i=1}^n y_i}{\lambda} - n \quad (\text{a.1.3.5})$$

Set the value to 0 and solve for λ :

$$\begin{aligned} \frac{\sum_{i=1}^n y_i}{\lambda} - n &= 0 \\ \Leftrightarrow \frac{\sum_{i=1}^n y_i}{\lambda} &= n \\ \Leftrightarrow \lambda &= \frac{1}{n} \sum_{i=1}^n y_i \end{aligned} \quad (\text{a.1.3.6})$$

derivations for equation (1.3.16)

$$\begin{aligned}
 & \lambda^{\sum_{i=1}^n y_i} e^{-n\lambda} \times \lambda^{a-1} e^{-\lambda/b} \\
 = & \lambda^{a+\sum_{i=1}^n y_i-1} e^{-\lambda(n+1/b)} \\
 = & \lambda^{a+\sum_{i=1}^n y_i-1} e^{-\lambda/(n+1/b)^{-1}}
 \end{aligned} \tag{a.1.3.7}$$

Now:

$$(n+1/b)^{-1} = \frac{1}{n+1/b} = \frac{b}{bn+1} \tag{a.1.3.8}$$

Hence:

$$\begin{aligned}
 & \lambda^{\sum_{i=1}^n y_i} e^{-n\lambda} \times \lambda^{a-1} e^{-\lambda/b} \\
 = & \lambda^{a+\sum_{i=1}^n y_i-1} e^{-\lambda/\frac{b}{bn+1}}
 \end{aligned} \tag{a.1.3.9}$$

derivations for equation (1.3.19)

$$\begin{aligned}
 & f(y|\mu) \\
 = & \prod_{i=1}^n (2\pi\sigma)^{-1/2} \exp\left(-\frac{1}{2} \frac{(y_i - \mu)^2}{\sigma}\right) \\
 = & \prod_{i=1}^n (2\pi\sigma)^{-1/2} \prod_{i=1}^n \exp\left(-\frac{1}{2} \frac{(y_i - \mu)^2}{\sigma}\right) \\
 = & (2\pi\sigma)^{-n/2} \exp\left(-\frac{1}{2} \sum_{i=1}^n \frac{(y_i - \mu)^2}{\sigma}\right)
 \end{aligned} \tag{a.1.3.10}$$

derivations for equation (1.3.21)

The derivative is given by:

$$\frac{d \log(f(y|\mu))}{d\mu} = \sum_{i=1}^n \frac{(y_i - \mu)}{\sigma} \tag{a.1.3.11}$$

Set the value to 0 and solve for μ :

$$\begin{aligned}
 & \sum_{i=1}^n \frac{(y_i - \mu)}{\sigma} = 0 \\
 \Leftrightarrow & \sum_{i=1}^n (y_i - \mu) = 0 \\
 \Leftrightarrow & \sum_{i=1}^n y_i - n\mu = 0 \\
 \Leftrightarrow & \sum_{i=1}^n y_i = n\mu \\
 \Leftrightarrow & \mu = \frac{1}{n} \sum_{i=1}^n y_i
 \end{aligned} \tag{a.1.3.12}$$

derivations for equation (1.3.24)

First group the exponential terms:

$$\pi(\mu|y) \propto \exp\left(-\frac{1}{2} \sum_{i=1}^n \frac{(y_i - \mu)^2}{\sigma}\right) \times \exp\left(-\frac{1}{2} \frac{(\mu - m)^2}{v}\right) = \exp\left(-\frac{1}{2} \left[\sum_{i=1}^n \frac{(y_i - \mu)^2}{\sigma} + \frac{(\mu - m)^2}{v} \right]\right) \quad (\text{a.1.3.13})$$

Develop the term within the square bracket:

$$\begin{aligned} & \sum_{i=1}^n \frac{(y_i - \mu)^2}{\sigma} + \frac{(\mu - m)^2}{v} \\ &= \frac{1}{\sigma} \sum_{i=1}^n (y_i^2 + \mu^2 - 2\mu y_i) + \frac{1}{v} (\mu^2 + m^2 - 2\mu m) \\ &= \frac{1}{\sigma} \left(\sum_{i=1}^n y_i^2 + n\mu^2 - 2\mu \sum_{i=1}^n y_i \right) + \frac{1}{v} (\mu^2 + m^2 - 2\mu m) \end{aligned} \quad (\text{a.1.3.14})$$

Group the terms:

$$= \mu^2 \left(\frac{n}{\sigma} + \frac{1}{v} \right) - 2\mu \left(\frac{1}{\sigma} \sum_{i=1}^n y_i + \frac{m}{v} \right) + \frac{1}{\sigma} \sum_{i=1}^n y_i^2 + \frac{m^2}{v} \quad (\text{a.1.3.15})$$

Set back in (a.1.3.13):

$$\pi(\mu|y) \propto \exp\left(-\frac{1}{2} \left[\mu^2 \left(\frac{n}{\sigma} + \frac{1}{v} \right) - 2\mu \left(\frac{1}{\sigma} \sum_{i=1}^n y_i + \frac{m}{v} \right) + \frac{1}{\sigma} \sum_{i=1}^n y_i^2 + \frac{m^2}{v} \right]\right) \quad (\text{a.1.3.16})$$

Further aspects of Bayesian priors and posteriors

derivations for equation (1.4.5)

First group the terms:

$$\begin{aligned}
 & \pi(\mu, \sigma|y) \\
 & \propto \sigma^{-n/2} \exp\left(-\frac{1}{2} \sum_{i=1}^n \frac{(y_i - \mu)^2}{\sigma}\right) \times \sigma^{-1/2} \exp\left(-\frac{1}{2} \frac{(\mu - m)^2}{v\sigma}\right) \times \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right) \\
 & = \sigma^{-(n+\alpha)/2-1} \times \sigma^{-1/2} \times \exp\left(-\frac{1}{2\sigma} \left[\sum_{i=1}^n (y_i - \mu)^2 + \frac{(\mu - m)^2}{v} + \delta \right] \right)
 \end{aligned} \tag{a.1.4.1}$$

Develop the term in the square bracket:

$$\begin{aligned}
 & \sum_{i=1}^n (y_i - \mu)^2 + \frac{(\mu - m)^2}{v} + \delta \\
 & = \sum_{i=1}^n (y_i^2 + \mu^2 - 2\mu y_i) + \frac{\mu^2}{v} + \frac{m^2}{v} - 2\mu \frac{m}{v} + \delta \\
 & = \sum_{i=1}^n y_i^2 + n\mu^2 - 2\mu \sum_{i=1}^n y_i + \frac{\mu^2}{v} + \frac{m^2}{v} - 2\mu \frac{m}{v} + \delta \\
 & = \mu^2 \left(n + \frac{1}{v} \right) - 2\mu \left(\sum_{i=1}^n y_i + \frac{m}{v} \right) + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} + \delta
 \end{aligned} \tag{a.1.4.2}$$

Complete the squares:

$$= \mu^2 \left(n + \frac{1}{v} \right) - 2\mu \frac{\bar{v}}{\bar{v}} \left(\sum_{i=1}^n y_i + \frac{m}{v} \right) + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} + \delta + \frac{\bar{m}^2}{\bar{v}} - \frac{\bar{m}^2}{\bar{v}} \tag{a.1.4.3}$$

Define:

$$\bar{v} = \left(n + \frac{1}{v} \right)^{-1} \quad \bar{m} = \bar{v} \left(\sum_{i=1}^n y_i + \frac{m}{v} \right) \tag{a.1.4.4}$$

Then (a.1.4.3) rewrites:

$$\begin{aligned}
 & = \frac{\mu^2}{\bar{v}} + \frac{\bar{m}^2}{\bar{v}} - 2\mu \frac{\bar{m}}{\bar{v}} + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} + \delta - \frac{\bar{m}^2}{\bar{v}} \\
 & = \frac{(\mu - \bar{m})^2}{\bar{v}} + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} + \delta - \frac{\bar{m}^2}{\bar{v}}
 \end{aligned} \tag{a.1.4.5}$$

Substituting back (a.1.4.5) in (a.1.4.1) eventually yields:

$$\begin{aligned}
& \pi(\mu, \sigma | y) \\
& \propto \sigma^{-(n+\alpha)/2-1} \times \sigma^{-1/2} \times \exp \left(-\frac{1}{2\sigma} \left[\frac{(\mu - \bar{m})^2}{\bar{v}} + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} + \delta - \frac{\bar{m}^2}{\bar{v}} \right] \right) \\
& = \sigma^{-(n+\alpha)/2-1} \times \sigma^{-1/2} \times \exp \left(-\frac{1}{2} \frac{(\mu - \bar{m})^2}{\sigma \bar{v}} \right) \times \exp \left(-\frac{1}{2\sigma} \left[\sum_{i=1}^n y_i^2 + \frac{m^2}{v} + \delta - \frac{\bar{m}^2}{\bar{v}} \right] \right) \\
& = \sigma^{-\bar{\alpha}/2-1} \times \sigma^{-1/2} \times \exp \left(-\frac{1}{2} \frac{(\mu - \bar{m})^2}{\sigma \bar{v}} \right) \times \exp \left(-\frac{\bar{\delta}}{2\sigma} \right) \tag{a.1.4.6}
\end{aligned}$$

with:

$$\bar{\alpha} = n + \alpha \quad \bar{\delta} = \sum_{i=1}^n y_i^2 + \frac{m^2}{v} + \delta - \frac{\bar{m}^2}{\bar{v}} \tag{a.1.4.7}$$

derivations for equation (1.4.10)

Rearrange the terms:

$$\begin{aligned}
& \pi(\mu | y) \\
& \propto \Gamma \left(\frac{\bar{\alpha} + 1}{2} \right) \left(\frac{\bar{\delta} + (\mu - \bar{m})^2 / \bar{v}}{2} \right)^{-\frac{\bar{\alpha}+1}{2}} \\
& \propto \left(\frac{\bar{\delta} + (\mu - \bar{m})^2 / \bar{v}}{2} \right)^{-\frac{\bar{\alpha}+1}{2}} \\
& \propto \left(\bar{\delta} + \frac{(\mu - \bar{m})^2}{\bar{v}} \right)^{-\frac{\bar{\alpha}+1}{2}} \\
& = \bar{\delta} \left(1 + \frac{(\mu - \bar{m})^2}{\bar{\delta} \bar{v}} \right)^{-\frac{\bar{\alpha}+1}{2}} \\
& \propto \left(1 + \frac{(\mu - \bar{m})^2}{\bar{\delta} \bar{v}} \right)^{-\frac{\bar{\alpha}+1}{2}} \\
& = \left(1 + \frac{1}{\bar{\alpha}} \frac{(\mu - \bar{m})^2}{\bar{\delta} \bar{v} / \bar{\alpha}} \right)^{-\frac{\bar{\alpha}+1}{2}} \tag{a.1.4.8}
\end{aligned}$$

derivations for equation (1.4.13)

Solve for the derivative:

$$\begin{aligned}
& 2 \int (\hat{\theta} - \theta) \pi(\theta | y) d\theta = 0 \\
& \Leftrightarrow \int (\hat{\theta} - \theta) \pi(\theta | y) d\theta = 0 \\
& \Leftrightarrow \int \hat{\theta} \pi(\theta | y) d\theta - \int \theta \pi(\theta | y) d\theta = 0 \\
& \Leftrightarrow \hat{\theta} \int \pi(\theta | y) d\theta = \int \theta \pi(\theta | y) d\theta \\
& \Leftrightarrow \hat{\theta} = \int \theta \pi(\theta | y) d\theta \tag{a.1.4.9}
\end{aligned}$$

derivations for equation (1.4.16)

Rearrange the expression:

$$\begin{aligned}
 f(y) &= \int \int (2\pi)^{-n/2} (2\pi)^{-1/2} v^{-1/2} \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \\
 &\quad \times \sigma^{-n/2} \exp\left(-\frac{1}{2} \sum_{i=1}^n \frac{(y_i - \mu)^2}{\sigma}\right) \times \sigma^{-1/2} \exp\left(-\frac{1}{2} \frac{(\mu - m)^2}{v\sigma}\right) \times \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right) d\mu d\sigma
 \end{aligned} \tag{a.1.4.10}$$

The second row can be recognised as equation (a.1.4.1). Using the same manipulations, one obtains equation (a.1.4.6), and thus the previous expression rewrites as:

$$\begin{aligned}
 f(y) &= \int \int (2\pi)^{-n/2} (2\pi)^{-1/2} v^{-1/2} \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \\
 &\quad \times \sigma^{-1/2} \exp\left(-\frac{1}{2} \frac{(\mu - \bar{m})^2}{\sigma \bar{v}}\right) \times \sigma^{-\bar{\alpha}/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right) d\mu d\sigma
 \end{aligned} \tag{a.1.4.11}$$

with $\bar{m}$, $\bar{v}$, $\bar{\alpha}$ and $\bar{\delta}$ defined as in (a.1.4.4) and (a.1.4.7). Now add multiplicative terms to obtain normal and inverse Gamma probability density functions, and take constants out of the integral:

$$\begin{aligned}
 f(y) &= (2\pi)^{-n/2} v^{-1/2} \bar{v}^{1/2} \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \frac{\Gamma(\bar{\alpha}/2)}{\bar{\delta}/2^{\bar{\alpha}/2}} \\
 &\quad \times \int \int (2\pi \bar{v} \sigma)^{-1/2} \exp\left(-\frac{1}{2} \frac{(\mu - \bar{m})^2}{\sigma \bar{v}}\right) \times \frac{\bar{\delta}/2^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)} \sigma^{-\bar{\alpha}/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right) d\mu d\sigma
 \end{aligned} \tag{a.1.4.12}$$

The expression can simplify further. Consider only the constant on the first line:

$$\begin{aligned}
 &(2\pi)^{-n/2} v^{-1/2} \bar{v}^{1/2} \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \frac{\Gamma(\bar{\alpha}/2)}{\bar{\delta}/2^{\bar{\alpha}/2}} \\
 &= 2^{-n/2} \pi^{-n/2} v^{-1/2} ((n+1/v)^{-1})^{1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{2^{\bar{\alpha}/2}}{2^{\alpha/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \\
 &= 2^{-n/2} \pi^{-n/2} v^{-1/2} (n+1/v)^{-1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{2^{(\alpha+n)/2}}{2^{\alpha/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \\
 &= \pi^{-n/2} (1+vn)^{-1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)}
 \end{aligned} \tag{a.1.4.13}$$

Substitute back in (a.1.4.12):

$$\begin{aligned}
 f(y) &= \pi^{-n/2} (1+vn)^{-1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \\
 &\quad \times \int \int (2\pi \bar{v} \sigma)^{-1/2} \exp\left(-\frac{1}{2} \frac{(\mu - \bar{m})^2}{\sigma \bar{v}}\right) \times \frac{\bar{\delta}/2^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)} \sigma^{-\bar{\alpha}/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right) d\mu d\sigma
 \end{aligned} \tag{a.1.4.14}$$

derivations for equation (1.4.19)

Rearrange the expression:

$$\begin{aligned}
 \mathbb{P}(M_i|y) &= \frac{f(y|M_i) \mathbb{P}(M_i)}{f(y)} \\
 \Leftrightarrow \mathbb{P}(M_i|y) &= \frac{f(y, M_i) / \pi(M_i) \mathbb{P}(M_i)}{f(y)} \\
 \Leftrightarrow \mathbb{P}(M_i|y) &= \frac{\int f(y, M_i, \theta_i) / \pi(M_i) d\theta_i \mathbb{P}(M_i)}{f(y)} \\
 \Leftrightarrow \mathbb{P}(M_i|y) &= \frac{\int \frac{f(y, M_i, \theta_i)}{\pi(M_i, \theta)} \frac{\pi(M_i, \theta)}{\pi(M_i)} d\theta_i \mathbb{P}(M_i)}{f(y)} \\
 \Leftrightarrow \mathbb{P}(M_i|y) &= \frac{\int f(y|M_i, \theta_i) \pi(\theta|M_i) d\theta_i \mathbb{P}(M_i)}{f(y)} \tag{a.1.4.15}
 \end{aligned}$$

derivations for equation (1.4.24)

Rearrange the expression to obtain:

$$\begin{aligned}
 &f(\hat{y}|y) \\
 &= \int \int \sigma^{-1/2} \exp\left(-\frac{1}{2} \frac{(\hat{y} - \mu)^2}{\sigma}\right) \times \sigma^{-n/2} \exp\left(-\frac{1}{2} \sum_{i=1}^n \frac{(y_i - \mu)^2}{\sigma}\right) \\
 &\quad \times \sigma^{-1/2} \exp\left(-\frac{1}{2} \frac{(\mu - m)^2}{\nu \sigma}\right) \times \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right) d\mu d\sigma \\
 &= \int \int \sigma^{-1/2} \exp\left(-\frac{1}{2\sigma} \left[(\hat{y} - \mu)^2 + \sum_{i=1}^n (y_i - \mu)^2 + \frac{(\mu - m)^2}{\nu} + \delta \right] \right) \sigma^{-(\alpha+n+1)/2-1} d\mu d\sigma \\
 &= \int \int \sigma^{-1/2} \exp\left(-\frac{1}{2\sigma} \left[(\hat{y} - \mu)^2 + \sum_{i=1}^n (y_i - \mu)^2 + \frac{(\mu - m)^2}{\nu} + \delta \right] \right) \sigma^{-\hat{\alpha}/2-1} d\mu d\sigma \tag{a.1.4.16}
 \end{aligned}$$

with:

$$\hat{\alpha} = \alpha + n + 1 \tag{a.1.4.17}$$

Consider the term in square brackets:

$$\begin{aligned}
& (\hat{y} - \mu)^2 + \sum_{i=1}^n (y_i - \mu)^2 + \frac{(\mu - m)^2}{v} + \delta \\
&= \hat{y}^2 + \mu^2 - 2\hat{y}\mu + \sum_{i=1}^n (y_i^2 + \mu^2 - 2y_i\mu) + \frac{\mu^2}{v} + \frac{m^2}{v} - 2\mu\frac{m}{v} + \delta \\
&= \hat{y}^2 + \mu^2 - 2\hat{y}\mu + \sum_{i=1}^n y_i^2 + n\mu^2 - 2\mu \sum_{i=1}^n y_i + \frac{\mu^2}{v} + \frac{m^2}{v} - 2\mu\frac{m}{v} + \delta \\
&= \left(\delta + \hat{y}^2 + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} \right) + \mu^2 \left(1 + n + \frac{1}{v} \right) - 2\mu \left(\hat{y} + \sum_{i=1}^n y_i + \frac{m}{v} \right) \\
&= \left(\delta + \hat{y}^2 + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} \right) + \mu^2 \left(1 + n + \frac{1}{v} \right) - 2\mu \frac{\hat{v}}{\hat{v}} \left(\hat{y} + \sum_{i=1}^n y_i + \frac{m}{v} \right) + \frac{\hat{m}^2}{\hat{v}} - \frac{\hat{m}^2}{\hat{v}} \\
&= \left(\delta + \hat{y}^2 + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \frac{\hat{m}^2}{\hat{v}} \right) + \mu^2 \left(1 + n + \frac{1}{v} \right) - 2\mu \frac{\hat{v}}{\hat{v}} \left(\hat{y} + \sum_{i=1}^n y_i + \frac{m}{v} \right) + \frac{\hat{m}^2}{\hat{v}} \tag{a.1.4.18}
\end{aligned}$$

Define:

$$\hat{\delta} = \left(\delta + \hat{y}^2 + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \frac{\hat{m}^2}{\hat{v}} \right) \quad \hat{v} = \left(1 + n + \frac{1}{v} \right)^{-1} \quad \hat{m} = \hat{v} \left(\hat{y} + \sum_{i=1}^n y_i + \frac{m}{v} \right) \tag{a.1.4.19}$$

Then (a.1.4.17) becomes:

$$\begin{aligned}
&= \hat{\delta} + \frac{\mu^2}{\hat{v}} - 2\mu \frac{\hat{m}}{\hat{v}} + \frac{\hat{m}^2}{\hat{v}} \\
&= \hat{\delta} + \frac{(\mu - \hat{m})^2}{\hat{v}} \tag{a.1.4.20}
\end{aligned}$$

Substitute back in (a.1.4.16):

$$\begin{aligned}
& f(\hat{y}|y) \\
&= \int \int \sigma^{-1/2} \exp \left(-\frac{1}{2\sigma} \left[\hat{\delta} + \frac{(\mu - \hat{m})^2}{\hat{v}} \right] \right) \sigma^{-\hat{\alpha}/2-1} d\mu d\sigma \\
&= \int \int \sigma^{-1/2} \exp \left(-\frac{1}{2} \frac{(\mu - \hat{m})^2}{\hat{v}\sigma} \right) \times \sigma^{-\hat{\alpha}/2-1} \exp \left(-\frac{\hat{\delta}}{2\sigma} \right) d\mu d\sigma \\
&= \int \sigma^{-\hat{\alpha}/2-1} \exp \left(-\frac{\hat{\delta}}{2\sigma} \right) \int \sigma^{-1/2} \exp \left(-\frac{1}{2} \frac{(\mu - \hat{m})^2}{\hat{v}\sigma} \right) d\mu d\sigma \tag{a.1.4.21}
\end{aligned}$$

The second integral contains the kernel of a normal distribution with mean $\hat{m}$ and variance $\hat{v}\sigma$. It thus integrates to a constant (not involving $\hat{y}$) and can be relegated to the normalization constant, yielding:

$$\propto \int \sigma^{-(\hat{\alpha}+1)/2-1} \exp \left(-\frac{\hat{\delta}}{2\sigma} \right) d\sigma \tag{a.1.4.22}$$

The remaining integral contains the kernel of an inverse Gamma distribution with shape $\hat{\alpha}$ and scale $\hat{\delta}$. It integrates to the reciprocal of the normalization constant of the inverse Gamma distribution (see book1, section 4.3), which does involve $\hat{y}$. The term must thus be retained, yielding:

$$\begin{aligned}
& f(\hat{y}|y) \\
& \propto \Gamma(\hat{\alpha})(\hat{\delta}/2)^{-\hat{\alpha}/2} \\
& \propto (\hat{\delta}/2)^{-\hat{\alpha}/2} \\
& \propto (\hat{\delta})^{-\hat{\alpha}/2} \\
& = \left(\delta + \hat{y}^2 + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \frac{\hat{m}^2}{\hat{v}} \right)^{-\hat{\alpha}/2} \\
& = \left(\delta + \hat{y}^2 + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \hat{v} \left[\hat{y} + \sum_{i=1}^n y_i + \frac{m}{v} \right]^2 \right)^{-\hat{\alpha}/2}
\end{aligned} \tag{a.1.4.23}$$

Define:

$$\tilde{m} = \sum_{i=1}^n y_i + \frac{m}{v} \tag{a.1.4.24}$$

Then (a.1.4.23) becomes:

$$\begin{aligned}
& = \left(\delta + \hat{y}^2 + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \hat{v}[\hat{y} + \tilde{m}]^2 \right)^{-\hat{\alpha}/2} \\
& = \left(\delta + \hat{y}^2 + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \hat{v}\hat{y}^2 - \hat{v}\tilde{m}^2 - 2\hat{v}\tilde{m}\hat{y} \right)^{-\hat{\alpha}/2} \\
& = \left(\delta + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \hat{v}\tilde{m}^2 + \hat{y}^2(1 - \hat{v}) - 2\hat{v}\tilde{m}\hat{y} \right)^{-\hat{\alpha}/2}
\end{aligned} \tag{a.1.4.25}$$

Complete the squares:

$$\begin{aligned}
& = \left(\delta + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \hat{v}\tilde{m}^2 + \hat{y}^2(1 - \hat{v}) - 2\hat{v}\frac{\ddot{v}}{\ddot{v}}\tilde{m}\hat{y} + \frac{\dot{m}^2}{\ddot{v}} - \frac{\ddot{m}^2}{\ddot{v}} \right)^{-\hat{\alpha}/2} \\
& = \left(\left[\delta + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \hat{v}\tilde{m}^2 - \frac{\ddot{m}^2}{\ddot{v}} \right] + \hat{y}^2(1 - \hat{v}) - 2\hat{v}\frac{\ddot{v}}{\ddot{v}}\tilde{m}\hat{y} + \frac{\dot{m}^2}{\ddot{v}} \right)^{-\hat{\alpha}/2}
\end{aligned} \tag{a.1.4.26}$$

Define:

$$\bar{\alpha} = \alpha + n \quad \bar{\delta} = \delta + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \hat{v}\tilde{m}^2 - \frac{\ddot{m}^2}{\ddot{v}} \quad \ddot{v} = (1 - \hat{v})^{-1} \quad \ddot{m} = \hat{v}\ddot{v}\tilde{m} \tag{a.1.4.27}$$

Then (a.1.4.26) becomes:

$$\begin{aligned}
&= \left(\ddot{\delta} + \frac{\hat{y}^2}{\hat{v}} - 2\hat{y}\frac{\dot{m}}{\hat{v}} + \frac{\dot{m}^2}{\hat{v}} \right)^{-(\bar{\alpha}+1)/2} \\
&= \left(\ddot{\delta} + \frac{(\hat{y} - \dot{m})^2}{\hat{v}} \right)^{-(\bar{\alpha}+1)/2} \\
&= \ddot{\delta}^{-(\bar{\alpha}+1)/2} \left(1 + \frac{(\hat{y} - \dot{m})^2}{\ddot{\delta}\hat{v}} \right)^{-(\bar{\alpha}+1)/2} \\
&\propto \left(1 + \frac{(\hat{y} - \dot{m})^2}{\ddot{\delta}\hat{v}} \right)^{-(\bar{\alpha}+1)/2} \\
&= \left(1 + \frac{1}{\bar{\alpha}} \frac{(\hat{y} - \dot{m})^2}{\ddot{\delta}\hat{v}/\bar{\alpha}} \right)^{-(\bar{\alpha}+1)/2} \tag{a.1.4.28}
\end{aligned}$$

Finally, reformulate all the messy terms:

$$\begin{aligned}
\hat{v} &= (1 - \hat{v})^{-1} = \frac{1}{1 - \hat{v}} = \frac{1}{1 - \frac{1}{1+n+\frac{1}{v}}} = \frac{1}{\frac{1+n+\frac{1}{v}-1}{1+n+\frac{1}{v}}} = \frac{1}{\frac{n+\frac{1}{v}}{1+n+\frac{1}{v}}} = \frac{1+n+\frac{1}{v}}{n+\frac{1}{v}} = \frac{v+vn+1}{vn+1} \\
&= 1 + \frac{v}{vn+1} = 1 + \frac{1}{n+1/v} = 1 + \left(n + \frac{1}{v} \right)^{-1} = 1 + \bar{v} \tag{a.1.4.29}
\end{aligned}$$

with $\bar{v}$ defined as in (a.1.4.4).

Also:

$$\frac{\hat{v}}{1 - \hat{v}} = \frac{\frac{1}{1+n+\frac{1}{v}}}{1 - \frac{1}{1+n+\frac{1}{v}}} = \frac{\frac{1}{1+n+\frac{1}{v}}}{\frac{1+n+\frac{1}{v}-1}{1+n+\frac{1}{v}}} = \frac{1}{\frac{n+\frac{1}{v}}{1+n+\frac{1}{v}}} = \frac{1}{n+\frac{1}{v}} = \left(n + \frac{1}{v} \right)^{-1} = \bar{v} \tag{a.1.4.30}$$

Then:

$$\ddot{m} = \hat{v}\ddot{m} = \frac{\hat{v}}{1 - \hat{v}}\ddot{m} = \bar{v}\ddot{m} = \bar{v} \left(\sum_{i=1}^n y_i + \frac{m}{v} \right) = \bar{m} \tag{a.1.4.31}$$

with $\bar{m}$ defined as in (a.1.4.4).

Finally:

$$\begin{aligned}
\hat{v}\ddot{m}^2 + \frac{\dot{m}^2}{\hat{v}} &= \hat{v}\ddot{m}^2 + (\hat{v}\ddot{m})^2/\hat{v} = \hat{v}\ddot{m}^2 + \hat{v}^2\ddot{m}^2 = \hat{v}\ddot{m}^2(1 + \hat{v}) = \hat{v}\ddot{m}^2 \left(1 + \frac{\hat{v}}{1 - \hat{v}} \right) \\
&= \hat{v}\ddot{m}^2 \left(\frac{1 - \hat{v} + \hat{v}}{1 - \hat{v}} \right) = \hat{v}\ddot{m}^2 \left(\frac{1}{1 - \hat{v}} \right) = \ddot{m}^2 \left(\frac{\hat{v}}{1 - \hat{v}} \right) = \ddot{m}^2 \bar{v} = \bar{m}\bar{m} = \frac{\bar{m}}{\bar{v}}\bar{m} = \frac{\bar{m}^2}{\bar{v}} \tag{a.1.4.32}
\end{aligned}$$

Substitute in (a.1.4.27) to obtain:

$$\ddot{\delta} = \delta + \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \frac{\bar{m}^2}{\bar{v}} = \bar{\delta} \tag{a.1.4.33}$$

with $\bar{\delta}$ defined as in (a.1.4.7).

Substitute (a.1.4.29), (a.1.4.31) and (a.1.4.33) in (a.1.4.28) to eventually obtain:

$$f(\hat{y}|y) \propto \left(1 + \frac{1}{\bar{\alpha}} \frac{(\hat{y} - \bar{m})^2}{\bar{\delta}(1 + \bar{v})/\bar{\alpha}} \right)^{-(\bar{\alpha}+1)/2} \tag{a.1.4.34}$$

Properties of Bayesian estimates

derivations for equation (1.5.1)

The mean of a Beta distribution with shapes a and b is given by $\frac{a}{a+b}$. Given the posterior hyperparameters $\bar{\alpha} = \alpha + m$ and $\bar{\beta} = \beta + n - m$, the posterior mean writes as:

$$\begin{aligned}
 & \mathbb{E}(p|y) \\
 &= \frac{\bar{\alpha}}{\bar{\alpha} + \bar{\beta}} \\
 &= \frac{\alpha + m}{\alpha + m + \beta + n - m} \\
 &= \frac{\alpha + m}{\alpha + \beta + n} \\
 &= \frac{\alpha}{\alpha + \beta + n} + \frac{m}{\alpha + \beta + n} \\
 &= \frac{\alpha}{\alpha + \beta} \frac{\alpha + \beta}{\alpha + \beta + n} + \frac{m}{n} \frac{n}{\alpha + \beta + n} \\
 &= \gamma \mathbb{E}(p) + (1 - \gamma) \hat{p}
 \end{aligned} \tag{a.1.5.1}$$

with:

$$\mathbb{E}(p) = \frac{\alpha}{\alpha + \beta} \quad \hat{p} = \frac{m}{n} \quad \gamma = \frac{\alpha + \beta}{\alpha + \beta + n} \tag{a.1.5.2}$$

derivations for equation (1.5.2)

The mean of a Gamma distribution with shape a and scale b is given by ab . Given the posterior hyperparameters $\bar{a} = a + \sum_{i=1}^n y_i$ and $\bar{b} = \frac{b}{bn+1}$, the posterior mean writes as:

$$\begin{aligned}
 & \mathbb{E}(\lambda|y) \\
 &= \frac{(a + \sum_{i=1}^n y_i)b}{bn + 1} \\
 &= \frac{ab}{bn + 1} + \frac{b \sum_{i=1}^n y_i}{bn + 1} \\
 &= ab \left(\frac{1}{bn + 1} \right) + \frac{\sum_{i=1}^n y_i}{n} \left(\frac{bn}{bn + 1} \right) \\
 &= \gamma \mathbb{E}(\lambda) + (1 - \gamma) \hat{\lambda}
 \end{aligned} \tag{a.1.5.3}$$

with:

$$\mathbb{E}(\lambda) = ab \quad \hat{\lambda} = \frac{\sum_{i=1}^n y_i}{n} \quad \gamma = \frac{1}{bn + 1} \tag{a.1.5.4}$$

derivations for equation (1.5.3)

The mean of a normal distribution with mean μ and variance σ is given by μ . Given the posterior hyperparameters $\bar{v} = \left(\frac{n}{\sigma} + \frac{1}{v}\right)^{-1}$ and $\bar{m} = \bar{v} \left(\frac{1}{\sigma} \sum_{i=1}^n y_i + \frac{m}{v}\right)$, the posterior variance writes as:

$$\bar{v} = \left(\frac{n}{\sigma} + \frac{1}{v}\right)^{-1} = \frac{1}{n/\sigma + 1/v} = \frac{\sigma}{n + \sigma/v} \quad (\text{a.1.5.5})$$

Then the posterior mean can be expressed as:

$$\begin{aligned} & \mathbb{E}(\mu|y) \\ &= \frac{\sigma}{n + \sigma/v} \left(\frac{1}{\sigma} \sum_{i=1}^n y_i + \frac{m}{v} \right) \\ &= \frac{1}{n + \sigma/v} \left(\sum_{i=1}^n y_i \right) + \frac{\sigma}{n + \sigma/v} \left(\frac{m}{v} \right) \\ &= \frac{n}{n + \sigma/v} \left(\frac{\sum_{i=1}^n y_i}{n} \right) + \frac{\sigma/v}{n + \sigma/v} m \\ &= \frac{vn}{vn + \sigma} \left(\frac{\sum_{i=1}^n y_i}{n} \right) + \frac{\sigma}{vn + \sigma} m \\ &= \gamma \mathbb{E}(\mu) + (1 - \gamma) \hat{\mu} \end{aligned} \quad (\text{a.1.5.6})$$

with:

$$\mathbb{E}(\mu) = m \quad \hat{\mu} = \frac{\sum_{i=1}^n y_i}{n} \quad \gamma = \frac{\sigma}{vn + \sigma} \quad (\text{a.1.5.7})$$

PART II

Simulation methods

The Gibbs sampling algorithm

derivations for equation (2.6.17)

Combine all the terms to obtain:

$$\begin{aligned}
 f(y) &\approx (2\pi\sigma)^{-n/2} \exp\left(-\frac{1}{2}\sum_{i=1}^n \frac{(y_i - \mu)^2}{\sigma}\right) \frac{(2\pi v)^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - m)^2}{v}\right)}{\frac{1}{J}\sum_{j=1}^J (2\pi\bar{v})^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - \bar{m})^2}{\bar{v}}\right)} \frac{\frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right)}{\frac{\bar{\delta}/2^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)} \sigma^{-\bar{\alpha}/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right)} \\
 &= (2\pi)^{-n/2} \sigma^{-n/2} \exp\left(-\frac{1}{2}\sum_{i=1}^n \frac{(y_i - \mu)^2}{\sigma}\right) \frac{v^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - m)^2}{v}\right)}{\frac{1}{J}\sum_{j=1}^J \bar{v}^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - \bar{m})^2}{\bar{v}}\right)} \frac{\frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right)}{\frac{\bar{\delta}/2^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)} \sigma^{-(\alpha+n)/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right)} \\
 &= (2\pi)^{-n/2} \exp \frac{v^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - m)^2}{v}\right)}{\frac{1}{J}\sum_{j=1}^J \bar{v}^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - \bar{m})^2}{\bar{v}}\right)} \frac{\frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \exp\left(-\frac{1}{2\sigma} [\delta + \sum_{i=1}^n (y_i - \mu)^2]\right)}{\frac{\bar{\delta}/2^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right)} \\
 &= (2\pi)^{-n/2} \exp \frac{v^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - m)^2}{v}\right)}{\frac{1}{J}\sum_{j=1}^J \bar{v}^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - \bar{m})^2}{\bar{v}}\right)} \frac{\frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)}}{\frac{\bar{\delta}/2^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)}} \\
 &= 2^{-n/2} \pi^{-n/2} \exp \frac{v^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - m)^2}{v}\right)}{\frac{1}{J}\sum_{j=1}^J \bar{v}^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - \bar{m})^2}{\bar{v}}\right)} \frac{\frac{\delta^{\alpha/2}}{\Gamma(\alpha/2)} 2^{\bar{\alpha}/2}}{\frac{\bar{\delta}^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)} 2^{\alpha/2}} \\
 &= \pi^{-n/2} \exp \frac{v^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - m)^2}{v}\right)}{\frac{1}{J}\sum_{j=1}^J \bar{v}^{-1/2} \exp\left(-\frac{1}{2}\frac{(\mu - \bar{m})^2}{\bar{v}}\right)} \frac{\frac{\delta^{\alpha/2}}{\Gamma(\alpha/2)}}{\frac{\bar{\delta}^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)}} \\
 &= \pi^{-n/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \frac{\exp\left(-\frac{1}{2}\frac{(\mu - m)^2}{v}\right)}{\frac{1}{J}\sum_{j=1}^J (v/\bar{v})^{1/2} \exp\left(-\frac{1}{2}\frac{(\mu - \bar{m})^2}{\bar{v}}\right)} \tag{a.2.6.1}
 \end{aligned}$$

Now:

$$v/\bar{v} = v(n/\sigma + 1/v) = vn/\sigma + 1 \tag{a.2.6.2}$$

Hence:

$$f(y) \approx \pi^{-n/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \frac{\exp\left(-\frac{1}{2}\frac{(\mu - m)^2}{v}\right)}{\frac{1}{J}\sum_{j=1}^J (1 + vn/\sigma)^{1/2} \exp\left(-\frac{1}{2}\frac{(\mu - \bar{m})^2}{\bar{v}}\right)} \tag{a.2.6.3}$$

The Metropolis-Hastings algorithm

derivations for equation (2.7.7)

Rearrange:

$$\begin{aligned}
 & \pi(\mu|y, \lambda) \\
 & \propto \exp\left(-\frac{1}{2} \sum_{i=1}^n \frac{(y_i - \mu)^2}{\exp(\lambda)}\right) \times \exp\left(-\frac{1}{2} \frac{(\mu - m)^2}{v}\right) \\
 & = \exp\left(-\frac{1}{2} \left[\sum_{i=1}^n \frac{(y_i - \mu)^2}{\exp(\lambda)} + \frac{(\mu - m)^2}{v} \right]\right)
 \end{aligned} \tag{a.2.7.1}$$

Develop the term within the square bracket:

$$\begin{aligned}
 & \sum_{i=1}^n \frac{(y_i - \mu)^2}{\exp(\lambda)} + \frac{(\mu - m)^2}{v} \\
 & = \frac{1}{\exp(\lambda)} \sum_{i=1}^n (y_i^2 + \mu^2 - 2\mu y_i) + \frac{1}{v} (\mu^2 + m^2 - 2\mu m) \\
 & = \frac{1}{\exp(\lambda)} \left(\sum_{i=1}^n y_i^2 + n\mu^2 - 2\mu \sum_{i=1}^n y_i \right) + \frac{1}{v} (\mu^2 + m^2 - 2\mu m)
 \end{aligned} \tag{a.2.7.2}$$

Group the terms and complete the squares:

$$\begin{aligned}
 & = \mu^2 \left(\frac{n}{\exp(\lambda)} + \frac{1}{v} \right) - 2\mu \left(\frac{1}{\exp(\lambda)} \sum_{i=1}^n y_i + \frac{m}{v} \right) + \frac{1}{\exp(\lambda)} \sum_{i=1}^n y_i^2 + \frac{m^2}{v} \\
 & = \mu^2 \left(\frac{n}{\exp(\lambda)} + \frac{1}{v} \right) - 2\mu \frac{\bar{v}}{\bar{v}} \left(\frac{1}{\exp(\lambda)} \sum_{i=1}^n y_i + \frac{m}{v} \right) + \frac{1}{\exp(\lambda)} \sum_{i=1}^n y_i^2 + \frac{m^2}{v} + \frac{\bar{m}^2}{\bar{v}} - \frac{\bar{m}^2}{\bar{v}}
 \end{aligned} \tag{a.2.7.3}$$

Define:

$$\bar{v} = \left(\frac{n}{\exp(\lambda)} + \frac{1}{v} \right)^{-1} \quad \bar{m} = \bar{v} \left(\frac{1}{\exp(\lambda)} \sum_{i=1}^n y_i + \frac{m}{v} \right) \tag{a.2.7.4}$$

Then (a.2.7.3) rewrites:

$$\begin{aligned}
 & = \frac{\mu^2}{\bar{v}} + \frac{\bar{m}^2}{\bar{v}} - 2\mu \frac{\bar{m}}{\bar{v}} + \frac{1}{\exp(\lambda)} \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \frac{\bar{m}^2}{\bar{v}} \\
 & = \frac{(\mu - \bar{m})^2}{\bar{v}} + \frac{1}{\exp(\lambda)} \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \frac{\bar{m}^2}{\bar{v}}
 \end{aligned} \tag{a.2.7.5}$$

Substitute back in (a.2.7.1):

$$\begin{aligned}
& \pi(\mu|y, \lambda) \\
& \propto \exp\left(-\frac{1}{2}\left[\frac{(\mu - \bar{m})^2}{\bar{v}} + \frac{1}{\exp(\lambda)} \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \frac{\bar{m}^2}{\bar{v}}\right]\right) \\
& = \exp\left(-\frac{1}{2}\frac{(\mu - \bar{m})^2}{\bar{v}}\right) \exp\left(-\frac{1}{2}\left[\frac{1}{\exp(\lambda)} \sum_{i=1}^n y_i^2 + \frac{m^2}{v} - \frac{\bar{m}^2}{\bar{v}}\right]\right) \\
& \propto \exp\left(-\frac{1}{2}\frac{(\mu - \bar{m})^2}{\bar{v}}\right)
\end{aligned} \tag{a.2.7.6}$$

derivations for equation (2.7.14)

$$\begin{aligned}
& \alpha(\lambda^{(j-1)}, \lambda^{(j)}) \\
& = \frac{\exp(\lambda^{(j)})^{-n/2}}{\exp(\lambda^{(j-1)})^{-n/2}} \frac{\exp\left(-\frac{1}{2}\sum_{i=1}^n \frac{(y_i - \mu)^2}{\exp(\lambda^{(j)})}\right)}{\exp\left(-\frac{1}{2}\sum_{i=1}^n \frac{(y_i - \mu)^2}{\exp(\lambda^{(j-1)})}\right)} \frac{\exp\left(-\frac{1}{2}\frac{(\lambda^{(j)} - g)^2}{z}\right)}{\exp\left(-\frac{1}{2}\frac{(\lambda^{(j-1)} - g)^2}{z}\right)}
\end{aligned} \tag{a.2.7.7}$$

Consider the first term:

$$\begin{aligned}
& \frac{\exp(\lambda^{(j)})^{-n/2}}{\exp(\lambda^{(j-1)})^{-n/2}} \\
& = \exp(\lambda^{(j)} - \lambda^{(j-1)})^{-n/2} \\
& = \exp\left(\frac{n}{2}(\lambda^{(j-1)} - \lambda^{(j)})\right)
\end{aligned} \tag{a.2.7.8}$$

Consider the second term:

$$\begin{aligned}
& \frac{\exp\left(-\frac{1}{2}\sum_{i=1}^n \frac{(y_i - \mu)^2}{\exp(\lambda^{(j)})}\right)}{\exp\left(-\frac{1}{2}\sum_{i=1}^n \frac{(y_i - \mu)^2}{\exp(\lambda^{(j-1)})}\right)} \\
& = \frac{\exp\left(-\frac{1}{2}\exp(-\lambda^{(j)})\sum_{i=1}^n (y_i - \mu)^2\right)}{\exp\left(-\frac{1}{2}\exp(-\lambda^{(j-1)})\sum_{i=1}^n (y_i - \mu)^2\right)} \\
& = \exp\left(-\frac{1}{2}\left[\exp(-\lambda^{(j)}) - \exp(-\lambda^{(j-1)})\right]\sum_{i=1}^n (y_i - \mu)^2\right) \\
& = \exp\left(\frac{1}{2}\left[\exp(-\lambda^{(j-1)}) - \exp(-\lambda^{(j)})\right]\sum_{i=1}^n (y_i - \mu)^2\right)
\end{aligned} \tag{a.2.7.9}$$

Consider the third term:

$$\begin{aligned}
& \frac{\exp\left(-\frac{1}{2}\frac{(\lambda^{(j)} - g)^2}{z}\right)}{\exp\left(-\frac{1}{2}\frac{(\lambda^{(j-1)} - g)^2}{z}\right)} \\
& = \exp\left(-\frac{1}{2}\left[\frac{(\lambda^{(j)} - g)^2 - (\lambda^{(j-1)} - g)^2}{z}\right]\right) \\
& = \exp\left(\frac{1}{2}\left[\frac{(\lambda^{(j-1)} - g)^2 - (\lambda^{(j)} - g)^2}{z}\right]\right)
\end{aligned} \tag{a.2.7.10}$$

Substitute back in (a.2.7.7):

$$\begin{aligned} & \alpha(\lambda^{(j-1)}, \lambda^{(j)}) \\ = & \exp \left(\frac{1}{2} \left[\frac{n(\lambda^{(j-1)} - \lambda^{(j)}) + [\exp(-\lambda^{(j-1)}) - \exp(-\lambda^{(j)})] \sum_{i=1}^n (y_i - \mu)^2}{(\lambda^{(j-1)} - g)^2 - (\lambda^{(j)} - g)^2} \right] \right) \end{aligned} \quad (\text{a.2.7.11})$$

derivations for equation (2.7.21)

Rearrange the expression:

$$\begin{aligned} & \frac{1}{f(y)} \\ \approx & \frac{1}{J} \sum_{j=1}^J \frac{g(\theta^{(j)})}{f(y|\mu^{(j)}, \lambda^{(j)}) \pi(\mu^{(j)}) \pi(\lambda^{(j)})} \\ = & \frac{1}{J} \sum_{j=1}^J \frac{\mathbb{1}(\theta \in \hat{\Theta}) \times \omega^{-1} (2\pi)^{-k/2} |\hat{\Sigma}|^{-1/2} \exp \left(-\frac{1}{2} (\theta - \hat{\theta})' \hat{\Sigma}^{-1} (\theta - \hat{\theta}) \right) \mathbb{1}(\theta \in \hat{\Theta})}{(2\pi \exp(\lambda))^{-n/2} \exp \left(-\frac{1}{2} \sum_{i=1}^n \frac{(y_i - \mu)^2}{\exp(\lambda)} \right) (2\pi v)^{-1/2} \exp \left(-\frac{1}{2} \frac{(\mu - m)^2}{v} \right) (2\pi z)^{-1/2} \exp \left(-\frac{1}{2} \frac{(\lambda - g)^2}{z} \right)} \\ = & \mathbb{1}(\theta \in \hat{\Theta}) \times \omega^{-1} (2\pi)^{(n+2-k)/2} |\hat{\Sigma}|^{-1/2} (vz)^{1/2} \\ & \times \frac{1}{J} \sum_{j=1}^J \exp \left(\frac{1}{2} \left[n\lambda + \sum_{i=1}^n \frac{(y_i - \mu)^2}{\exp(\lambda)} + \frac{(\mu - m)^2}{v} + \frac{(\lambda - g)^2}{z} - (\theta - \hat{\theta})' \hat{\Sigma}^{-1} (\theta - \hat{\theta}) \right] \right) \\ = & \mathbb{1}(\theta \in \hat{\Theta}) \times (\omega J)^{-1} (2\pi)^{n/2} |\hat{\Sigma}|^{-1/2} (vz)^{1/2} \\ & \times \sum_{j=1}^J \exp \left(\frac{1}{2} \left[n\lambda + \sum_{i=1}^n \frac{(y_i - \mu)^2}{\exp(\lambda)} + \frac{(\mu - m)^2}{v} + \frac{(\lambda - g)^2}{z} - (\theta - \hat{\theta})' \hat{\Sigma}^{-1} (\theta - \hat{\theta}) \right] \right) \end{aligned} \quad (\text{a.2.7.12})$$

Mathematical theory

derivations for equation (2.8.11)

The definition of an invariant distribution implies that:

$$(\pi_1 \quad \pi_2 \quad \pi_3 \quad \pi_4 \quad \cdots) \begin{pmatrix} p+q & r & 0 & 0 & 0 & \cdots \\ p & q & r & 0 & 0 & \cdots \\ 0 & p & q & r & 0 & \cdots \\ 0 & 0 & p & q & r & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} = (\pi_1 \quad \pi_2 \quad \pi_3 \quad \pi_4 \quad \cdots) \quad (\text{a.2.8.1})$$

The product with the first column of P yields:

$$\begin{aligned} \pi_1(p+q) + \pi_2p &= \pi_1 \\ \Leftrightarrow \pi_2p &= \pi_1(1-p-q) \\ \Leftrightarrow \pi_2p &= \pi_1r \\ \Leftrightarrow \pi_2 &= (r/p)\pi_1 \end{aligned} \quad (\text{a.2.8.2})$$

The second column yields:

$$\begin{aligned} \pi_1r + \pi_2q + \pi_3p &= \pi_2 \\ \Leftrightarrow \pi_1r + \pi_3p &= \pi_2(1-q) \\ \Leftrightarrow \pi_1r + \pi_3p &= \pi_1(r/p)(1-q) \\ \Leftrightarrow \pi_1 + \pi_3(p/r) &= \pi_1(1-q)/p \\ \Leftrightarrow \pi_3(p/r) &= \pi_1(1-q-p)/p \\ \Leftrightarrow \pi_3(p/r) &= \pi_1(r/p) \\ \Leftrightarrow \pi_3(p/r) &= \pi_2 \\ \Leftrightarrow \pi_3 &= (r/p)\pi_2 \\ \Leftrightarrow \pi_3 &= (r/p)^2\pi_1 \end{aligned} \quad (\text{a.2.8.3})$$

The third column yields:

$$\begin{aligned}
& \pi_2 r + \pi_3 q + \pi_4 p = \pi_3 \\
\Leftrightarrow & \pi_2 r + \pi_4 p = \pi_3(1 - q) \\
\Leftrightarrow & \pi_2 r + \pi_4 p = \pi_2(r/p)(1 - q) \\
\Leftrightarrow & \pi_2 + \pi_4(p/r) = \pi_2(1 - q)/p \\
\Leftrightarrow & \pi_4(p/r) = \pi_2(1 - q - p)/p \\
\Leftrightarrow & \pi_4(p/r) = \pi_2(r/p) \\
\Leftrightarrow & \pi_4(p/r) = \pi_3 \\
\Leftrightarrow & \pi_4 = (r/p)\pi_3 \\
\Leftrightarrow & \pi_4 = (r/p)^2\pi_2 \\
\Leftrightarrow & \pi_4 = (r/p)^3\pi_1
\end{aligned} \tag{a.2.8.4}$$

Continuing this way, one obtains in general that $\pi_j = (r/p)^{j-1}\pi_1$. If an invariant distribution exists, we must have $\pi_1 + \pi_2 + \pi_3 + \dots = 1$. Hence:

$$\begin{aligned}
& \pi_1 + \pi_2 + \pi_3 + \dots = 1 \\
\Leftrightarrow & \pi_1 + (r/p)\pi_1 + (r/p)^2\pi_1 + \dots = 1 \\
\Leftrightarrow & \pi_1(1 + (r/p) + (r/p)^2 + \dots) = 1 \\
\Leftrightarrow & \pi_1 \frac{1}{1 - r/p} = 1 \\
\Leftrightarrow & \pi_1 = 1 - r/p
\end{aligned} \tag{a.2.8.5}$$

derivations for equation (2.8.16)

Start from the definition and rearrange:

$$\begin{aligned}
\pi(y_t) &= \int \pi(y_{t-1}) p(y_{t-1}, y_t) dy_{t-1} \\
&\propto \int \exp\left(-\frac{1}{2} \frac{(y_{t-1} - \mu)^2}{\sigma}\right) \exp\left(-\frac{1}{2} \frac{(y_t - c - \gamma y_{t-1})^2}{s}\right) dy_{t-1} \\
&= \int \exp\left(-\frac{1}{2} \left[\frac{(y_{t-1} - \mu)^2}{\sigma} + \frac{(y_t - c - \gamma y_{t-1})^2}{s} \right]\right) dy_{t-1} \\
&= \int \exp\left(-\frac{1}{2} \left[\frac{(y_{t-1} - \mu)^2(1 - \gamma^2)}{s} + \frac{(y_t - c - \gamma y_{t-1})^2}{s} \right]\right) dy_{t-1} \\
&= \int \exp\left(-\frac{1}{2s} [(y_{t-1} - \mu)^2(1 - \gamma^2) + (y_t - \mu(1 - \gamma) - \gamma y_{t-1})^2]\right) dy_{t-1}
\end{aligned} \tag{a.2.8.6}$$

Consider the term within the square brackets:

$$\begin{aligned}
& (y_{t-1} - \mu)^2(1 - \gamma^2) + (y_t - \mu(1 - \gamma) - \gamma y_{t-1})^2 \\
&= (1 - \gamma^2)y_{t-1}^2 + (1 - \gamma^2)\mu^2 - 2(1 - \gamma^2)\mu y_{t-1} + y_t^2 + \mu^2(1 - \gamma)^2 + \gamma^2 y_{t-1}^2 \\
&\quad - 2\mu(1 - \gamma)y_t - 2\gamma y_t y_{t-1} + 2\mu\gamma(1 - \gamma)y_{t-1} \\
&= y_{t-1}^2 + (1 - \gamma^2)\mu^2 - 2(1 - \gamma^2)\mu y_{t-1} + y_t^2 + \mu^2(1 - \gamma)^2 \\
&\quad - 2\mu(1 - \gamma)y_t - 2\gamma y_t y_{t-1} + 2\mu\gamma(1 - \gamma)y_{t-1} \\
&= y_{t-1}^2 + (1 - \gamma^2)\mu^2 - 2\mu y_{t-1} + 2\gamma^2 \mu y_{t-1} + y_t^2 + \mu^2(1 - \gamma)^2 \\
&\quad - 2\mu(1 - \gamma)y_t - 2\gamma y_t y_{t-1} + 2\mu\gamma y_{t-1} - 2\mu\gamma^2 y_{t-1}
\end{aligned} \tag{a.2.8.7}$$

$$\begin{aligned}
&= y_{t-1}^2 + (1 - \gamma^2)\mu^2 - 2\mu y_{t-1} + y_t^2 + \mu^2(1 - \gamma)^2 - 2\mu(1 - \gamma)y_t - 2\gamma y_t y_{t-1} + 2\mu\gamma y_{t-1} \\
&= y_{t-1}^2 + (1 - \gamma^2)\mu^2 - 2\mu y_{t-1}(1 - \gamma) + y_t^2 + \mu^2(1 - \gamma)^2 - 2\mu(1 - \gamma)y_t - 2\gamma y_t y_{t-1} \\
&= y_{t-1}^2 + (1 - \gamma^2)\mu^2 - 2\mu y_{t-1}(1 - \gamma) + y_t^2 + \mu^2(1 - \gamma)^2 - 2\mu(1 - \gamma^2)y_t + 2\mu(1 - \gamma)\gamma y_t - 2\gamma y_t y_{t-1} \\
&= y_{t-1}^2 + (1 - \gamma^2)\mu^2 - 2\mu y_{t-1}(1 - \gamma) + (1 - \gamma^2)y_t^2 + \gamma^2 y_t^2 + \mu^2(1 - \gamma)^2 \\
&\quad - 2\mu(1 - \gamma^2)y_t + 2\mu(1 - \gamma)\gamma y_t - 2\gamma y_t y_{t-1} \\
&= (1 - \gamma^2)y_t^2 + (1 - \gamma^2)\mu^2 - 2\mu(1 - \gamma^2)y_t \\
&\quad + y_{t-1}^2 + \mu^2(1 - \gamma)^2 + \gamma^2 y_t^2 - 2\gamma y_t y_{t-1} - 2\mu y_{t-1}(1 - \gamma) + 2\mu(1 - \gamma)\gamma y_t \\
&= (1 - \gamma^2)y_t^2 + (1 - \gamma^2)\mu^2 - 2\mu(1 - \gamma^2)y_t \\
&\quad + y_{t-1}^2 + c^2 + \gamma^2 y_t^2 - 2\gamma y_t y_{t-1} - 2c y_{t-1} + 2c\gamma y_t \\
&= (1 - \gamma^2)(y_t - \mu)^2 + (y_{t-1} - c - \gamma y_t)^2 \tag{a.2.8.8}
\end{aligned}$$

Substitute back in (a.2.8.6):

$$\pi(y_t) = \int \exp\left(-\frac{1}{2s} [(1 - \gamma^2)(y_t - \mu)^2 + (y_{t-1} - c - \gamma y_t)^2]\right) dy_{t-1} \tag{a.2.8.9}$$

and this eventually reformulates as:

$$\pi(y_t) = \exp\left(-\frac{1}{2} \frac{(y_t - \mu)^2}{\sigma}\right) \int \exp\left(-\frac{1}{2} \frac{(y_{t-1} - c - \gamma y_t)^2}{s}\right) dy_{t-1} \tag{a.2.8.10}$$

PART III

Econometrics

The linear regression model

derivations for equation (3.9.7)

Consider first β . To do so, rewrite the likelihood function as:

$$\log(f(y|\beta, \sigma)) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma) - \frac{1}{2\sigma} (y'y + \beta'X'X\beta - 2\beta'X'y) \quad (\text{a.3.9.1})$$

Then solve for the partial derivative:

$$\begin{aligned} \frac{\partial \log(f(y|\beta, \sigma))}{\partial \beta} &= 0 \\ \Leftrightarrow -\frac{1}{2\sigma} (2\beta'X'X - 2y'X) &= 0 \\ \Leftrightarrow \beta'X'X - y'X &= 0 \\ \Leftrightarrow \beta'X'X &= y'X \\ \Leftrightarrow X'X\beta &= X'y \\ \Leftrightarrow \beta &= (X'X)^{-1}X'y \end{aligned} \quad (\text{a.3.9.2})$$

Hence the estimate is $\hat{\beta} = (X'X)^{-1}X'y$. Consider now σ . Solve for the partial derivative:

$$\begin{aligned} \frac{\partial \log(f(y|\beta, \sigma))}{\partial \sigma} &= 0 \\ \Leftrightarrow -\frac{n}{2} \frac{1}{\sigma} + \frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma^2} &= 0 \\ \Leftrightarrow -n + \frac{(y - X\beta)'(y - X\beta)}{\sigma} &= 0 \\ \Leftrightarrow \frac{(y - X\beta)'(y - X\beta)}{\sigma} &= n \\ \Leftrightarrow \sigma &= \frac{(y - X\hat{\beta})'(y - X\hat{\beta})}{n} \end{aligned} \quad (\text{a.3.9.3})$$

This expression gives the optimum for any value of β . To obtain a global maximum we must choose the value of β that maximizes the likelihood, namely $\hat{\beta}$. Therefore, the estimate for σ is given by $\hat{\sigma} = (y - X\hat{\beta})'(y - X\hat{\beta})/n$.

derivations for equation (3.9.12)

Develop and group:

$$\begin{aligned} &\exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \times \exp\left(-\frac{1}{2} (\beta - b)'V^{-1}(\beta - b)\right) \\ &= \exp\left(-\frac{1}{2} [(y - X\beta)' \sigma^{-1} (y - X\beta) + (\beta - b)'V^{-1}(\beta - b)]\right) \end{aligned} \quad (\text{a.3.9.4})$$

Consider the term in square brackets:

$$\begin{aligned}
& (y - X\beta)' \sigma^{-1} (y - X\beta) + (\beta - b)' V^{-1} (\beta - b) \\
&= y' \sigma^{-1} y + \beta' X' \sigma^{-1} X \beta - 2\beta' X' \sigma^{-1} y + \beta' V^{-1} \beta + b' V^{-1} b - 2\beta' V^{-1} b \\
&= \beta' (V^{-1} + \sigma^{-1} X' X) \beta - 2\beta' (V^{-1} b + \sigma^{-1} X' y) + b' V^{-1} b + y' \sigma^{-1} y
\end{aligned} \tag{a.3.9.5}$$

Substitute back in (a.3.9.4):

$$= \exp \left(-\frac{1}{2} [\beta' (V^{-1} + \sigma^{-1} X' X) \beta - 2\beta' (V^{-1} b + \sigma^{-1} X' y) + b' V^{-1} b + y' \sigma^{-1} y] \right) \tag{a.3.9.6}$$

derivations for equation (3.9.23)

Group and rearrange:

$$\begin{aligned}
& \pi(\beta, \sigma | y) \\
&\propto \sigma^{-n/2} \exp \left(-\frac{1}{2} \frac{(y - X\beta)' (y - X\beta)}{\sigma} \right) \times |\sigma V|^{-1/2} \exp \left(-\frac{1}{2} (\beta - b)' (\sigma V)^{-1} (\beta - b) \right) \\
&\times \sigma^{-\alpha/2-1} \exp \left(-\frac{\delta}{2\sigma} \right) \\
&= \sigma^{-k/2} \exp \left(-\frac{1}{2\sigma} [(y - X\beta)' (y - X\beta) + (\beta - b)' V^{-1} (\beta - b)] \right) \sigma^{-(\alpha+n)/2-1} \exp \left(-\frac{\delta}{2\sigma} \right)
\end{aligned} \tag{a.3.9.7}$$

Define:

$$\bar{\alpha} = \alpha + n \tag{a.3.9.8}$$

Then:

$$= \sigma^{-k/2} \exp \left(-\frac{1}{2\sigma} [(y - X\beta)' (y - X\beta) + (\beta - b)' V^{-1} (\beta - b)] \right) \sigma^{-\bar{\alpha}/2-1} \exp \left(-\frac{\delta}{2\sigma} \right) \tag{a.3.9.9}$$

Consider the term between the square brackets:

$$\begin{aligned}
& (y - X\beta)' (y - X\beta) + (\beta - b)' V^{-1} (\beta - b) \\
&= y' y + \beta' X' X \beta - 2\beta' X' y + \beta' V^{-1} \beta + b' V^{-1} b - 2\beta' V^{-1} b \\
&= \beta' (V^{-1} + X' X) \beta - 2\beta' (V^{-1} b + X' y) + y' y + b' V^{-1} b \\
&= \beta' (V^{-1} + X' X) \beta - 2\beta' \bar{V}^{-1} \bar{V} (V^{-1} b + X' y) + y' y + b' V^{-1} b + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b}
\end{aligned} \tag{a.3.9.10}$$

Define:

$$\bar{V} = (V^{-1} + X' X)^{-1} \quad \bar{b} = \bar{V} (V^{-1} b + X' y) \tag{a.3.9.11}$$

Then (a.3.9.10) becomes:

$$\begin{aligned}
&= \beta' \bar{V}^{-1} \beta - 2\beta' \bar{V}^{-1} \bar{b} + y' y + b' V^{-1} b + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b} \\
&= (\beta' \bar{V}^{-1} \beta - 2\beta' \bar{V}^{-1} \bar{b} + \bar{b}' \bar{V}^{-1} \bar{b}) + y' y + b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} \\
&= (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + y' y + b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b}
\end{aligned} \tag{a.3.9.12}$$

Substitute back in (a.3.9.9):

$$= \sigma^{-k/2} \exp \left(-\frac{1}{2} (\beta - \bar{b})' (\sigma \bar{V})^{-1} (\beta - \bar{b}) \right) \sigma^{-\bar{\alpha}/2-1} \exp \left(-\frac{\delta + y' y + b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b}}{2\sigma} \right) \tag{a.3.9.13}$$

Define:

$$\bar{\delta} = \delta + y'y + b'V^{-1}b - \bar{b}'\bar{V}^{-1}\bar{b} \quad (\text{a.3.9.14})$$

Then (a.3.9.13) eventually rewrites:

$$\pi(\beta, \sigma|y) \propto \sigma^{-k/2} \exp\left(-\frac{1}{2}(\beta - \bar{b})'(\sigma\bar{V})^{-1}(\beta - \bar{b})\right) \sigma^{-\bar{\alpha}/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right) \quad (\text{a.3.9.15})$$

derivations for equation (3.9.28)

Rearrange:

$$\begin{aligned} & \Gamma\left(\frac{\bar{\alpha}+k}{2}\right) \left(\frac{\bar{\delta} + (\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})}{2}\right)^{-\frac{\bar{\alpha}+k}{2}} \\ & \propto \left(\frac{\bar{\delta} + (\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})}{2}\right)^{-\frac{\bar{\alpha}+k}{2}} \\ & \propto (\bar{\delta} + (\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b}))^{-\frac{\bar{\alpha}+k}{2}} \\ & \propto (1 + (\beta - \bar{b})'(\bar{\delta}\bar{V})^{-1}(\beta - \bar{b}))^{-\frac{\bar{\alpha}+k}{2}} \\ & \propto \left(1 + \frac{1}{\bar{\alpha}}(\beta - \bar{b})'(\bar{\delta}\bar{V}/\bar{\alpha})^{-1}(\beta - \bar{b})\right)^{-\frac{\bar{\alpha}+k}{2}} \end{aligned} \quad (\text{a.3.9.16})$$

derivations for equation (3.9.38)

Rearrange the likelihood function:

$$\begin{aligned} & f(y|\beta, \sigma) \\ &= (2\pi)^{-n/2} |\sigma W|^{-1/2} \exp\left(-\frac{1}{2}(y - X\beta)'(\sigma W)^{-1}(y - X\beta)\right) \\ &= (2\pi\sigma)^{-n/2} |W|^{-1/2} \exp\left(-\frac{1}{2}\frac{(y - X\beta)'W^{-1}(y - X\beta)}{\sigma}\right) \end{aligned} \quad (\text{a.3.9.17})$$

This reformulates further as:

$$\begin{aligned} &= (2\pi\sigma)^{-n/2} \left(\prod_{i=1}^n w_i^{-1/2}\right) \exp\left(-\frac{1}{2}\frac{(y - X\beta)' \text{diag}(\exp(-Z\gamma)) (y - X\beta)}{\sigma}\right) \\ &= (2\pi\sigma)^{-n/2} \left(\prod_{i=1}^n \exp(z_i'\gamma)\right)^{-1/2} \exp\left(-\frac{1}{2}\frac{(y - X\beta)' \text{diag}(\exp(-Z\gamma)) (y - X\beta)}{\sigma}\right) \\ &= (2\pi\sigma)^{-n/2} \left(\exp\left(\sum_{i=1}^n z_i'\gamma\right)\right)^{-1/2} \exp\left(-\frac{1}{2}\frac{(y - X\beta)' \text{diag}(\exp(-Z\gamma)) (y - X\beta)}{\sigma}\right) \\ &= (2\pi\sigma)^{-n/2} (\exp(1_n'Z\gamma))^{-1/2} \exp\left(-\frac{1}{2}\frac{(y - X\beta)' \text{diag}(\exp(-Z\gamma)) (y - X\beta)}{\sigma}\right) \\ &= (2\pi\sigma)^{-n/2} \exp\left(-\frac{1}{2}1_n'Z\gamma\right) \exp\left(-\frac{1}{2}\frac{(y - X\beta)' \text{diag}(\exp(-Z\gamma)) (y - X\beta)}{\sigma}\right) \\ &= (2\pi\sigma)^{-n/2} \exp\left(-\frac{1}{2}\left[1_n'Z\gamma + (y - X\beta)' \text{diag}(\exp(-Z\gamma)) (y - X\beta)/\sigma\right]\right) \end{aligned} \quad (\text{a.3.9.18})$$

derivations for equation (3.9.44)

Rearrange the terms:

$$\begin{aligned}
& \pi(\beta|y, \sigma, w) \\
&= \exp\left(-\frac{1}{2} \frac{(y - X\beta)'W^{-1}(y - X\beta)}{\sigma}\right) \times \exp\left(-\frac{1}{2}(\beta - b)'V^{-1}(\beta - b)\right) \\
&= \exp\left(-\frac{1}{2}[(y - X\beta)'(\sigma W)^{-1}(y - X\beta) + (\beta - b)'V^{-1}(\beta - b)]\right) \tag{a.3.9.19}
\end{aligned}$$

Consider the term in square brackets and complete the squares:

$$\begin{aligned}
& (y - X\beta)'(\sigma W)^{-1}(y - X\beta) + (\beta - b)'V^{-1}(\beta - b) \\
&= y'(\sigma W)^{-1}y + \beta'X'(\sigma W)^{-1}X\beta - 2\beta'X'(\sigma W)^{-1}y + \beta'V^{-1}\beta + b'V^{-1}b - 2\beta'V^{-1}b \\
&= \beta'(V^{-1} + \sigma^{-1}X'W^{-1}X)\beta - 2\beta'(V^{-1}b + \sigma^{-1}X'W^{-1}y) + y'(\sigma W)^{-1}y + b'V^{-1}b \\
&= \beta'(V^{-1} + \sigma^{-1}X'W^{-1}X)\beta - 2\beta'\bar{V}^{-1}\bar{V}(V^{-1}b + \sigma^{-1}X'W^{-1}y) \\
&\quad + y'(\sigma W)^{-1}y + b'V^{-1}b + \bar{b}'\bar{V}^{-1}\bar{b} - \bar{b}'\bar{V}^{-1}\bar{b} \tag{a.3.9.20}
\end{aligned}$$

Define:

$$\bar{V} = (V^{-1} + \sigma^{-1}X'W^{-1}X)^{-1} \quad \bar{b} = \bar{V}(V^{-1}b + \sigma^{-1}X'W^{-1}y) \tag{a.3.9.21}$$

Then (a.3.9.20) rewrites:

$$\begin{aligned}
&= \beta'\bar{V}^{-1}\beta - 2\beta'\bar{V}^{-1}\bar{b} + \bar{b}'\bar{V}^{-1}\bar{b} + y'(\sigma W)^{-1}y + b'V^{-1}b - \bar{b}'\bar{V}^{-1}\bar{b} \\
&= (\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b}) + y'(\sigma W)^{-1}y + b'V^{-1}b - \bar{b}'\bar{V}^{-1}\bar{b} \tag{a.3.9.22}
\end{aligned}$$

Substitute back in (a.3.9.19):

$$\begin{aligned}
& \pi(\beta|y, \sigma, w) \\
&= \exp\left(-\frac{1}{2}[(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b}) + y'(\sigma W)^{-1}y + b'V^{-1}b - \bar{b}'\bar{V}^{-1}\bar{b}]\right) \\
&= \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right) \exp\left(-\frac{1}{2}[y'(\sigma W)^{-1}y + b'V^{-1}b - \bar{b}'\bar{V}^{-1}\bar{b}]\right) \\
&\propto \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right) \tag{a.3.9.23}
\end{aligned}$$

derivations for equation (3.9.69)

Rearrange the terms:

$$\begin{aligned}
& \pi(\phi|y, \beta, \sigma) \\
&\propto \exp\left(-\frac{1}{2} \frac{(\varepsilon - E\phi)'(\varepsilon - E\phi)}{\sigma}\right) \times \exp\left(-\frac{1}{2}(\phi - p)'H^{-1}(\phi - p)\right) \\
&= \exp\left(-\frac{1}{2}[(\varepsilon - E\phi)'\sigma^{-1}(\varepsilon - E\phi) + (\phi - p)'H^{-1}(\phi - p)]\right) \tag{a.3.9.24}
\end{aligned}$$

Consider the term in square brackets and complete the squares:

$$\begin{aligned}
& (\varepsilon - E\phi)' \sigma^{-1} (\varepsilon - E\phi) + (\phi - p)' H^{-1} (\phi - p) \\
&= \varepsilon' \sigma^{-1} \varepsilon + \phi' E' \sigma^{-1} E \phi - 2\phi' E' \sigma^{-1} \varepsilon + \phi' H^{-1} \phi + p' H^{-1} p - 2\phi' H^{-1} p \\
&= \phi' (H^{-1} + \sigma^{-1} E' E) \phi - 2\phi' (H^{-1} p + \sigma^{-1} E' \varepsilon) + \varepsilon' \sigma^{-1} \varepsilon + p' H^{-1} p \\
&= \phi' (H^{-1} + \sigma^{-1} E' E) \phi - 2\phi' \bar{H}^{-1} \bar{H} (H^{-1} p + \sigma^{-1} E' \varepsilon) + \varepsilon' \sigma^{-1} \varepsilon + p' H^{-1} p + \bar{p}' \bar{H}^{-1} \bar{p} - \bar{p}' \bar{H}^{-1} \bar{p}
\end{aligned} \tag{a.3.9.25}$$

Define:

$$\bar{H} = (H^{-1} + \sigma^{-1} E' E)^{-1} \quad \bar{p} = \bar{H} (H^{-1} p + \sigma^{-1} E' \varepsilon) \tag{a.3.9.26}$$

Then (a.3.9.25) becomes:

$$\begin{aligned}
&= \phi' \bar{H}^{-1} \phi - 2\phi' \bar{H}^{-1} \bar{p} + \bar{p}' \bar{H}^{-1} \bar{p} + \varepsilon' \sigma^{-1} \varepsilon + p' H^{-1} p - \bar{p}' \bar{H}^{-1} \bar{p} \\
&= (\phi - \bar{p})' \bar{H}^{-1} (\phi - \bar{p}) + \varepsilon' \sigma^{-1} \varepsilon + p' H^{-1} p - \bar{p}' \bar{H}^{-1} \bar{p}
\end{aligned} \tag{a.3.9.27}$$

Substitute back in (a.3.9.24) to obtain:

$$\begin{aligned}
& \pi(\phi|y, \beta, \sigma) \\
&= \exp \left(-\frac{1}{2} [(\phi - \bar{p})' \bar{H}^{-1} (\phi - \bar{p}) + \varepsilon' \sigma^{-1} \varepsilon + p' H^{-1} p - \bar{p}' \bar{H}^{-1} \bar{p}] \right) \\
&= \exp \left(-\frac{1}{2} (\phi - \bar{p})' \bar{H}^{-1} (\phi - \bar{p}) \right) \exp \left(-\frac{1}{2} [\varepsilon' \sigma^{-1} \varepsilon + p' H^{-1} p - \bar{p}' \bar{H}^{-1} \bar{p}] \right) \\
&\propto \exp \left(-\frac{1}{2} (\phi - \bar{p})' \bar{H}^{-1} (\phi - \bar{p}) \right)
\end{aligned} \tag{a.3.9.28}$$

Applications with the linear regression model

derivations for equation (3.10.4)

Rearrange the expression:

$$\begin{aligned}
 & f(\hat{y}|y) \\
 & \propto \int \exp\left(-\frac{1}{2} \frac{(\hat{y} - \hat{X}\beta)'(\hat{y} - \hat{X}\beta)}{\sigma}\right) \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right) d\beta \\
 & = \int \exp\left(-\frac{1}{2} [\sigma^{-1}(\hat{y} - \hat{X}\beta)'(\hat{y} - \hat{X}\beta) + (\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})]\right) d\beta \tag{a.3.10.1}
 \end{aligned}$$

Consider the term in square brackets:

$$\begin{aligned}
 & \sigma^{-1}(\hat{y} - \hat{X}\beta)'(\hat{y} - \hat{X}\beta) + (\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b}) \\
 & = \sigma^{-1}\hat{y}'\hat{y} + \sigma^{-1}\beta'\hat{X}'\hat{X}\beta - 2\sigma^{-1}\beta'\hat{X}'\hat{y} + \beta'\bar{V}^{-1}\beta + \bar{b}'\bar{V}^{-1}\bar{b} - 2\beta'\bar{V}^{-1}\bar{b} \\
 & = \beta'(\bar{V}^{-1} + \sigma^{-1}\hat{X}'\hat{X})\beta - 2\beta'(\bar{V}^{-1}\bar{b} + \sigma^{-1}\hat{X}'\hat{y}) + \sigma^{-1}\hat{y}'\hat{y} + \bar{b}'\bar{V}^{-1}\bar{b} \\
 & = \beta'(\bar{V}^{-1} + \sigma^{-1}\hat{X}'\hat{X})\beta - 2\beta'\hat{V}^{-1}\hat{V}(\bar{V}^{-1}\bar{b} + \sigma^{-1}\hat{X}'\hat{y}) + \sigma^{-1}\hat{y}'\hat{y} + \bar{b}'\bar{V}^{-1}\bar{b} + \hat{b}'\hat{V}^{-1}\hat{b} - \hat{b}'\hat{V}^{-1}\hat{b} \tag{a.3.10.2}
 \end{aligned}$$

Define:

$$\hat{V} = (\bar{V}^{-1} + \sigma^{-1}\hat{X}'\hat{X})^{-1} \quad \hat{b} = \hat{V}(\bar{V}^{-1}\bar{b} + \sigma^{-1}\hat{X}'\hat{y}) \tag{a.3.10.3}$$

Then (a.3.10.2) becomes:

$$\begin{aligned}
 & = \beta'\hat{V}^{-1}\beta - 2\beta'\hat{V}^{-1}\hat{b} + \hat{b}'\hat{V}^{-1}\hat{b} + \sigma^{-1}\hat{y}'\hat{y} + \bar{b}'\bar{V}^{-1}\bar{b} - \hat{b}'\hat{V}^{-1}\hat{b} \\
 & = (\beta - \hat{b})'\hat{V}^{-1}(\beta - \hat{b}) + \sigma^{-1}\hat{y}'\hat{y} + \bar{b}'\bar{V}^{-1}\bar{b} - \hat{b}'\hat{V}^{-1}\hat{b} \tag{a.3.10.4}
 \end{aligned}$$

Substituting back in (a.3.10.1):

$$\begin{aligned}
 & f(\hat{y}|y) \\
 & \propto \int \exp\left(-\frac{1}{2} [(\beta - \hat{b})'\hat{V}^{-1}(\beta - \hat{b}) + \sigma^{-1}\hat{y}'\hat{y} + \bar{b}'\bar{V}^{-1}\bar{b} - \hat{b}'\hat{V}^{-1}\hat{b}]\right) d\beta \\
 & = \int \exp\left(-\frac{1}{2}(\beta - \hat{b})'\hat{V}^{-1}(\beta - \hat{b})\right) \exp\left(-\frac{1}{2} [\sigma^{-1}\hat{y}'\hat{y} + \bar{b}'\bar{V}^{-1}\bar{b} - \hat{b}'\hat{V}^{-1}\hat{b}]\right) d\beta \\
 & = \exp\left(-\frac{1}{2} [\sigma^{-1}\hat{y}'\hat{y} + \bar{b}'\bar{V}^{-1}\bar{b} - \hat{b}'\hat{V}^{-1}\hat{b}]\right) \int \exp\left(-\frac{1}{2}(\beta - \hat{b})'\hat{V}^{-1}(\beta - \hat{b})\right) d\beta \\
 & \propto \exp\left(-\frac{1}{2} [\sigma^{-1}\hat{y}'\hat{y} + \bar{b}'\bar{V}^{-1}\bar{b} - \hat{b}'\hat{V}^{-1}\hat{b}]\right) \tag{a.3.10.5}
 \end{aligned}$$

Consider the term in square brackets:

$$\begin{aligned}
& \sigma^{-1} \hat{y}' \hat{y} + \bar{b}' \bar{V}^{-1} \bar{b} - \hat{b}' \hat{V}^{-1} \hat{b} \\
&= \sigma^{-1} \hat{y}' \hat{y} + \bar{b}' \bar{V}^{-1} \bar{b} - (\bar{V}^{-1} \bar{b} + \sigma^{-1} \hat{X}' \hat{y})' \hat{V} \hat{V}^{-1} \hat{V} (\bar{V}^{-1} \bar{b} + \sigma^{-1} \hat{X}' \hat{y}) \\
&= \sigma^{-1} \hat{y}' \hat{y} + \bar{b}' \bar{V}^{-1} \bar{b} - (\bar{V}^{-1} \bar{b} + \sigma^{-1} \hat{X}' \hat{y})' \hat{V} (\bar{V}^{-1} \bar{b} + \sigma^{-1} \hat{X}' \hat{y}) \\
&= \sigma^{-1} \hat{y}' \hat{y} + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \hat{V} \bar{V}^{-1} \bar{b} - \sigma^{-2} \hat{y}' \hat{X} \hat{V} \hat{X}' \hat{y} - 2 \sigma^{-1} \hat{y}' \hat{X} \hat{V} \bar{V}^{-1} \bar{b} \\
&= \hat{y}' (\sigma^{-1} I_m - \sigma^{-2} \hat{X} \hat{V} \hat{X}') \hat{y} - \bar{b}' (\bar{V}^{-1} - \bar{V}^{-1} \hat{V} \bar{V}^{-1}) \bar{b} - 2 \sigma^{-1} \hat{y}' \hat{X} \hat{V} \bar{V}^{-1} \bar{b}
\end{aligned} \tag{a.3.10.6}$$

In what follows, we make use of property m.13 (the Sherman-Woodbury-Morrison identity):
 $(A + BDC)^{-1} = A^{-1} - A^{-1}B(D^{-1} + CA^{-1}B)^{-1}CA^{-1}$.

Consider the central part of the second term in (a.3.10.6). Rearrange and use the identity twice to obtain:

$$\begin{aligned}
& \bar{V}^{-1} - \bar{V}^{-1} \hat{V} \bar{V}^{-1} \\
&= \bar{V}^{-1} - \bar{V}^{-1} (\bar{V}^{-1} + \sigma^{-1} \hat{X}' \hat{X})^{-1} \bar{V}^{-1} \\
&= (\bar{V} + \sigma (\hat{X}' \hat{X})^{-1})^{-1} \\
&= \sigma^{-1} \hat{X}' \hat{X} - \sigma^{-1} \hat{X}' \hat{X} (\sigma^{-1} \hat{X}' \hat{X} + \bar{V}^{-1})^{-1} \sigma^{-1} \hat{X}' \hat{X} \\
&= \sigma^{-1} \hat{X}' \hat{X} - \sigma^{-2} \hat{X}' \hat{X} \hat{V} \hat{X}' \hat{X} \\
&= \hat{X}' (\sigma^{-1} I_m - \sigma^{-2} \hat{X} \hat{V} \hat{X}') \hat{X}
\end{aligned} \tag{a.3.10.7}$$

Consider finally the contral part of the third term in (a.3.10.6). We note that $\hat{V}(\bar{V}^{-1} + \sigma^{-1} \hat{X}' \hat{X}) = I_m$ so $\hat{V} \bar{V}^{-1} = I_m - \hat{V} \sigma^{-1} \hat{X}' \hat{X}$. Following:

$$\hat{X} \hat{V} \bar{V}^{-1} = \hat{X} - \hat{X} \hat{V} \sigma^{-1} \hat{X}' \hat{X} = (I_m - \sigma^{-1} \hat{X} \hat{V} \hat{X}') \hat{X} \tag{a.3.10.8}$$

Substitute (a.3.10.7) and (a.3.10.8) back in (a.3.10.6) to obtain:

$$\begin{aligned}
&= \hat{y}' (\sigma^{-1} I_m - \sigma^{-2} \hat{X} \hat{V} \hat{X}') \hat{y} - \bar{b}' \hat{X}' (\sigma^{-1} I_m - \sigma^{-2} \hat{X} \hat{V} \hat{X}') \hat{X} \bar{b} - 2 \sigma^{-1} \hat{y}' (I_m - \sigma^{-1} \hat{X} \hat{V} \hat{X}') \hat{X} \bar{b} \\
&= \sigma^{-1} \hat{y}' (I_m - \sigma^{-1} \hat{X} \hat{V} \hat{X}') \hat{y} - \sigma^{-1} \bar{b}' \hat{X}' (I_m - \sigma^{-1} \hat{X} \hat{V} \hat{X}') \hat{X} \bar{b} - 2 \sigma^{-1} \hat{y}' (I_m - \sigma^{-1} \hat{X} \hat{V} \hat{X}') \hat{X} \bar{b} \\
&= \sigma^{-1} [\hat{y}' (I_m - \sigma^{-1} \hat{X} \hat{V} \hat{X}') \hat{y} - \bar{b}' \hat{X}' (I_m - \sigma^{-1} \hat{X} \hat{V} \hat{X}') \hat{X} \bar{b} - 2 \hat{y}' (I_m - \sigma^{-1} \hat{X} \hat{V} \hat{X}') \hat{X} \bar{b}] \\
&= \sigma^{-1} (\hat{y} - \hat{X} \bar{b})' (I_m - \sigma^{-1} \hat{X} \hat{V} \hat{X}') (\hat{y} - \hat{X} \bar{b})
\end{aligned} \tag{a.3.10.9}$$

We use one last time the Sherman-Woodbury-Morrison identity on the central term to obtain:

$$I_m - \sigma^{-1} \hat{X} \hat{V} \hat{X}' = I_m - \sigma^{-1} \hat{X} (\bar{V}^{-1} + \sigma^{-1} \hat{X}' \hat{X})^{-1} \hat{X}' = (I_m + \sigma^{-1} \hat{X} \bar{V} \hat{X}')^{-1} \tag{a.3.10.10}$$

Substituting back in (a.3.10.9):

$$\begin{aligned}
&= \sigma^{-1} (\hat{y} - \hat{X} \bar{b})' (I_m + \sigma^{-1} \hat{X} \bar{V} \hat{X}')^{-1} (\hat{y} - \hat{X} \bar{b}) \\
&= (\hat{y} - \hat{X} \bar{b})' (\sigma I_m + \hat{X} \bar{V} \hat{X}')^{-1} (\hat{y} - \hat{X} \bar{b})
\end{aligned} \tag{a.3.10.11}$$

Eventually substituting back in (a.3.10.5), we conclude:

$$f(\hat{y}|y) \propto \exp \left(-\frac{1}{2} (\hat{y} - \hat{X} \bar{b})' (\sigma I_m + \hat{X} \bar{V} \hat{X}')^{-1} (\hat{y} - \hat{X} \bar{b}) \right) \tag{a.3.10.12}$$

derivations for equation (3.10.6)

Rearrange the expression:

$$\begin{aligned}
& f(\hat{y}|y) \\
& \propto \int \int \sigma^{-m/2} \exp\left(-\frac{1}{2} \frac{(\hat{y} - \hat{X}\beta)'(\hat{y} - \hat{X}\beta)}{\sigma}\right) \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \\
& \times \sigma^{-k/2} \exp\left(-\frac{1}{2}(\beta - b)'(\sigma V)^{-1}(\beta - b)\right) \times \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right) \\
& = \int \int \sigma^{-k/2} \exp\left(-\frac{1}{2\sigma} [(\hat{y} - \hat{X}\beta)'(\hat{y} - \hat{X}\beta) + (y - X\beta)'(y - X\beta) + (\beta - b)'(\sigma V)^{-1}(\beta - b) + \delta]\right) \\
& \times \sigma^{-(\alpha+n+m)/2-1} d\beta d\sigma \tag{a.3.10.13}
\end{aligned}$$

Consider the term in the square bracket:

$$\begin{aligned}
& (\hat{y} - \hat{X}\beta)'(\hat{y} - \hat{X}\beta) + (y - X\beta)'(y - X\beta) + (\beta - b)'(\sigma V)^{-1}(\beta - b) + \delta \\
& = \hat{y}'\hat{y} + \beta'\hat{X}'\hat{X}\beta - 2\beta'\hat{X}'\hat{y} + y'y + \beta'X'X\beta - 2\beta'X'y + \beta'V^{-1}\beta + b'V^{-1}b - 2\beta'V^{-1}b + \delta \\
& = (\delta + \hat{y}'\hat{y} + y'y + b'V^{-1}b) + \beta'(V^{-1} + X'X + \hat{X}'\hat{X})\beta - 2\beta'(V^{-1}b + X'y + \hat{X}'\hat{y}) \\
& = (\delta + \hat{y}'\hat{y} + y'y + b'V^{-1}b) + \beta'(V^{-1} + X'X + \hat{X}'\hat{X})\beta - 2\beta'\hat{V}^{-1}\hat{V}(V^{-1}b + X'y + \hat{X}'\hat{y}) + \hat{b}'\hat{V}^{-1}\hat{b} - \hat{b}'\hat{V}^{-1}\hat{b} \\
& = (\delta + \hat{y}'\hat{y} + y'y + b'V^{-1}b - \hat{b}'\hat{V}^{-1}\hat{b}) + \beta'(V^{-1} + X'X + \hat{X}'\hat{X})\beta - 2\beta'\hat{V}^{-1}\hat{V}(V^{-1}b + X'y + \hat{X}'\hat{y}) + \hat{b}'\hat{V}^{-1}\hat{b} \tag{a.3.10.14}
\end{aligned}$$

Define:

$$\hat{\delta} = \delta + \hat{y}'\hat{y} + y'y + b'V^{-1}b - \hat{b}'\hat{V}^{-1}\hat{b} \quad \hat{V} = (V^{-1} + X'X + \hat{X}'\hat{X})^{-1} \quad \hat{b} = \hat{V}(V^{-1}b + X'y + \hat{X}'\hat{y}) \tag{a.3.10.15}$$

Then (a.3.10.14) rewrites:

$$\begin{aligned}
& = \hat{\delta} + \beta'\hat{V}^{-1}\beta - 2\beta'\hat{V}^{-1}\hat{b} + \hat{b}'\hat{V}^{-1}\hat{b} \\
& = \hat{\delta} + (\beta - \hat{b})'\hat{V}^{-1}(\beta - \hat{b}) \tag{a.3.10.16}
\end{aligned}$$

Substitute back in (a.3.10.13):

$$\begin{aligned}
& f(\hat{y}|y) \\
& \propto \int \int \sigma^{-k/2} \exp\left(-\frac{1}{2\sigma} [\hat{\delta} + (\beta - \hat{b})'\hat{V}^{-1}(\beta - \hat{b})]\right) \sigma^{-(\alpha+n+m)/2-1} d\beta d\sigma \\
& = \int \int \sigma^{-k/2} \exp\left(-\frac{1}{2\sigma} (\beta - \hat{b})'\hat{V}^{-1}(\beta - \hat{b})\right) d\beta \sigma^{-(\alpha+n+m)/2-1} \exp\left(-\frac{\hat{\delta}}{2\sigma}\right) d\sigma \\
& = \int \int \sigma^{-k/2} \exp\left(-\frac{1}{2\sigma} (\beta - \hat{b})'\hat{V}^{-1}(\beta - \hat{b})\right) d\beta \sigma^{-(\bar{\alpha}+m)/2-1} \exp\left(-\frac{\hat{\delta}}{2\sigma}\right) d\sigma \tag{a.3.10.17}
\end{aligned}$$

with:

$$\hat{\alpha} = \alpha + n + m \tag{a.3.10.18}$$

The first term is the kernel of a multivariate normal distribution; integration hence yields a constant:

$$= \int \sigma^{-\hat{\alpha}/2-1} \exp\left(-\frac{\hat{\delta}}{2\sigma}\right) d\sigma \tag{a.3.10.19}$$

The remaining term is the kernel of an inverse gamma distribution; integration thus yields the reciprocal of the normalization constant:

$$\begin{aligned}
&= \Gamma(\hat{\alpha}/2)(\hat{\delta}/2)^{-\hat{\alpha}/2} \\
&\propto (\hat{\delta}/2)^{-\hat{\alpha}/2} \\
&\propto \hat{\delta}^{-\hat{\alpha}/2} \\
&= (\delta + \hat{y}'\hat{y} + y'y + b'V^{-1}b - \hat{b}'\hat{V}^{-1}\hat{b})^{-\hat{\alpha}/2} \\
&= (\delta + \hat{y}'\hat{y} + y'y + b'V^{-1}b - (V^{-1}b + X'y + \hat{X}'\hat{y})'\hat{V}^{-1}\hat{V}(V^{-1}b + X'y + \hat{X}'\hat{y}))^{-\hat{\alpha}/2} \\
&= (\delta + \hat{y}'\hat{y} + y'y + b'V^{-1}b - (V^{-1}b + X'y + \hat{X}'\hat{y})'\hat{V}(V^{-1}b + X'y + \hat{X}'\hat{y}))^{-\hat{\alpha}/2} \tag{a.3.10.20}
\end{aligned}$$

Define:

$$\tilde{b} = V^{-1}b + X'y \tag{a.3.10.21}$$

Then (a.3.10.20) rewrites:

$$\begin{aligned}
&= (\delta + \hat{y}'\hat{y} + y'y + b'V^{-1}b - (\tilde{b} + \hat{X}'\hat{y})'\hat{V}(\tilde{b} + \hat{X}'\hat{y}))^{-\hat{\alpha}/2} \\
&= (\delta + \hat{y}'\hat{y} + y'y + b'V^{-1}b - \tilde{b}'\hat{V}\tilde{b} - \hat{y}'\hat{X}\hat{V}\hat{X}'\hat{y} - 2\hat{y}'\hat{X}\hat{V}\tilde{b})^{-\hat{\alpha}/2} \\
&= (\delta + y'y + b'V^{-1}b + \hat{y}'(I_m - \hat{X}\hat{V}\hat{X}')\hat{y} - \tilde{b}'\hat{V}\tilde{b} - 2\hat{y}'\hat{X}\hat{V}\tilde{b})^{-\hat{\alpha}/2} \\
&= ([\delta + y'y + b'V^{-1}b - \tilde{b}'\hat{V}\tilde{b} - \hat{y}'\hat{V}^{-1}\hat{y}] + \hat{y}'(I_m - \hat{X}\hat{V}\hat{X}')\hat{y} - 2\hat{y}'\hat{V}^{-1}\hat{V}\hat{X}\hat{V}\tilde{b} + \hat{y}'\hat{V}^{-1}\hat{y})^{-\hat{\alpha}/2} \tag{a.3.10.22}
\end{aligned}$$

Define:

$$\ddot{\delta} = \delta + y'y + b'V^{-1}b - \tilde{b}'\hat{V}\tilde{b} - \hat{y}'\hat{V}^{-1}\hat{y} \quad \ddot{V} = (I_m - \hat{X}\hat{V}\hat{X}')^{-1} \quad \ddot{y} = \hat{V}\hat{X}\hat{V}\tilde{b} \tag{a.3.10.23}$$

Then (a.3.10.22) rewrites:

$$\begin{aligned}
&= (\ddot{\delta} + \hat{y}'\hat{V}^{-1}\hat{y} - 2\hat{y}'\hat{V}^{-1}\hat{y} + \hat{y}'\hat{V}^{-1}\hat{y})^{-\hat{\alpha}/2} \\
&= (\ddot{\delta} + (\hat{y} - \ddot{y})'\hat{V}^{-1}(\hat{y} - \ddot{y}))^{-\hat{\alpha}/2} \\
&= \ddot{\delta}^{-\hat{\alpha}/2}(1 + (\hat{y} - \ddot{y})'[\ddot{\delta}\hat{V}]^{-1}(\hat{y} - \ddot{y}))^{-\hat{\alpha}/2} \\
&\propto (1 + (\hat{y} - \ddot{y})'[\ddot{\delta}\hat{V}]^{-1}(\hat{y} - \ddot{y}))^{-\hat{\alpha}/2} \\
&= \left(1 + \frac{1}{\bar{\alpha}}(\hat{y} - \ddot{y})'[\ddot{\delta}\hat{V}/\bar{\alpha}]^{-1}(\hat{y} - \ddot{y})\right)^{-(\bar{\alpha}+m)/2} \tag{a.3.10.24}
\end{aligned}$$

Thus we finally conclude:

$$f(\hat{y}|y) \propto \left(1 + \frac{1}{\bar{\alpha}}(\hat{y} - \ddot{y})'[\ddot{\delta}\hat{V}/\bar{\alpha}]^{-1}(\hat{y} - \ddot{y})\right)^{-(\bar{\alpha}+m)/2} \tag{a.3.10.25}$$

Finally, reformulate the messy terms. First, reformulate $\ddot{V}$. For this, we make again use of property m.13 (the Sherman-Woodbury-Morrison identity): $(A + BDC)^{-1} = A^{-1} - A^{-1}B(D^{-1} + CA^{-1}B)^{-1}CA^{-1}$.

Then, starting from (a.3.10.23):

$$\begin{aligned}
&\ddot{V} \\
&= (I_m - \hat{X}\hat{V}\hat{X}')^{-1} \\
&= (I_m - \hat{X}(V^{-1} + X'X + \hat{X}'\hat{X})^{-1}\hat{X}')^{-1} \\
&= I_m + \hat{X}(V^{-1} + X'X)^{-1}\hat{X}' \\
&= I_m + \hat{X}\bar{V}\hat{X}' \tag{a.3.10.26}
\end{aligned}$$

Now consider the term $\ddot{y}$. Start from:

$$\begin{aligned}
 & \dot{\hat{V}}\hat{X}\hat{V} \\
 = & (I_m + \hat{X}\bar{V}\hat{X}')\hat{X}\hat{V} \\
 = & \hat{X}\hat{V} + \hat{X}\bar{V}\hat{X}'\hat{X}\hat{V} \\
 = & \hat{X}(\hat{V} + \bar{V}\hat{X}'\hat{X}\hat{V}) \\
 = & \hat{X}(I_m + \bar{V}\hat{X}'\hat{X})\hat{V}
 \end{aligned} \tag{a.3.10.27}$$

We then note that (a.3.10.15) implies:

$$\hat{V} = (V^{-1} + X'X + \hat{X}'\hat{X})^{-1} \Leftrightarrow \hat{V} = (\bar{V}^{-1} + \hat{X}'\hat{X})^{-1} \tag{a.3.10.28}$$

Hence:

$$\begin{aligned}
 & = \hat{X}(I_m + \bar{V}\hat{X}'\hat{X})(\bar{V}^{-1} + \hat{X}'\hat{X})^{-1} \\
 & = \hat{X}\bar{V}(\bar{V}^{-1} + \hat{X}'\hat{X})(\bar{V}^{-1} + \hat{X}'\hat{X})^{-1} \\
 & = \hat{X}\bar{V}
 \end{aligned} \tag{a.3.10.29}$$

Using this result in (a.3.10.23), and combining with definition (a.3.10.21), we obtain:

$$\ddot{y} = \dot{\hat{V}}\hat{X}\hat{V}\tilde{b} = \hat{X}\bar{V}\tilde{b} = \hat{X}\bar{b} \tag{a.3.10.30}$$

Finally, reformulate $\ddot{\delta}$. First, note that:

$$\begin{aligned}
 & \tilde{b}'\hat{V}\tilde{b} + \dot{y}'\bar{V}^{-1}\ddot{y} \\
 = & \tilde{b}'\bar{V}^{-1}\hat{V}\bar{V}^{-1}\tilde{b} + \tilde{b}'\hat{X}'(I_m - \hat{X}\hat{V}\hat{X}')\hat{X}\bar{b} \\
 = & \tilde{b}'[(\hat{V}^{-1} - \hat{X}'\hat{X})'\hat{V}(\hat{V}^{-1} - \hat{X}'\hat{X}) + \hat{X}'\hat{X} - \hat{X}'\hat{X}\hat{V}\hat{X}'\hat{X}]\bar{b} \\
 = & \tilde{b}'[(\hat{V}^{-1} - \hat{X}'\hat{X})'(I_k - \hat{V}\hat{X}'\hat{X}) + \hat{X}'\hat{X} - \hat{X}'\hat{X}\hat{V}\hat{X}'\hat{X}]\bar{b} \\
 = & \tilde{b}'[\hat{V}^{-1} - \hat{X}'\hat{X} - \hat{X}'\hat{X} + \hat{X}'\hat{X}\hat{V}\hat{X}'\hat{X} + \hat{X}'\hat{X} - \hat{X}'\hat{X}\hat{V}\hat{X}'\hat{X}]\bar{b} \\
 = & \tilde{b}'[\hat{V}^{-1} - \hat{X}'\hat{X}]\bar{b} \\
 = & \tilde{b}'[\bar{V}^{-1} + \hat{X}'\hat{X} - \hat{X}'\hat{X}]\bar{b} \\
 = & \tilde{b}'\bar{V}^{-1}\bar{b}
 \end{aligned} \tag{a.3.10.31}$$

Substituting this in (a.3.10.23) to obtain:

$$\ddot{\delta} = \delta + y'y + b'V^{-1}b - \tilde{b}'\hat{V}\tilde{b} - \dot{y}'\bar{V}^{-1}\ddot{y} = \delta + y'y + b'V^{-1}b - \tilde{b}'\bar{V}^{-1}\bar{b} = \bar{\delta} \tag{a.3.10.32}$$

Eventually substituting for (a.3.10.26), (a.3.10.30) and (a.3.10.32) in (a.3.10.25) yields:

$$f(\hat{y}|y) \propto \left(1 + \frac{1}{\bar{\alpha}}(\hat{y} - \hat{X}\bar{b})'[\bar{\delta}(I_m + \hat{X}\bar{V}\hat{X}')/\bar{\alpha}]^{-1}(\hat{y} - \hat{X}\bar{b})\right)^{-(\bar{\alpha}+m)/2} \tag{a.3.10.33}$$

derivations for equation (3.10.10)

The log likelihood function is given by:

$$\log(f(y|\beta, \sigma)) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma) - \frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma} \quad (\text{a.3.10.34})$$

The function is estimated at the maximum likelihood values. Hence $\beta = \hat{\beta}$ and $\sigma = \hat{\sigma} = \frac{\hat{\epsilon}'\hat{\epsilon}}{n}$.

Substituting in (a.3.10.34):

$$\begin{aligned} & \log(f(y|\hat{\beta}, \hat{\sigma})) \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\hat{\sigma}) - \frac{1}{2} \frac{(y - X\hat{\beta})'(y - X\hat{\beta})}{\hat{\sigma}} \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\hat{\sigma}) - \frac{1}{2} \frac{n \hat{\epsilon}'\hat{\epsilon}}{\hat{\epsilon}'\hat{\epsilon}} \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\hat{\sigma}) - \frac{n}{2} \\ &= -\frac{n}{2} [\log(2\pi) + \log(\hat{\sigma}) + 1] \end{aligned} \quad (\text{a.3.10.35})$$

Then AIC obtains as:

$$\begin{aligned} AIC &= 2k/n - 2 \hat{L}/n \\ &= 2k/n - 2 \left(-\frac{n}{2} [\log(2\pi) + \log(\hat{\sigma}) + 1] \right) / n \\ &= 2k/n + \log(2\pi) + \log(\hat{\sigma}) + 1 \end{aligned} \quad (\text{a.3.10.36})$$

Using similar calculations, BIC immediately obtains as:

$$BIC = k \log(n)/n + \log(2\pi) + \log(\hat{\sigma}) + 1 \quad (\text{a.3.10.37})$$

Removing the constants that make the value invariant to the number of coefficients:

$$AIC = 2k/n + \log(\hat{\sigma}) \quad BIC = k \log(n)/n + \log(\hat{\sigma}) \quad (\text{a.3.10.38})$$

derivations for equation (3.10.19)

Rearrange:

$$\begin{aligned}
 & f(y) \\
 &= \int (2\pi\sigma)^{-n/2} \exp\left(-\frac{1}{2} \frac{(y-X\beta)'(y-X\beta)}{\sigma}\right) \times (2\pi)^{-k/2} |V|^{-1/2} \exp\left(-\frac{1}{2} (\beta-b)'V^{-1}(\beta-b)\right) d\beta \\
 &= \int (2\pi)^{-(n+k)/2} \sigma^{-n/2} |V|^{-1/2} \times \exp\left(-\frac{1}{2} [(y-X\beta)'\sigma^{-1}(y-X\beta) + (\beta-b)'V^{-1}(\beta-b)]\right) d\beta
 \end{aligned} \tag{a.3.10.39}$$

Consider the term square brackets:

$$\begin{aligned}
 & (y-X\beta)'\sigma^{-1}(y-X\beta) + (\beta-b)'V^{-1}(\beta-b) \\
 &= y'\sigma^{-1}y + \beta'X'\sigma^{-1}X\beta - 2\beta'X'\sigma^{-1}y + \beta'V^{-1}\beta + b'V^{-1}b - 2\beta'V^{-1}b \\
 &= \beta'(V^{-1} + \sigma^{-1}X'X)\beta - 2\beta'(V^{-1}b + \sigma^{-1}X'y) + y'\sigma^{-1}y + b'V^{-1}b \\
 &= \beta'(V^{-1} + \sigma^{-1}X'X)\beta - 2\beta'\bar{V}^{-1}\bar{V}(V^{-1}b + \sigma^{-1}X'y) + \bar{b}\bar{V}^{-1}\bar{b} + y'\sigma^{-1}y + b'V^{-1}b - \bar{b}\bar{V}^{-1}\bar{b}
 \end{aligned} \tag{a.3.10.40}$$

Define:

$$\bar{V} = (V^{-1} + \sigma^{-1}X'X)^{-1} \quad \bar{b} = \bar{V}(V^{-1}b + \sigma^{-1}X'y) \tag{a.3.10.41}$$

Then (a.3.10.40) reformulates:

$$\begin{aligned}
 &= \beta'\bar{V}^{-1}\beta - 2\beta'\bar{V}^{-1}\bar{b} + \bar{b}\bar{V}^{-1}\bar{b} + y'\sigma^{-1}y + b'V^{-1}b - \bar{b}\bar{V}^{-1}\bar{b} \\
 &= (\beta-\bar{b})'\bar{V}^{-1}(\beta-\bar{b}) + y'\sigma^{-1}y + b'V^{-1}b - \bar{b}\bar{V}^{-1}\bar{b}
 \end{aligned} \tag{a.3.10.42}$$

Substitute back in (a.3.10.39):

$$\begin{aligned}
 & f(y) \\
 &= \int (2\pi)^{-(n+k)/2} \sigma^{-n/2} |V|^{-1/2} \times \exp\left(-\frac{1}{2} [(\beta-\bar{b})'\bar{V}^{-1}(\beta-\bar{b}) + y'\sigma^{-1}y + b'V^{-1}b - \bar{b}\bar{V}^{-1}\bar{b}]\right) d\beta \\
 &= (2\pi)^{-(n+k)/2} \sigma^{-n/2} |V|^{-1/2} (2\pi)^{k/2} |\bar{V}|^{1/2} \times \exp\left(-\frac{1}{2} [y'\sigma^{-1}y + b'V^{-1}b - \bar{b}\bar{V}^{-1}\bar{b}]\right) \\
 &\times \int (2\pi)^{-k/2} |\bar{V}|^{-1/2} \exp\left(-\frac{1}{2} (\beta-\bar{b})'\bar{V}^{-1}(\beta-\bar{b})\right) d\beta \\
 &= (2\pi)^{-n/2} \sigma^{-n/2} |\bar{V}|^{1/2} |V|^{-1/2} \times \exp\left(-\frac{1}{2} [y'\sigma^{-1}y + b'V^{-1}b - \bar{b}\bar{V}^{-1}\bar{b}]\right) \\
 &\times \int (2\pi)^{-k/2} |\bar{V}|^{-1/2} \exp\left(-\frac{1}{2} (\beta-\bar{b})'\bar{V}^{-1}(\beta-\bar{b})\right) d\beta
 \end{aligned} \tag{a.3.10.43}$$

derivations for equation (3.10.21)

Consider the term:

$$\begin{aligned}
 & (2\pi)^{-n/2} \sigma^{-n/2} |\bar{V}|^{1/2} |V|^{-1/2} \\
 &= (2\pi)^{-n/2} \sigma^{-n/2} |(V^{-1} + \sigma^{-1} X'X)^{-1}|^{1/2} |V|^{-1/2} \\
 &= (2\pi)^{-n/2} \sigma^{-n/2} |V^{-1} + \sigma^{-1} X'X|^{-1/2} |V|^{-1/2} \\
 &= (2\pi)^{-n/2} \sigma^{-n/2} (|V| |V^{-1} + \sigma^{-1} X'X|)^{-1/2} \\
 &= (2\pi)^{-n/2} \sigma^{-n/2} |I_k + \sigma^{-1} V X'X|^{-1/2} \tag{a.3.10.44}
 \end{aligned}$$

Hence:

$$f(y) = (2\pi)^{-n/2} \sigma^{-n/2} |I_k + \sigma^{-1} V X'X|^{-1/2} \exp \left(-\frac{1}{2} [y' \sigma^{-1} y + b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b}] \right) \tag{a.3.10.45}$$

derivations for equation (3.10.23)

Rearrange the expression:

$$\begin{aligned}
 & f(y) \\
 &= \int \int (2\pi\sigma)^{-n/2} \exp \left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma} \right) \\
 &\times (2\pi)^{-k/2} |\sigma V|^{-1/2} \exp \left(-\frac{1}{2} (\beta - b)' (\sigma V)^{-1} (\beta - b) \right) \times \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \sigma^{-\alpha/2-1} \exp \left(-\frac{\delta}{2\sigma} \right) d\beta d\sigma \\
 &= \int \int (2\pi\sigma)^{-n/2} (2\pi)^{-k/2} |\sigma V|^{-1/2} \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \sigma^{-\alpha/2-1} \\
 &\times \exp \left(-\frac{1}{2\sigma} [(y - X\beta)'(y - X\beta) + (\beta - b)' V^{-1} (\beta - b) + \delta] \right) d\beta d\sigma \tag{a.3.10.46}
 \end{aligned}$$

Consider the term in square brackets and complete the squares:

$$\begin{aligned}
 & (y - X\beta)'(y - X\beta) + (\beta - b)' V^{-1} (\beta - b) + \delta \\
 &= y'y + \beta' X' X \beta - 2\beta' X'y + \beta' V^{-1} \beta + b' V^{-1} b - 2\beta' V^{-1} b + \delta \\
 &= \beta' (V^{-1} + X'X) \beta - 2\beta' (V^{-1} b + X'y) + \delta + y'y + b' V^{-1} b \\
 &= \beta' (V^{-1} + X'X) \beta - 2\beta' \bar{V}^{-1} \bar{V} (V^{-1} b + X'y) + \delta + y'y + b' V^{-1} b + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b} \tag{a.3.10.47}
 \end{aligned}$$

Define:

$$\bar{V} = (V^{-1} + X'X)^{-1} \quad \bar{b} = \bar{V} (V^{-1} b + X'y) \quad \bar{\delta} = \delta + y'y + b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} \tag{a.3.10.48}$$

Then (a.3.10.47) rewrites:

$$\begin{aligned}
 &= \beta' \bar{V}^{-1} \beta - 2\beta' \bar{V}^{-1} \bar{b} + \bar{b}' \bar{V}^{-1} \bar{b} + \bar{\delta} \\
 &= (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + \bar{\delta} \tag{a.3.10.49}
 \end{aligned}$$

Substituting back in (a.3.10.46):

$$\begin{aligned}
& f(y) \\
&= \int \int (2\pi\sigma)^{-n/2} (2\pi)^{-k/2} |\sigma V|^{-1/2} \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \sigma^{-\alpha/2-1} \\
&\times \exp\left(-\frac{1}{2\sigma} [(\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + \bar{\delta}]\right) d\beta d\sigma \\
&= \int \int (2\pi)^{-n/2} (2\pi)^{-k/2} |\sigma V|^{-1/2} \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \\
&\times \sigma^{-(\alpha+n)/2-1} \exp\left(-\frac{1}{2\sigma} [(\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + \bar{\delta}]\right) d\beta d\sigma
\end{aligned} \tag{a.3.10.50}$$

define:

$$\bar{\alpha} = \alpha + n \tag{a.3.10.51}$$

Then (a.3.10.50) rewrites:

$$\begin{aligned}
&= \int \int (2\pi)^{-n/2} (2\pi)^{-k/2} |\sigma V|^{-1/2} \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \\
&\times \sigma^{-\bar{\alpha}/2-1} \exp\left(-\frac{1}{2\sigma} [(\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + \bar{\delta}]\right) d\beta d\sigma \\
&= (2\pi)^{-n/2} |\sigma V|^{-1/2} |\sigma \bar{V}|^{1/2} \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \frac{\Gamma(\bar{\alpha}/2)}{\bar{\delta}/2^{\bar{\alpha}/2}} \\
&\times \int \int (2\pi)^{-k/2} |\sigma \bar{V}|^{-1/2} \exp\left(-\frac{1}{2} (\beta - \bar{b})' (\sigma \bar{V})^{-1} (\beta - \bar{b})\right) \times \frac{\bar{\delta}/2^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)} \sigma^{-\bar{\alpha}/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right) d\beta d\sigma \\
&= 2^{-n/2} \pi^{-n/2} |V|^{-1/2} |\bar{V}|^{1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{2^{(\alpha+n)/2}}{2^{\alpha/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \\
&\times \int \int (2\pi)^{-k/2} |\sigma \bar{V}|^{-1/2} \exp\left(-\frac{1}{2} (\beta - \bar{b})' (\sigma \bar{V})^{-1} (\beta - \bar{b})\right) \times \frac{\bar{\delta}/2^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)} \sigma^{-\bar{\alpha}/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right) d\beta d\sigma \\
&= \pi^{-n/2} |V|^{-1/2} |\bar{V}|^{1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \\
&\times \int \int (2\pi)^{-k/2} |\sigma \bar{V}|^{-1/2} \exp\left(-\frac{1}{2} (\beta - \bar{b})' (\sigma \bar{V})^{-1} (\beta - \bar{b})\right) \times \frac{\bar{\delta}/2^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)} \sigma^{-\bar{\alpha}/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right) d\beta d\sigma
\end{aligned} \tag{a.3.10.52}$$

derivations for equation (3.10.25)

Reformulate the expression:

$$\begin{aligned}
& \pi^{-n/2} |V|^{-1/2} |\bar{V}|^{1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \\
= & \pi^{-n/2} |V|^{-1/2} |(V^{-1} + X'X)^{-1}|^{1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \\
= & \pi^{-n/2} |V|^{-1/2} |(V^{-1} + X'X)|^{-1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \\
= & \pi^{-n/2} |V(V^{-1} + X'X)|^{-1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \\
= & \pi^{-n/2} |I_k + VX'X|^{-1/2} \frac{\delta^{\alpha/2}}{\bar{\delta}^{\bar{\alpha}/2}} \frac{\Gamma(\bar{\alpha}/2)}{\Gamma(\alpha/2)} \tag{a.3.10.53}
\end{aligned}$$

derivations for equation (3.10.27)

Rearrange the expression:

$$\begin{aligned}
& f(y) \\
& \approx \frac{f(y|\beta^*, \sigma^*)\pi(\beta^*, \sigma^*)}{\pi(\sigma^*|y, \beta^*) \times \frac{1}{J} \sum_{j=1}^J \pi(\beta^*|\sigma^{(j)}, y)} \\
& = (2\pi\sigma)^{-n/2} \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \times \frac{(2\pi)^{-k/2} |V|^{-1/2} \exp\left(-\frac{1}{2}(\beta - b)'V^{-1}(\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J (2\pi)^{-k/2} |\bar{V}|^{-1/2} \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right)} \\
& \times \frac{\frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right)}{\frac{\bar{\delta}/2^{\bar{\alpha}/2}}{\Gamma(\bar{\alpha}/2)} \sigma^{-\bar{\alpha}/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right)} \\
& = 2^{-n/2} \pi^{-n/2} \sigma^{-n/2} \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \times \frac{(2\pi)^{-k/2} |V|^{-1/2} \exp\left(-\frac{1}{2}(\beta - b)'V^{-1}(\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J (2\pi)^{-k/2} |\bar{V}|^{-1/2} \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right)} \\
& \times \frac{\Gamma(\bar{\alpha}/2) 2^{(\alpha+n)/2}}{\Gamma(\alpha/2) 2^{\alpha/2}} \frac{\delta^{\alpha/2} \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right)}{\bar{\delta}^{\bar{\alpha}/2} \sigma^{-(\alpha+n)/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right)} \\
& = 2^{-n/2} \pi^{-n/2} \sigma^{-n/2} \times \frac{(2\pi)^{-k/2} |V|^{-1/2} \exp\left(-\frac{1}{2}(\beta - b)'V^{-1}(\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J (2\pi)^{-k/2} |\bar{V}|^{-1/2} \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right)} \\
& \times \frac{\Gamma(\bar{\alpha}/2) 2^{(\alpha+n)/2}}{\Gamma(\alpha/2) 2^{\alpha/2}} \frac{\delta^{\alpha/2} \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta + (y - X\beta)'(y - X\beta)}{2\sigma}\right)}{\bar{\delta}^{\bar{\alpha}/2} \sigma^{-(\alpha+n)/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right)} \\
& = 2^{-n/2} \pi^{-n/2} \sigma^{-n/2} \times \frac{(2\pi)^{-k/2} |V|^{-1/2} \exp\left(-\frac{1}{2}(\beta - b)'V^{-1}(\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J (2\pi)^{-k/2} |\bar{V}|^{-1/2} \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right)} \\
& \times \frac{\Gamma(\bar{\alpha}/2) 2^{(\alpha+n)/2}}{\Gamma(\alpha/2) 2^{\alpha/2}} \frac{\delta^{\alpha/2} \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right)}{\bar{\delta}^{\bar{\alpha}/2} \sigma^{-(\alpha+n)/2-1} \exp\left(-\frac{\bar{\delta}}{2\sigma}\right)} \\
& = \pi^{-n/2} \frac{|V|^{-1/2} \exp\left(-\frac{1}{2}(\beta - b)'V^{-1}(\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J |\bar{V}|^{-1/2} \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right)} \frac{\Gamma(\bar{\alpha}/2) \delta^{\alpha/2}}{\Gamma(\alpha/2) \bar{\delta}^{\bar{\alpha}/2}} \\
& = \pi^{-n/2} \frac{\exp\left(-\frac{1}{2}(\beta - b)'V^{-1}(\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J |V|^{1/2} |\bar{V}|^{-1/2} \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right)} \frac{\Gamma(\bar{\alpha}/2) \delta^{\alpha/2}}{\Gamma(\alpha/2) \bar{\delta}^{\bar{\alpha}/2}} \\
& = \pi^{-n/2} \frac{\exp\left(-\frac{1}{2}(\beta - b)'V^{-1}(\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J |V|^{1/2} |V^{-1} + \sigma^{-1}X'X|^{-1/2} \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right)} \frac{\Gamma(\bar{\alpha}/2) \delta^{\alpha/2}}{\Gamma(\alpha/2) \bar{\delta}^{\bar{\alpha}/2}} \\
& = \pi^{-n/2} \frac{\exp\left(-\frac{1}{2}(\beta - b)'V^{-1}(\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J |V|^{1/2} |V^{-1} + \sigma^{-1}X'X|^{1/2} \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right)} \frac{\Gamma(\bar{\alpha}/2) \delta^{\alpha/2}}{\Gamma(\alpha/2) \bar{\delta}^{\bar{\alpha}/2}} \\
& = \pi^{-n/2} \frac{\exp\left(-\frac{1}{2}(\beta - b)'V^{-1}(\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J |I_k + \sigma^{-1}VX'X|^{1/2} \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right)} \frac{\Gamma(\bar{\alpha}/2) \delta^{\alpha/2}}{\Gamma(\alpha/2) \bar{\delta}^{\bar{\alpha}/2}} \tag{a.3.10.54}
\end{aligned}$$

derivations for equation (3.10.29)

Substitute for the functions and rearrange:

$$\begin{aligned}
& \frac{1}{f(y)} \\
& \approx \frac{1}{J} \sum_{j=1}^J \frac{g(\theta^{(j)})}{f(y|\beta^{(j)}, \sigma^{(j)}, \gamma^{(j)}) \pi(\beta^{(j)}) \pi(\sigma^{(j)}) \pi(\gamma^{(j)})} \\
& = \frac{1}{J} \sum_{j=1}^J \frac{\omega^{-1} (2\pi)^{-(k+h+1)/2} |\hat{\Sigma}|^{-1/2} \exp\left(-\frac{1}{2}(\theta - \hat{\theta})' \hat{\Sigma}^{-1}(\theta - \hat{\theta})\right) \mathbb{1}(\theta \in \hat{\Theta})}{\left[(2\pi\sigma)^{-n/2} |W|^{-1/2} \exp\left(-\frac{1}{2} \frac{(y-X\beta)' W^{-1}(y-X\beta)}{\sigma}\right) \times (2\pi)^{-k/2} |V|^{-1/2} \exp\left(-\frac{1}{2}(\beta - b)' V^{-1}(\beta - b)\right) \right.} \\
& \quad \left. \times \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right) \times (2\pi)^{-h/2} |Q|^{-1/2} \exp\left(-\frac{1}{2}(\gamma - g)' Q^{-1}(\gamma - g)\right) \right] } \\
& = \frac{1}{J} \sum_{j=1}^J \mathbb{1}(\theta \in \hat{\Theta}) \omega^{-1} (2\pi)^{(n+k+h-(k+h+1))/2} |\hat{\Sigma}|^{-1/2} |W|^{1/2} |V|^{1/2} |Q|^{1/2} \frac{\Gamma(\alpha/2)}{\delta/2^{\alpha/2}} \sigma^{(\alpha+n)/2+1} \\
& \quad \times \exp\left(\frac{1}{2} \left[(y-X\beta)'(\sigma W)^{-1}(y-X\beta) + (\beta - b)' V^{-1}(\beta - b) + \delta \sigma^{-1} \right. \right. \\
& \quad \left. \left. + (\gamma - g)' Q^{-1}(\gamma - g) - (\theta - \hat{\theta})' \hat{\Sigma}^{-1}(\theta - \hat{\theta}) \right] \right) \\
& = (\omega J)^{-1} (2\pi)^{(n-1)/2} |\hat{\Sigma}|^{-1/2} |V|^{1/2} |Q|^{1/2} \frac{\Gamma(\alpha/2)}{\delta/2^{\alpha/2}} \\
& \quad \times \sum_{j=1}^J \mathbb{1}(\theta \in \hat{\Theta}) |W|^{1/2} \sigma^{(\alpha+n)/2+1} \exp\left(\frac{1}{2} \left[(y-X\beta)'(\sigma W)^{-1}(y-X\beta) + (\beta - b)' V^{-1}(\beta - b) \right. \right. \\
& \quad \left. \left. + \delta \sigma^{-1} + (\gamma - g)' Q^{-1}(\gamma - g) - (\theta - \hat{\theta})' \hat{\Sigma}^{-1}(\theta - \hat{\theta}) \right] \right) \tag{a.3.10.55}
\end{aligned}$$

Using logs on both sides yields:

$$\begin{aligned}
& -\log(f(y)) \approx \log\left((\omega J)^{-1} (2\pi)^{(n-1)/2} |\hat{\Sigma}|^{-1/2} |V|^{1/2} |Q|^{1/2} \frac{\Gamma(\alpha/2)}{\delta/2^{\alpha/2}}\right) \\
& + \log\left(\sum_{j=1}^J \mathbb{1}(\theta \in \hat{\Theta}) |W|^{1/2} \sigma^{(\alpha+n)/2+1} \exp\left(\frac{1}{2} \left[(y-X\beta)'(\sigma W)^{-1}(y-X\beta) + (\beta - b)' V^{-1}(\beta - b) \right. \right. \right. \\
& \quad \left. \left. + \delta \sigma^{-1} + (\gamma - g)' Q^{-1}(\gamma - g) - (\theta - \hat{\theta})' \hat{\Sigma}^{-1}(\theta - \hat{\theta}) \right] \right)\right) \tag{a.3.10.56}
\end{aligned}$$

or:

$$\begin{aligned}
& \log(f(y)) \approx -\log\left((\omega J)^{-1} (2\pi)^{(n-1)/2} |\hat{\Sigma}|^{-1/2} |V|^{1/2} |Q|^{1/2} \frac{\Gamma(\alpha/2)}{\delta/2^{\alpha/2}}\right) \\
& -\log\left(\sum_{j=1}^J \mathbb{1}(\theta \in \hat{\Theta}) |W|^{1/2} \sigma^{(\alpha+n)/2+1} \exp\left(\frac{1}{2} \left[(y-X\beta)'(\sigma W)^{-1}(y-X\beta) + (\beta - b)' V^{-1}(\beta - b) \right. \right. \right. \\
& \quad \left. \left. + \delta \sigma^{-1} + (\gamma - g)' Q^{-1}(\gamma - g) - (\theta - \hat{\theta})' \hat{\Sigma}^{-1}(\theta - \hat{\theta}) \right] \right)\right) \tag{a.3.10.57}
\end{aligned}$$

derivations for equation (3.10.32)

Substitute for the functions and rearrange:

$$\begin{aligned}
& \frac{1}{f(y)} \\
& \approx \frac{1}{J} \sum_{j=1}^J \frac{g(\theta^{(j)})}{f(y|\beta^{(j)}, \sigma^{(j)}, \phi^{(j)}) \pi(\beta^{(j)}) \pi(\sigma^{(j)}) \pi(\phi^{(j)})} \\
& = \frac{1}{J} \sum_{j=1}^J \frac{\omega^{-1} (2\pi)^{-(k+q+1)/2} |\hat{\Sigma}|^{-1/2} \exp\left(-\frac{1}{2}(\theta - \hat{\theta})' \hat{\Sigma}^{-1}(\theta - \hat{\theta})\right) \mathbb{1}(\theta \in \hat{\Theta})}{\left[(2\pi\sigma)^{-T/2} \exp\left(-\frac{1}{2}(\varepsilon - E\phi)' \sigma^{-1}(\varepsilon - E\phi)\right) \times (2\pi)^{-k/2} |V|^{-1/2} \exp\left(-\frac{1}{2}(\beta - b)' V^{-1}(\beta - b)\right) \right.} \\
& \quad \left. \times \frac{\delta/2^{\alpha/2}}{\Gamma(\alpha/2)} \sigma^{-\alpha/2-1} \exp\left(-\frac{\delta}{2\sigma}\right) \times (2\pi)^{-q/2} |Z|^{-1/2} \exp\left(-\frac{1}{2}(\phi - p)' Z^{-1}(\phi - p)\right) \right] } \\
& = \frac{1}{J} \sum_{j=1}^J \mathbb{1}(\theta \in \hat{\Theta}) \omega^{-1} (2\pi)^{(T+k+q-(k+q+1))/2} |\hat{\Sigma}|^{-1/2} |V|^{1/2} |Z|^{1/2} \frac{\Gamma(\alpha/2)}{\delta/2^{\alpha/2}} \sigma^{(\alpha+T)/2+1} \\
& \quad \times \exp\left(\frac{1}{2} \left[(\varepsilon - E\phi)' \sigma^{-1}(\varepsilon - E\phi) + (\beta - b)' V^{-1}(\beta - b) \right. \right. \\
& \quad \left. \left. + \delta \sigma^{-1} + (\phi - p)' Z^{-1}(\phi - p) - (\theta - \hat{\theta})' \hat{\Sigma}^{-1}(\theta - \hat{\theta}) \right] \right) \\
& = (\omega J)^{-1} (2\pi)^{(T-1)/2} |\hat{\Sigma}|^{-1/2} |V|^{1/2} |Z|^{1/2} \frac{\Gamma(\alpha/2)}{\delta/2^{\alpha/2}} \\
& \quad \times \sum_{j=1}^J \mathbb{1}(\theta \in \hat{\Theta}) \sigma^{(\alpha+T)/2+1} \exp\left(\frac{1}{2} \left[(\varepsilon - E\phi)' \sigma^{-1}(\varepsilon - E\phi) + (\beta - b)' V^{-1}(\beta - b) \right. \right. \\
& \quad \left. \left. + \delta \sigma^{-1} + (\phi - p)' Z^{-1}(\phi - p) - (\theta - \hat{\theta})' \hat{\Sigma}^{-1}(\theta - \hat{\theta}) \right] \right)
\end{aligned} \tag{a.3.10.58}$$

Using logs on both sides yields:

$$\begin{aligned}
& -\log(f(y)) \approx \log\left((\omega J)^{-1} (2\pi)^{(T-1)/2} |\hat{\Sigma}|^{-1/2} |V|^{1/2} |Z|^{1/2} \frac{\Gamma(\alpha/2)}{\delta/2^{\alpha/2}}\right) \\
& + \log\left(\sum_{j=1}^J \mathbb{1}(\theta \in \hat{\Theta}) \sigma^{(\alpha+T)/2+1} \exp\left(\frac{1}{2} \left[(\varepsilon - E\phi)' \sigma^{-1}(\varepsilon - E\phi) + (\beta - b)' V^{-1}(\beta - b) \right. \right. \right. \\
& \quad \left. \left. + \delta \sigma^{-1} + (\phi - p)' Z^{-1}(\phi - p) - (\theta - \hat{\theta})' \hat{\Sigma}^{-1}(\theta - \hat{\theta}) \right] \right)\right)
\end{aligned} \tag{a.3.10.59}$$

or:

$$\begin{aligned}
& \log(f(y)) \approx -\log\left((\omega J)^{-1} (2\pi)^{(T-1)/2} |\hat{\Sigma}|^{-1/2} |V|^{1/2} |Z|^{1/2} \frac{\Gamma(\alpha/2)}{\delta/2^{\alpha/2}}\right) \\
& -\log\left(\sum_{j=1}^J \mathbb{1}(\theta \in \hat{\Theta}) \sigma^{(\alpha+T)/2+1} \exp\left(\frac{1}{2} \left[(\varepsilon - E\phi)' \sigma^{-1}(\varepsilon - E\phi) + (\beta - b)' V^{-1}(\beta - b) \right. \right. \right. \\
& \quad \left. \left. + \delta \sigma^{-1} + (\phi - p)' Z^{-1}(\phi - p) - (\theta - \hat{\theta})' \hat{\Sigma}^{-1}(\theta - \hat{\theta}) \right] \right)\right)
\end{aligned} \tag{a.3.10.60}$$

PART IV

Vector autoregressions

Vector autoregressions

derivations for equation (4.11.9)

Consider first β . To do so, rewrite the likelihood function as:

$$\log(f(y|\beta, \Sigma)) = -\frac{nT}{2} \log(2\pi) - \frac{1}{2} \log(|\bar{\Sigma}|) - \frac{1}{2} (y' \bar{\Sigma}^{-1} y + \beta' \bar{X}' \bar{\Sigma}^{-1} \bar{X} \beta - 2\beta' \bar{X}' \bar{\Sigma}^{-1} y) \quad (\text{a.4.11.1})$$

Then solve for the partial derivative:

$$\begin{aligned} \frac{\partial \log(f(y|\beta, \Sigma))}{\partial \beta} &= 0 \\ \Leftrightarrow -\frac{1}{2} (2\beta' \bar{X}' \bar{\Sigma}^{-1} \bar{X} - 2y' \bar{\Sigma}^{-1} \bar{X}' y) &= 0 \\ \Leftrightarrow \beta' \bar{X}' \bar{\Sigma}^{-1} \bar{X} - y' \bar{\Sigma}^{-1} \bar{X}' y &= 0 \\ \Leftrightarrow \beta' \bar{X}' \bar{\Sigma}^{-1} \bar{X} &= y' \bar{\Sigma}^{-1} \bar{X}' y \\ \Leftrightarrow \bar{X}' \bar{\Sigma}^{-1} \bar{X} \beta &= \bar{X}' \bar{\Sigma}^{-1} y \\ \Leftrightarrow \beta &= (\bar{X}' \bar{\Sigma}^{-1} \bar{X})^{-1} \bar{X}' \bar{\Sigma}^{-1} y \end{aligned} \quad (\text{a.4.11.2})$$

The formula can simplify further. Note first that:

$$\begin{aligned} \bar{X}' \bar{\Sigma}^{-1} &= (I_n \otimes X)' (\Sigma \otimes I_T)^{-1} \\ &= (I_n \otimes X') (\Sigma^{-1} \otimes I_T) \\ &= \Sigma^{-1} \otimes X' \end{aligned} \quad (\text{a.4.11.3})$$

Substituting for (a.4.11.3) in (a.4.11.2):

$$\begin{aligned} \beta &= (\bar{X}' \bar{\Sigma}^{-1} \bar{X})^{-1} \bar{X}' \bar{\Sigma}^{-1} y \\ &= [(\Sigma^{-1} \otimes X') (I_n \otimes X)]^{-1} [(\Sigma^{-1} \otimes X') y] \\ &= (\Sigma^{-1} \otimes X' X)^{-1} (\Sigma^{-1} \otimes X') \text{vec}(Y) \\ &= (\Sigma \otimes (X' X)^{-1}) (\Sigma^{-1} \otimes X') \text{vec}(Y) \\ &= (I_n \otimes (X' X)^{-1} X') \text{vec}(Y) \\ &= \text{vec}((X' X)^{-1} X' Y) \\ &= \text{vec}(\hat{B}) \end{aligned} \quad (\text{a.4.11.4})$$

with:

$$\hat{B} = (X' X)^{-1} X' Y \quad (\text{a.4.11.5})$$

Hence the maximum likelihood estimate is $\hat{\beta} = \text{vec}(\hat{B})$.

To obtain the maximum likelihood estimate for Σ , it is convenient to use property d.7 that states the equivalence between the multivariate normal and matrix normal distributions. Doing so, the likelihood function of the VAR model rewrites as:

$$f(y|\beta, \Sigma) = (2\pi)^{-nT/2} |\Sigma|^{-T/2} \exp \left(-\frac{1}{2} \text{tr} [\Sigma^{-1} (Y - X\beta)' (Y - X\beta)] \right) \quad (\text{a.4.11.6})$$

And the log-likelihood becomes:

$$\log(f(y|\beta, \Sigma)) = -\frac{nT}{2} \log(2\pi) - \frac{T}{2} \log(|\Sigma|) - \frac{1}{2} \text{tr} [\Sigma^{-1} (Y - X\beta)' (Y - X\beta)] \quad (\text{a.4.11.7})$$

Then solve for the partial derivative:

$$\begin{aligned} \frac{\partial \log(f(y|\beta, \Sigma))}{\partial \Sigma} &= 0 \\ \Leftrightarrow -\frac{T}{2} \Sigma^{-1} - \frac{1}{2} (Y - X\beta)' (Y - X\beta) (-\Sigma^{-1} \Sigma^{-1}) &= 0 \\ \Leftrightarrow -T \Sigma^{-1} + (Y - X\beta)' (Y - X\beta) (\Sigma^{-1} \Sigma^{-1}) &= 0 \\ \Leftrightarrow T \Sigma^{-1} &= (Y - X\beta)' (Y - X\beta) (\Sigma^{-1} \Sigma^{-1}) \\ \Leftrightarrow T \Sigma &= (Y - X\beta)' (Y - X\beta) \\ \Leftrightarrow \Sigma &= \frac{1}{T} (Y - X\beta)' (Y - X\beta) \end{aligned} \quad (\text{a.4.11.8})$$

Replacing β with its maximum likelihood estimate $\hat{\beta}$, we conclude that the maximum likelihood estimate for Σ is $\hat{\Sigma} = \frac{1}{T} (Y - X\hat{\beta})' (Y - X\hat{\beta})$.

derivations for equation (4.11.14)

Group terms:

$$\begin{aligned} \pi(\beta|y) &\propto \exp \left(-\frac{1}{2} (y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) \right) \times \exp \left(-\frac{1}{2} (\beta - b)' V^{-1} (\beta - b) \right) \\ &= \exp \left(-\frac{1}{2} [(y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) + (\beta - b)' V^{-1} (\beta - b)] \right) \end{aligned} \quad (\text{a.4.11.9})$$

Consider the terms in square brackets:

$$\begin{aligned} &(y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) + (\beta - b)' V^{-1} (\beta - b) \\ &= y' \bar{\Sigma}^{-1} y + \beta' \bar{X}' \bar{\Sigma}^{-1} \bar{X} \beta - 2\beta' \bar{X}' \bar{\Sigma}^{-1} y + \beta' V^{-1} \beta + b' V^{-1} b - 2\beta' V^{-1} b \\ &= \beta' (V^{-1} + \bar{X}' \bar{\Sigma}^{-1} \bar{X}) \beta - 2\beta' (V^{-1} b + \bar{X}' \bar{\Sigma}^{-1} y) + b' V^{-1} b + y' \bar{\Sigma}^{-1} y \end{aligned} \quad (\text{a.4.11.10})$$

Complete the squares:

$$= \beta' (V^{-1} + \bar{X}' \bar{\Sigma}^{-1} \bar{X}) \beta - 2\beta' \bar{V}^{-1} \bar{V} (V^{-1} b + \bar{X}' \bar{\Sigma}^{-1} y) + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b} + b' V^{-1} b + y' \bar{\Sigma}^{-1} y \quad (\text{a.4.11.11})$$

Define:

$$\bar{V} = (V^{-1} + \bar{X}' \bar{\Sigma}^{-1} \bar{X})^{-1} \quad \bar{b} = \bar{V} (V^{-1} b + \bar{X}' \bar{\Sigma}^{-1} y) \quad (\text{a.4.11.12})$$

Then (a.4.11.11) rewrites:

$$\begin{aligned}
 &= \beta' \bar{V}^{-1} \beta - 2\beta' \bar{V}^{-1} \bar{b} + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b} + b' V^{-1} b + y' \bar{\Sigma}^{-1} y \\
 &= (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + (b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + y' \bar{\Sigma}^{-1} y)
 \end{aligned} \tag{a.4.11.13}$$

Substitute (a.4.11.13) back in (a.4.11.9):

$$\begin{aligned}
 &\pi(\beta|y) \\
 &= \exp \left(-\frac{1}{2} [(\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + (b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + y' \bar{\Sigma}^{-1} y)] \right) \\
 &= \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right) \exp \left(-\frac{1}{2} (b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + y' \bar{\Sigma}^{-1} y) \right) \\
 &\propto \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right)
 \end{aligned} \tag{a.4.11.14}$$

Where the last line obtains by noting that the second term in row 2 does not involve β and can hence be relegated to the normalization constant.

The terms in (a.4.11.12) simplify. Note first that:

$$\begin{aligned}
 &\bar{X}' \bar{\Sigma}^{-1} \bar{X} \\
 &= (I_n \otimes X)' (\Sigma \otimes I_T)^{-1} (I_n \otimes X) \\
 &= (I_n \otimes X') (\Sigma^{-1} \otimes I_T) (I_n \otimes X) \\
 &= (\Sigma^{-1} \otimes X') (I_n \otimes X) \\
 &= \Sigma^{-1} \otimes X' X
 \end{aligned} \tag{a.4.11.15}$$

Similarly:

$$\begin{aligned}
 &\bar{X}' \bar{\Sigma}^{-1} y \\
 &= (I_n \otimes X)' (\Sigma \otimes I_T)^{-1} \text{vec}(Y) \\
 &= (I_n \otimes X') (\Sigma^{-1} \otimes I_T) \text{vec}(Y) \\
 &= (\Sigma^{-1} \otimes X') \text{vec}(Y) \\
 &= \text{vec}(X' Y \Sigma^{-1})
 \end{aligned} \tag{a.4.11.16}$$

Then (a.4.11.12) rewrites:

$$\bar{V} = (V^{-1} + \Sigma^{-1} \otimes X' X)^{-1} \quad \bar{b} = \bar{V} (V^{-1} b + \text{vec}(X' Y \Sigma^{-1})) \tag{a.4.11.17}$$

derivations for equation (4.11.23)

Start from the vectorized likelihood function:

$$f(y|\beta, \Sigma) = (2\pi)^{-nT/2} |\bar{\Sigma}|^{-1/2} \exp \left(-\frac{1}{2} (y - \bar{X} \beta)' \bar{\Sigma}^{-1} (y - \bar{X} \beta) \right) \tag{a.4.11.18}$$

Use then property d.7 that establishes the equivalence between the multivariate normal and matrix normal distributions to reformulate the likelihood in vectorized form as:

$$f(y|\beta, \Sigma) = (2\pi)^{-nT/2} |\Sigma|^{-T/2} \exp \left(-\frac{1}{2} \text{tr} [\Sigma^{-1} (Y - X \beta)' (Y - X \beta)] \right) \tag{a.4.11.19}$$

Consider the quadratic term:

$$\begin{aligned}
& (Y - X\beta)'(Y - X\beta) \\
&= Y'Y + \beta'X'XB - 2\beta'X'Y \\
&= Y'Y + \beta'X'XB - 2\beta'X'Y + 2\hat{\beta}X'Y - 2\hat{\beta}X'Y \\
&= Y'Y + \beta'X'XB - 2\beta'(X'X)(X'X)^{-1}X'Y + 2\hat{\beta}(X'X)(X'X)^{-1}X'Y - 2\hat{\beta}X'Y \\
&= Y'Y + \beta'X'XB - 2\beta'(X'X)\hat{\beta} + 2\hat{\beta}(X'X)\hat{\beta} - 2\hat{\beta}X'Y \\
&= Y'Y + \beta'X'XB - 2\beta'(X'X)\hat{\beta} + \hat{\beta}(X'X)\hat{\beta} + \hat{\beta}(X'X)\hat{\beta} - 2\hat{\beta}X'Y \\
&= (\beta'(X'X)\hat{\beta} + \hat{\beta}(X'X)\hat{\beta} - 2\beta'(X'X)\hat{\beta}) + (Y'Y + \hat{\beta}(X'X)\hat{\beta} - 2\hat{\beta}X'Y) \\
&= (B - \hat{B})'(X'X)(B - \hat{B}) + (Y - X\hat{B})'(Y - X\hat{B})
\end{aligned} \tag{a.4.11.20}$$

Hence (a.4.11.19) rewrites:

$$\begin{aligned}
f(y|\beta, \Sigma) &= (2\pi)^{-nT/2} |\Sigma|^{-T/2} \exp \left(-\frac{1}{2} \text{tr} [\Sigma^{-1} (B - \hat{B})'(X'X)(B - \hat{B}) + \Sigma^{-1} (Y - X\hat{B})'(Y - X\hat{B})] \right) \\
&= (2\pi)^{-nT/2} |\Sigma|^{-T/2} \exp \left(-\frac{1}{2} \text{tr} [\Sigma^{-1} (B - \hat{B})'(X'X)(B - \hat{B})] \right) \\
&\quad \times \exp \left(-\frac{1}{2} \text{tr} [\Sigma^{-1} (Y - X\hat{B})'(Y - X\hat{B})] \right)
\end{aligned} \tag{a.4.11.21}$$

Also, it follows from property m.55 that:

$$\text{tr} [\Sigma^{-1} (B - \hat{B})'(X'X)(B - \hat{B})] = (\beta - \hat{\beta})' (\Sigma \otimes (X'X)^{-1})^{-1} (\beta - \hat{\beta}) \tag{a.4.11.22}$$

Substituting back in (a.4.11.21) yields:

$$\begin{aligned}
f(y|\beta, \Sigma) &= (2\pi)^{-nT/2} |\Sigma|^{-T/2} \exp \left(-\frac{1}{2} (\beta - \hat{\beta})' (\Sigma \otimes (X'X)^{-1})^{-1} (\beta - \hat{\beta}) \right) \\
&\quad \times \exp \left(-\frac{1}{2} \text{tr} [\Sigma^{-1} (Y - X\hat{B})'(Y - X\hat{B})] \right)
\end{aligned} \tag{a.4.11.23}$$

derivations for equation (4.11.32)

Start from the joint posterior:

$$\begin{aligned}
\pi(\beta, \Sigma|y) &\propto |\Sigma|^{-T/2} \exp \left(-\frac{1}{2} (\beta - \hat{\beta})' (\Sigma \otimes (X'X)^{-1})^{-1} (\beta - \hat{\beta}) \right) \\
&\quad \times \exp \left(-\frac{1}{2} \text{tr} [\Sigma^{-1} (Y - X\hat{B})'(Y - X\hat{B})] \right) \\
&\quad \times |\Sigma \otimes W|^{-1/2} \exp \left(-\frac{1}{2} (\beta - b)' (\Sigma \otimes W)^{-1} (\beta - b) \right) \\
&\quad \times |\Sigma|^{-(\alpha+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} S \} \right)
\end{aligned} \tag{a.4.11.24}$$

Note first that:

$$|\Sigma \otimes W|^{-1/2} = |\Sigma|^{-k/2} |W|^{-n/2} \tag{a.4.11.25}$$

Hence, substituting back in (a.4.11.24) and rearranging:

$$\begin{aligned}
\pi(\beta, \Sigma|y) &\propto \exp\left(-\frac{1}{2}(\beta - \hat{\beta})' (\Sigma \otimes (X'X)^{-1})^{-1} (\beta - \hat{\beta})\right) \\
&\times |\Sigma|^{-T/2} \exp\left(-\frac{1}{2} \text{tr} [\Sigma^{-1} (Y - X\hat{B})' (Y - X\hat{B})]\right) \\
&\times |\Sigma|^{-k/2} \exp\left(-\frac{1}{2}(\beta - b)' (\Sigma \otimes W)^{-1} (\beta - b)\right) \\
&\times |\Sigma|^{-(\alpha+n+1)/2} \exp\left(-\frac{1}{2} \text{tr} \{\Sigma^{-1} S\}\right)
\end{aligned} \tag{a.4.11.26}$$

After regrouping, one obtains:

$$\begin{aligned}
\pi(\beta, \Sigma|y) &\propto |\Sigma|^{-k/2} \exp\left(-\frac{1}{2} \left[(\beta - b)' (\Sigma \otimes W)^{-1} (\beta - b) + (\beta - \hat{\beta})' (\Sigma \otimes (X'X)^{-1})^{-1} (\beta - \hat{\beta}) \right]\right) \\
&\times |\Sigma|^{-(\alpha+T+n+1)/2} \exp\left(-\frac{1}{2} \text{tr} \{\Sigma^{-1} [S + (Y - X\hat{B})' (Y - X\hat{B})]\}\right)
\end{aligned} \tag{a.4.11.27}$$

Consider the term within the curly brackets in the first row:

$$\begin{aligned}
&(\beta - b)' (\Sigma \otimes W)^{-1} (\beta - b) + (\beta - \hat{\beta})' (\Sigma \otimes (X'X)^{-1})^{-1} (\beta - \hat{\beta}) \\
&= \text{tr} \{\Sigma^{-1} (\mathcal{B} - B)' W^{-1} (\mathcal{B} - B)\} + \text{tr} \{\Sigma^{-1} (\mathcal{B} - \hat{B})' (X'X) (\mathcal{B} - \hat{B})\} \\
&= \text{tr} \{\Sigma^{-1} [(\mathcal{B} - B)' W^{-1} (\mathcal{B} - B) + (\mathcal{B} - \hat{B})' (X'X) (\mathcal{B} - \hat{B})]\} \\
&= \text{tr} \{\Sigma^{-1} [(\mathcal{B} - B)' W^{-1} (\mathcal{B} - B) + (\mathcal{B} - \hat{B})' (X'X) (\mathcal{B} - \hat{B})]\} \\
&= \text{tr} \{\Sigma^{-1} [\mathcal{B}' W^{-1} \mathcal{B} + \mathcal{B}' W^{-1} B - 2 \mathcal{B}' W^{-1} B + \mathcal{B}' (X'X) \mathcal{B} + \hat{B}' (X'X) \hat{B} - 2 \mathcal{B}' (X'X) \hat{B}]\} \\
&= \text{tr} \{\Sigma^{-1} [\mathcal{B}' (W^{-1} + X'X) \mathcal{B} - 2 \mathcal{B}' (W^{-1} B + X'X \hat{B}) + \mathcal{B}' W^{-1} B + \hat{B}' (X'X) \hat{B}]\}
\end{aligned} \tag{a.4.11.28}$$

Complete the squares:

$$= \text{tr} \{\Sigma^{-1} [\mathcal{B}' (W^{-1} + X'X) \mathcal{B} - 2 \mathcal{B}' \bar{W}^{-1} \bar{W} (W^{-1} B + X'X \hat{B}) + \bar{B}' \bar{W}^{-1} \bar{B} - \bar{B}' \bar{W}^{-1} \bar{B} + B' W^{-1} B + \hat{B}' (X'X) \hat{B}]\} \tag{a.4.11.29}$$

Define:

$$\bar{W} = (W^{-1} + X'X)^{-1} \quad \bar{B} = \bar{W} (W^{-1} B + X'X \hat{B}) \tag{a.4.11.30}$$

then the expression becomes:

$$\begin{aligned}
&= \text{tr} \{\Sigma^{-1} [\mathcal{B}' \bar{W}^{-1} \mathcal{B} - 2 \mathcal{B}' \bar{W}^{-1} \bar{B} + \bar{B}' \bar{W}^{-1} \bar{B} - \bar{B}' \bar{W}^{-1} \bar{B} + B' W^{-1} B + \hat{B}' (X'X) \hat{B}]\} \\
&= \text{tr} \{\Sigma^{-1} [(\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B}) - \bar{B}' \bar{W}^{-1} \bar{B} + B' W^{-1} B + \hat{B}' (X'X) \hat{B}]\} \\
&= \text{tr} \{\Sigma^{-1} (\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B})\} + \text{tr} \{\Sigma^{-1} [B' W^{-1} B + \hat{B}' (X'X) \hat{B} - \bar{B}' \bar{W}^{-1} \bar{B}]\}
\end{aligned} \tag{a.4.11.31}$$

Substituting (a.4.11.31) in (a.4.11.27) yields:

$$\begin{aligned}
\pi(\beta, \Sigma|y) &\propto |\Sigma|^{-k/2} \exp \left(-\frac{1}{2} \left[\text{tr} \{ \Sigma^{-1} (\mathcal{B}' - \bar{\mathcal{B}})' \bar{W}^{-1} (\mathcal{B}' - \bar{\mathcal{B}}) \} + \text{tr} \{ \Sigma^{-1} [B' W^{-1} B + \hat{B}' (X' X) \hat{B} - \bar{B}' \bar{W}^{-1} \bar{B}] \} \right] \right) \\
&\quad \times |\Sigma|^{-(\alpha+T+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} [S + (Y - X \hat{B})' (Y - X \hat{B})] \} \right) \\
&= |\Sigma|^{-k/2} \exp \left(-\frac{1}{2} \left[\text{tr} \{ \Sigma^{-1} (\mathcal{B}' - \bar{\mathcal{B}})' \bar{W}^{-1} (\mathcal{B}' - \bar{\mathcal{B}}) \} \right] \right) \\
&\quad \times |\Sigma|^{-(\alpha+T+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} [S + (Y - X \hat{B})' (Y - X \hat{B}) + B' W^{-1} B + \hat{B}' (X' X) \hat{B} - \bar{B}' \bar{W}^{-1} \bar{B}] \} \right)
\end{aligned} \tag{a.4.11.32}$$

Define:

$$\bar{\alpha} = \alpha + T \quad \bar{S} = S + (Y - X \hat{B})' (Y - X \hat{B}) + B' W^{-1} B + \hat{B}' (X' X) \hat{B} - \bar{B}' \bar{W}^{-1} \bar{B} \tag{a.4.11.33}$$

Then the expression becomes:

$$\begin{aligned}
\pi(\beta, \Sigma|y) &\propto |\Sigma|^{-k/2} \exp \left(-\frac{1}{2} \left[\text{tr} \{ \Sigma^{-1} (\mathcal{B}' - \bar{\mathcal{B}})' \bar{V}^{-1} (\mathcal{B}' - \bar{\mathcal{B}}) \} \right] \right) \\
&\quad \times |\Sigma|^{-(\bar{\alpha}+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} \bar{S} \} \right)
\end{aligned} \tag{a.4.11.34}$$

Some of the expressions simplify. Consider first (a.4.11.30):

$$\bar{B} = \bar{W} (W^{-1} B + X' X \hat{B}) = \bar{W} (W^{-1} B + X' X (X' X)^{-1} X' Y) = \bar{W} (W^{-1} B + X' Y) \tag{a.4.11.35}$$

Consider then:

$$\begin{aligned}
&(Y - X \hat{B})' (Y - X \hat{B}) + \hat{B}' X' X \hat{B} \\
&= Y' Y + \hat{B}' X' X \hat{B} - \hat{B}' X' Y - Y' X \hat{B} + \hat{B}' X' X \hat{B} \\
&= Y' Y + 2 \hat{B}' X' X \hat{B} - \hat{B}' X' Y - Y' X \hat{B} \\
&= Y' Y + 2 Y' X (X' X)^{-1} X' X (X' X)^{-1} X' Y - Y' X (X' X)^{-1} X' Y - Y' X (X' X)^{-1} X' Y \\
&= Y' Y + 2 Y' X (X' X)^{-1} X' Y - Y' X (X' X)^{-1} X' Y - Y' X (X' X)^{-1} X' Y \\
&= Y' Y
\end{aligned} \tag{a.4.11.36}$$

Substitute back in (a.4.11.33):

$$\bar{S} = S + Y' Y + B' W^{-1} B - \bar{B}' \bar{W}^{-1} \bar{B} \tag{a.4.11.37}$$

derivations for equation (4.11.37)

Start from the initial equation and reformulate:

$$\begin{aligned}
& \pi(\mathcal{B}|y) \\
& \propto \left| \bar{S} + (\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B}) \right|^{-\frac{\bar{\alpha}+k}{2}} \\
& = \left| \bar{S} \{ I_n + \bar{S}^{-1} (\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B}) \} \right|^{-\frac{\bar{\alpha}+k}{2}} \\
& = \left| \bar{S} \right|^{-\frac{\bar{\alpha}+k}{2}} \left| I_n + \bar{S}^{-1} (\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B}) \right|^{-\frac{\bar{\alpha}+k}{2}} \\
& = \left| I_n + \bar{S}^{-1} (\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B}) \right|^{-\frac{\bar{\alpha}+k}{2}} \\
& = \left| I_n + \bar{S}^{-1} (\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B}) \right|^{-\frac{\alpha+T+k}{2}} \\
& = \left| I_n + \bar{S}^{-1} (\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B}) \right|^{-\frac{(\alpha+T-n+1)+k+n-1}{2}}
\end{aligned} \tag{a.4.11.38}$$

Define:

$$\hat{\alpha} = \alpha + T - n + 1 \tag{a.4.11.39}$$

Then:

$$\begin{aligned}
& = \left| I_n + \bar{S}^{-1} (\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B}) \right|^{-\frac{\hat{\alpha}+k+n-1}{2}} \\
& = \left| I_n + \frac{1}{\hat{\alpha}} (\bar{S}/\hat{\alpha})^{-1} (\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B}) \right|^{-\frac{\hat{\alpha}+k+n-1}{2}}
\end{aligned} \tag{a.4.11.40}$$

Define:

$$\hat{S} = \bar{S}/\hat{\alpha} \tag{a.4.11.41}$$

Then:

$$\pi(\mathcal{B}|y) \propto \left| I_n + \frac{1}{\hat{\alpha}} \hat{S}^{-1} (\mathcal{B}' - \bar{B})' \bar{W}^{-1} (\mathcal{B}' - \bar{B}) \right|^{-\frac{\hat{\alpha}+k+n-1}{2}} \tag{a.4.11.42}$$

derivations for equation (4.11.45)

Note that:

$$|\bar{\Sigma}|^{-1/2} = |\Sigma \otimes I_T|^{-1/2} = |\Sigma|^{-T/2} |I_T|^{-n/2} = |\Sigma|^{-T/2} \tag{a.4.11.43}$$

Also:

$$(y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) = (y - (I_n \otimes X)\beta)' (\Sigma \otimes I_T)^{-1} (y - (I_n \otimes X)\beta) = tr \{ \Sigma^{-1} (Y - X\mathcal{B})' (Y - X\mathcal{B}) \} \tag{a.4.11.44}$$

Then substituting in the original expression:

$$\begin{aligned}
\pi(\Sigma|y, \beta) &\propto |\Sigma|^{-T/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}(Y - X\beta)'(Y - X\beta)\}\right) \times |\Sigma|^{-(\alpha+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}S\}\right) \\
&= |\Sigma|^{-(\alpha+T+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}[S + (Y - X\beta)'(Y - X\beta)]\}\right) \\
&= |\Sigma|^{-(\bar{\alpha}+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}\bar{S}\}\right)
\end{aligned} \tag{a.4.11.45}$$

with:

$$\bar{\alpha} = \alpha + T \quad \bar{S} = S + (Y - X\beta)'(Y - X\beta) \tag{a.4.11.46}$$

derivations for equation (4.11.49)

Start from Bayes rule and rearrange:

$$\begin{aligned}
&\pi(\beta, \Sigma|y) \\
&\propto f(y|\beta, \Sigma) \pi(\beta) \pi(\Sigma) \\
&\propto |\Sigma|^{-T/2} \exp\left(-\frac{1}{2}(\beta - \hat{\beta})'(\Sigma \otimes (X'X)^{-1})^{-1}(\beta - \hat{\beta})\right) \\
&\times \exp\left(-\frac{1}{2} \text{tr}[\Sigma^{-1}(Y - X\hat{\beta})'(Y - X\hat{\beta})]\right) \times |\Sigma|^{-(\alpha+1)/2} \\
&= |\Sigma|^{-k/2} \exp\left(-\frac{1}{2}(\beta - \hat{\beta})'(\Sigma \otimes (X'X)^{-1})^{-1}(\beta - \hat{\beta})\right) \\
&\times |\Sigma|^{-(T-k+\alpha+1)/2} \exp\left(-\frac{1}{2} \text{tr}[\Sigma^{-1}(Y - X\hat{\beta})'(Y - X\hat{\beta})]\right) \\
&= |\Sigma|^{-k/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}(\mathcal{B}' - \hat{B})'(X'X)(\mathcal{B}' - \hat{B})\}\right) \\
&\times |\Sigma|^{-(T-k+n+3)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}(Y - X\hat{\beta})'(Y - X\hat{\beta})\}\right) \\
&= |\Sigma|^{-k/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}(\mathcal{B}' - \hat{B})'\hat{W}^{-1}(\mathcal{B}' - \hat{B})\}\right) \\
&\times |\Sigma|^{-(\hat{\alpha}+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}\hat{S}\}\right)
\end{aligned} \tag{a.4.11.47}$$

with:

$$\hat{W} = (X'X)^{-1} \quad \hat{\alpha} = T - k + 2 \quad \hat{S} = (Y - X\hat{\beta})'(Y - X\hat{\beta}) \tag{a.4.11.48}$$

derivations for equation (4.11.52)

Start from the joint posterior, group the terms and integrate:

$$\pi(\mathcal{B}|y) = \int \pi(\mathcal{B}, \Sigma|y) d\Sigma \propto \int |\Sigma|^{-(\hat{\alpha}+k+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}[\hat{S} + (\mathcal{B}' - \hat{B})'\hat{W}^{-1}(\mathcal{B}' - \hat{B})]\}\right) d\Sigma \tag{a.4.11.49}$$

This is the kernel of an inverse Wishart distribution with degrees of freedom $(\hat{\alpha} + k)$ and scale $\hat{S} + (\mathcal{B}' - \hat{B})'\hat{W}^{-1}(\mathcal{B}' - \hat{B})$, and integration yields the reciprocal of the normalization constant of the distribution. Hence:

$$\pi(\mathcal{B}|y) \propto \Gamma_n \left(\frac{\hat{\alpha} + k}{2} \right) 2^{(\hat{\alpha} + k)n/2} |\hat{S} + (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B})|^{-\frac{\hat{\alpha} + k}{2}} \propto |\hat{S} + (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B})|^{-\frac{\hat{\alpha} + k}{2}} \quad (\text{a.4.11.50})$$

Rearrange:

$$\begin{aligned} \pi(\mathcal{B}|y) &\propto |\hat{S} + (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B})|^{-\frac{\hat{\alpha} + k}{2}} \\ &= |\hat{S} \{I_n + \hat{S}^{-1} (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B})\}|^{-\frac{\hat{\alpha} + k}{2}} \\ &= |\hat{S}|^{-\frac{\hat{\alpha} + k}{2}} |I_n + \hat{S}^{-1} (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B})|^{-\frac{\hat{\alpha} + k}{2}} \\ &= |I_n + \hat{S}^{-1} (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B})|^{-\frac{\hat{\alpha} + k}{2}} \\ &= |I_n + \hat{S}^{-1} (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B})|^{-\frac{T+2}{2}} \\ &= |I_n + \hat{S}^{-1} (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B})|^{-\frac{(T-n-k+3)+k+n-1}{2}} \end{aligned} \quad (\text{a.4.11.51})$$

Define:

$$\tilde{\alpha} = T - n - k + 3 \quad (\text{a.4.11.52})$$

Then:

$$\begin{aligned} &= |I_n + \hat{S}^{-1} (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B})|^{-\frac{\tilde{\alpha} + k + n - 1}{2}} \\ &= \left| I_n + \frac{1}{\tilde{\alpha}} (\hat{S}/\tilde{\alpha})^{-1} (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B}) \right|^{-\frac{\tilde{\alpha} + k + n - 1}{2}} \end{aligned} \quad (\text{a.4.11.53})$$

Define:

$$\tilde{S} = \hat{S}/\tilde{\alpha} \quad (\text{a.4.11.54})$$

Then:

$$\pi(\mathcal{B}|y) \propto \left| I_n + \frac{1}{\tilde{\alpha}} \tilde{S}^{-1} (\mathcal{B}' - \hat{B})' \hat{W}^{-1} (\mathcal{B}' - \hat{B}) \right|^{-\frac{\tilde{\alpha} + k + n - 1}{2}} \quad (\text{a.4.11.55})$$

derivations for equation (4.11.68)

Start from the original likelihood function:

$$f(y|\beta, \Sigma) = (2\pi)^{-nT/2} |\bar{\Sigma}|^{-1/2} \exp \left(-\frac{1}{2} (y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) \right) \quad (\text{a.4.11.56})$$

Consider first the determinant term in (a.4.11.56) and rearrange:

$$\begin{aligned}
& |\bar{\Sigma}|^{-1/2} \\
&= |\Sigma \otimes I_T|^{-1/2} \\
&= |\Sigma|^{-T/2} |I_T|^{-k/2} \\
&= |\Sigma|^{-T/2} \\
&= |\Phi^{-1} \Lambda \Phi^{-1'}|^{-T/2} \\
&= |\Phi^{-1}|^{-T/2} |\Lambda|^{-T/2} |\Phi^{-1'}|^{-T/2} \\
&= |\Phi|^{T/2} |\Lambda|^{-T/2} |\Phi'|^{T/2} \\
&= |\Lambda|^{-T/2} \\
&= \prod_{i=1}^n \lambda_i^{-T/2}
\end{aligned} \tag{a.4.11.57}$$

Consider then the quadratic form in (a.4.11.56):

$$\begin{aligned}
& (y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) \\
&= (y - (I_n \otimes X)\beta)' (\Sigma \otimes I_T)^{-1} (y - (I_n \otimes X)\beta) \\
&= tr [\Sigma^{-1} (Y - X\mathcal{B})' I_T^{-1} (Y - X\mathcal{B})] \\
&= tr [\Sigma^{-1} (Y - X\mathcal{B})' (Y - X\mathcal{B})] \\
&= tr [\Sigma^{-1} \mathcal{E}' \mathcal{E}] \\
&= tr [\mathcal{E} \Sigma^{-1} \mathcal{E}'] \\
&= tr \left[\begin{pmatrix} \varepsilon_1' \Sigma^{-1} \varepsilon_1 & \varepsilon_1' \Sigma^{-1} \varepsilon_2 & \cdots & \varepsilon_1' \Sigma^{-1} \varepsilon_T \\ \varepsilon_2' \Sigma^{-1} \varepsilon_1 & \varepsilon_2' \Sigma^{-1} \varepsilon_2 & \cdots & \varepsilon_2' \Sigma^{-1} \varepsilon_T \\ \vdots & \vdots & \ddots & \vdots \\ \varepsilon_T' \Sigma^{-1} \varepsilon_1 & \varepsilon_T' \Sigma^{-1} \varepsilon_2 & \cdots & \varepsilon_T' \Sigma^{-1} \varepsilon_T \end{pmatrix} \right] \\
&= \sum_{t=1}^T \varepsilon_t' \Sigma^{-1} \varepsilon_t \\
&= \sum_{t=1}^T \varepsilon_t' (\Phi^{-1} \Lambda \Phi^{-1'})^{-1} \varepsilon_t \\
&= \sum_{t=1}^T \varepsilon_t' (\Phi' \Lambda^{-1} \Phi) \varepsilon_t \\
&= \sum_{t=1}^T (\Phi \varepsilon_t)' \Lambda^{-1} (\Phi \varepsilon_t)
\end{aligned} \tag{a.4.11.58}$$

Consider the first term in the quadratic form:

$$\begin{aligned}
& \Phi \varepsilon_t \\
&= \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \phi_{21} & 1 & \ddots & \cdots \\ \vdots & \ddots & \ddots & 0 \\ \phi_{n1} & \cdots & \phi_{n(n-1)} & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \vdots \\ \varepsilon_{n,t} \end{pmatrix} \\
&= \begin{pmatrix} \varepsilon_{1,t} \\ \phi_2 \varepsilon_{-2,t} + \varepsilon_{2,t} \\ \vdots \\ \phi_n \varepsilon_{-n,t} + \varepsilon_{n,t} \end{pmatrix} \tag{a.4.11.59}
\end{aligned}$$

with:

$$\varepsilon_{-i,t} = (\varepsilon_{1,t} \ \varepsilon_{2,t} \ \cdots \ \varepsilon_{i-1,t})' \tag{a.4.11.60}$$

Substitute (a.4.11.59) back in the quadratic form (a.4.11.58):

$$\begin{aligned}
& (\Phi \varepsilon_t)' \Lambda^{-1} (\Phi \varepsilon_t) \\
&= (\varepsilon_{1,t} \ \phi_2 \varepsilon_{-2,t} + \varepsilon_{2,t} \ \cdots \ \phi_n \varepsilon_{-n,t} + \varepsilon_{n,t}) \begin{pmatrix} \lambda_1^{-1} & 0 & \cdots & 0 \\ 0 & \lambda_2^{-1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n^{-1} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \phi_2 \varepsilon_{-2,t} + \varepsilon_{2,t} \\ \vdots \\ \phi_n \varepsilon_{-n,t} + \varepsilon_{n,t} \end{pmatrix} \\
&= (\lambda_1^{-1} \varepsilon_{1,t} \ \lambda_2^{-1} (\phi_2 \varepsilon_{-2,t} + \varepsilon_{2,t}) \ \cdots \ \lambda_n^{-1} (\phi_n \varepsilon_{-n,t} + \varepsilon_{n,t})) \begin{pmatrix} \varepsilon_{1,t} \\ \phi_2 \varepsilon_{-2,t} + \varepsilon_{2,t} \\ \vdots \\ \phi_n \varepsilon_{-n,t} + \varepsilon_{n,t} \end{pmatrix} \\
&= \sum_{i=1}^n \lambda_i^{-1} (\varepsilon_{i,t} + \phi_i \varepsilon_{-i,t})^2 \tag{a.4.11.61}
\end{aligned}$$

Substitute (a.4.11.61) back in (a.4.11.58) to obtain:

$$\begin{aligned}
& \sum_{t=1}^T (\Phi \varepsilon_t)' \Lambda^{-1} (\Phi \varepsilon_t) \\
&= \sum_{t=1}^T \sum_{i=1}^n \lambda_i^{-1} (\varepsilon_{i,t} + \phi_i \varepsilon_{-i,t})^2 \\
&= \sum_{i=1}^n \lambda_i^{-1} \left(\sum_{t=1}^T (\varepsilon_{i,t} + \phi_i \varepsilon_{-i,t})^2 \right) \tag{a.4.11.62}
\end{aligned}$$

Then, note that:

$$\begin{aligned}
& \sum_{t=1}^T (\varepsilon_{i,t} + \phi_i \varepsilon_{-i,t})^2 \\
&= (\varepsilon_{i,1} + \phi_i' \varepsilon_{-i,1} \quad \varepsilon_{i,2} + \phi_i' \varepsilon_{-i,2} \quad \cdots \quad \varepsilon_{i,T} + \phi_i' \varepsilon_{-i,T}) \begin{pmatrix} \varepsilon_{i,1} + \phi_i' \varepsilon_{-i,1} \\ \varepsilon_{i,2} + \phi_i' \varepsilon_{-i,2} \\ \vdots \\ \varepsilon_{i,T} + \phi_i' \varepsilon_{-i,T} \end{pmatrix} \\
&= (\varepsilon_{i,1} + \varepsilon_{-i,1}' \phi_i \quad \varepsilon_{i,2} + \varepsilon_{-i,2}' \phi_i \quad \cdots \quad \varepsilon_{i,T} + \varepsilon_{-i,T}' \phi_i) \begin{pmatrix} \varepsilon_{i,1} + \varepsilon_{-i,1}' \phi_i \\ \varepsilon_{i,2} + \varepsilon_{-i,2}' \phi_i \\ \vdots \\ \varepsilon_{i,T} + \varepsilon_{-i,T}' \phi_i \end{pmatrix} \\
&= (\mathcal{E}_i + \mathcal{E}_{-i}' \phi_i)' (\mathcal{E}_i + \mathcal{E}_{-i} \phi_i) \tag{a.4.11.63}
\end{aligned}$$

with:

$$\mathcal{E}_{-i} = (\mathcal{E}_1 \quad \mathcal{E}_2 \quad \cdots \quad \mathcal{E}_{i-1}) \tag{a.4.11.64}$$

Also, noting that $\mathcal{E}_i = Y_i - X\beta_i$:

$$\sum_{t=1}^T (\varepsilon_{i,t} + \phi_i \varepsilon_{-i,t})^2 = (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i)' (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i) \tag{a.4.11.65}$$

Substitute (a.4.11.65) back in (a.4.11.62):

$$\sum_{t=1}^T (\Phi \varepsilon_t)' \Lambda^{-1} (\Phi \varepsilon_t) = \sum_{i=1}^n \lambda_i^{-1} (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i)' (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i) \tag{a.4.11.66}$$

Therefore, substituting (a.4.11.66) back in (a.4.11.58):

$$(y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) = \sum_{i=1}^n \lambda_i^{-1} (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i)' (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i) \tag{a.4.11.67}$$

Finally, replacing (a.4.11.57) and (a.4.11.67) in (a.4.11.56) yields:

$$f(y|\beta, \lambda, \phi) = (2\pi)^{-nT/2} \left(\prod_{i=1}^n \lambda_i^{-T/2} \right) \exp \left(-\frac{1}{2} \sum_{i=1}^n \lambda_i^{-1} (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i)' (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i) \right) \tag{a.4.11.68}$$

derivations for equation (4.11.75)

Start from:

$$\pi(\beta_i|y, \beta_{-i}) \propto \exp \left(-\frac{1}{2} \lambda_i^{-1} (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i)' (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i) \right) \times \exp \left(-\frac{1}{2} (\beta_i - b_i)' V_i^{-1} (\beta_i - b_i) \right) \tag{a.4.11.69}$$

Develop the first quadratic form:

$$\begin{aligned}
& (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i)' (Y_i - X\beta_i + \mathcal{E}_{-i} \phi_i) \\
&= Y_i' Y_i + \beta_i' X' X \beta_i + \phi_i' \mathcal{E}_{-i}' \mathcal{E}_{-i} \phi_i - 2\beta_i' X' (Y_i + \mathcal{E}_{-i} \phi_i) + 2Y_i' \mathcal{E}_{-i} \phi_i \tag{a.4.11.70}
\end{aligned}$$

Develop the second quadratic form:

$$\begin{aligned}
& (\beta_i - b_i)' V_i^{-1} (\beta_i - b_i) \\
&= \beta_i' V_i^{-1} \beta_i + b_i' V_i^{-1} b_i - 2\beta_i' V_i^{-1} b_i
\end{aligned} \tag{a.4.11.71}$$

Substitute (a.4.11.70) and (a.4.11.71) back in (a.4.11.69) to obtain:

$$\begin{aligned}
& \pi(\beta_i | y, \beta_{-i}) \\
&\propto \exp \left(-\frac{1}{2} \lambda_i^{-1} (Y_i' Y_i + \beta_i' X' X \beta_i + \phi_i' \mathcal{E}_{-i}' \mathcal{E}_{-i} \phi_i - 2\beta_i' X' (Y_i + \mathcal{E}_{-i} \phi_i) + 2Y_i' \mathcal{E}_{-i} \phi_i) \right) \\
&\times \exp \left(-\frac{1}{2} (\beta_i' V_i^{-1} \beta_i + b_i' V_i^{-1} b_i - 2\beta_i' V_i^{-1} b_i) \right) \\
&= \exp \left(-\frac{1}{2} (\lambda_i^{-1} \beta_i' X' X \beta_i - 2\lambda_i^{-1} \beta_i' X' (Y_i + \mathcal{E}_{-i} \phi_i) + \beta_i' V_i^{-1} \beta_i - 2\beta_i' V_i^{-1} b_i) \right) \\
&\times \exp \left(-\frac{1}{2} (\lambda_i^{-1} Y_i' Y_i + \lambda_i^{-1} \phi_i' \mathcal{E}_{-i}' \mathcal{E}_{-i} \phi_i + 2\lambda_i^{-1} Y_i' \mathcal{E}_{-i} \phi_i + b_i' V_i^{-1} b_i) \right) \\
&\propto \exp \left(-\frac{1}{2} (\lambda_i^{-1} \beta_i' X' X \beta_i - 2\lambda_i^{-1} \beta_i' X' (Y_i + \mathcal{E}_{-i} \phi_i) + \beta_i' V_i^{-1} \beta_i - 2\beta_i' V_i^{-1} b_i) \right) \\
&\propto \exp \left(-\frac{1}{2} [\beta_i' (\lambda_i^{-1} X' X + V_i^{-1}) \beta_i - 2\beta_i' (V_i^{-1} b_i + \lambda_i^{-1} X' [Y_i + \mathcal{E}_{-i} \phi_i])] \right)
\end{aligned} \tag{a.4.11.72}$$

Consider the term within the square brackets and complete the squares:

$$\begin{aligned}
& \beta_i' (\lambda_i^{-1} X' X + V_i^{-1}) \beta_i - 2\beta_i' (V_i^{-1} b_i + \lambda_i^{-1} X' [Y_i + \mathcal{E}_{-i} \phi_i]) \\
&= \beta_i' (\lambda_i^{-1} X' X + V_i^{-1}) \beta_i - 2\beta_i' \bar{V}_i^{-1} \bar{V}_i (V_i^{-1} b_i + \lambda_i^{-1} X' [Y_i + \mathcal{E}_{-i} \phi_i]) + \bar{b}_i' \bar{V}_i^{-1} \bar{b}_i - \bar{b}_i' \bar{V}_i^{-1} \bar{b}_i
\end{aligned} \tag{a.4.11.73}$$

Define:

$$\bar{V}_i = (\lambda_i^{-1} X' X + V_i^{-1})^{-1} \quad \bar{b}_i = \bar{V}_i (V_i^{-1} b_i + \lambda_i^{-1} X' [Y_i + \mathcal{E}_{-i} \phi_i]) \tag{a.4.11.74}$$

Then (a.4.11.73) rewrites:

$$\begin{aligned}
& \beta_i' (\lambda_i^{-1} X' X + V_i^{-1}) \beta_i - 2\beta_i' (V_i^{-1} b_i + \lambda_i^{-1} X' [Y_i + \mathcal{E}_{-i} \phi_i]) \\
&= \beta_i' \bar{V}_i^{-1} \beta_i - 2\beta_i' \bar{V}_i^{-1} \bar{b}_i + \bar{b}_i' \bar{V}_i^{-1} \bar{b}_i - \bar{b}_i' \bar{V}_i^{-1} \bar{b}_i \\
&= (\beta_i - \bar{b}_i)' \bar{V}_i^{-1} (\beta_i - \bar{b}_i) - \bar{b}_i' \bar{V}_i^{-1} \bar{b}_i
\end{aligned} \tag{a.4.11.75}$$

Substitute (a.4.11.75) back in (a.4.11.72) to obtain:

$$\begin{aligned}
& \pi(\beta_i | y, \beta_{-i}) \\
&\propto \exp \left(-\frac{1}{2} [(\beta_i - \bar{b}_i)' \bar{V}_i^{-1} (\beta_i - \bar{b}_i) - \bar{b}_i' \bar{V}_i^{-1} \bar{b}_i] \right) \\
&= \exp \left(-\frac{1}{2} (\beta_i - \bar{b}_i)' \bar{V}_i^{-1} (\beta_i - \bar{b}_i) \right) \exp \left(-\frac{1}{2} (-\bar{b}_i' \bar{V}_i^{-1} \bar{b}_i) \right) \\
&\propto \exp \left(-\frac{1}{2} (\beta_i - \bar{b}_i)' \bar{V}_i^{-1} (\beta_i - \bar{b}_i) \right)
\end{aligned} \tag{a.4.11.76}$$

derivations for equation (4.11.81)

Rearrange the initial formula:

$$\begin{aligned}
 & \pi(\phi_i|y, \phi_{-i}) \\
 & \propto \exp\left(-\frac{1}{2}\lambda_i^{-1}(Y_i - X\beta_i + \mathcal{E}_{-i}\phi_i)'(Y_i - X\beta_i + \mathcal{E}_{-i}\phi_i)\right) \times \exp\left(-\frac{1}{2}\tau^{-1}\phi_i'\phi_i\right) \\
 & = \exp\left(-\frac{1}{2}\left[\lambda_i^{-1}(Y_i - X\beta_i + \mathcal{E}_{-i}\phi_i)'(Y_i - X\beta_i + \mathcal{E}_{-i}\phi_i) + \tau^{-1}\phi_i'\phi_i\right]\right) \quad (\text{a.4.11.77})
 \end{aligned}$$

Consider the term within the square brackets:

$$\begin{aligned}
 & \lambda_i^{-1}(Y_i - X\beta_i + \mathcal{E}_{-i}\phi_i)'(Y_i - X\beta_i + \mathcal{E}_{-i}\phi_i) + \tau^{-1}\phi_i'\phi_i \\
 & = \lambda_i^{-1}(Y_i'Y_i + \beta_i'X'X\beta_i + \phi_i'\mathcal{E}_{-i}'\mathcal{E}_{-i}\phi_i + 2\phi_i'\mathcal{E}_{-i}'[Y_i - X\beta_i] - 2Y_i'X\beta_i) + \tau^{-1}\phi_i'\phi_i \\
 & = \lambda_i^{-1}(Y_i'Y_i + \beta_i'X'X\beta_i + \phi_i'\mathcal{E}_{-i}'\mathcal{E}_{-i}\phi_i - 2\phi_i'(-\mathcal{E}_{-i}'[Y_i - X\beta_i]) - 2Y_i'X\beta_i) + \tau^{-1}\phi_i'\phi_i \\
 & = \phi_i'(\tau^{-1}I_{i-1} + \lambda_i^{-1}\mathcal{E}_{-i}'\mathcal{E}_{-i})\phi_i - 2\phi_i'(-\lambda_i^{-1}\mathcal{E}_{-i}'[Y_i - X\beta_i]) + \lambda_i^{-1}(Y_i'Y_i + \beta_i'X'X\beta_i - 2Y_i'X\beta_i) \quad (\text{a.4.11.78})
 \end{aligned}$$

Complete the squares:

$$\begin{aligned}
 & = \phi_i'(\tau^{-1}I_{i-1} + \lambda_i^{-1}\mathcal{E}_{-i}'\mathcal{E}_{-i})\phi_i - 2\phi_i'\bar{Z}_i^{-1}\bar{Z}_i(-\lambda_i^{-1}\mathcal{E}_{-i}'[Y_i - X\beta_i]) + \lambda_i^{-1}(Y_i'Y_i + \beta_i'X'X\beta_i - 2Y_i'X\beta_i) \\
 & + \bar{f}_i'\bar{Z}_i^{-1}\bar{f}_i - \bar{f}_i'\bar{Z}_i^{-1}\bar{f}_i \quad (\text{a.4.11.79})
 \end{aligned}$$

Define:

$$\bar{Z}_i = (\tau^{-1}I_{i-1} + \lambda_i^{-1}\mathcal{E}_{-i}'\mathcal{E}_{-i})^{-1} \quad \bar{f}_i = \bar{Z}_i(-\lambda_i^{-1}\mathcal{E}_{-i}'[Y_i - X\beta_i]) \quad (\text{a.4.11.80})$$

Then (a.4.11.79) rewrites:

$$\begin{aligned}
 & = \phi_i'\bar{Z}_i^{-1}\phi_i - 2\phi_i'\bar{Z}_i^{-1}\bar{f}_i + \bar{f}_i'\bar{Z}_i^{-1}\bar{f}_i + \lambda_i^{-1}(Y_i'Y_i + \beta_i'X'X\beta_i - 2Y_i'X\beta_i) - \bar{f}_i'\bar{Z}_i^{-1}\bar{f}_i \\
 & = (\phi_i - \bar{f}_i)'\bar{Z}_i^{-1}(\phi_i - \bar{f}_i) + \lambda_i^{-1}(Y_i'Y_i + \beta_i'X'X\beta_i - 2Y_i'X\beta_i) - \bar{f}_i'\bar{Z}_i^{-1}\bar{f}_i \quad (\text{a.4.11.81})
 \end{aligned}$$

Substitute (a.4.11.81) back in (a.4.11.77):

$$\begin{aligned}
 & \pi(\phi_i|y, \phi_{-i}) \\
 & \propto \exp\left(-\frac{1}{2}\left[(\phi_i - \bar{f}_i)'\bar{Z}_i^{-1}(\phi_i - \bar{f}_i) + \lambda_i^{-1}(Y_i'Y_i + \beta_i'X'X\beta_i - 2Y_i'X\beta_i) - \bar{f}_i'\bar{Z}_i^{-1}\bar{f}_i\right]\right) \\
 & = \exp\left(-\frac{1}{2}(\phi_i - \bar{f}_i)'\bar{Z}_i^{-1}(\phi_i - \bar{f}_i)\right) \exp\left(-\frac{1}{2}\left[\lambda_i^{-1}(Y_i'Y_i + \beta_i'X'X\beta_i - 2Y_i'X\beta_i) - \bar{f}_i'\bar{Z}_i^{-1}\bar{f}_i\right]\right) \\
 & \propto \exp\left(-\frac{1}{2}(\phi_i - \bar{f}_i)'\bar{Z}_i^{-1}(\phi_i - \bar{f}_i)\right) \quad (\text{a.4.11.82})
 \end{aligned}$$

Eventually, notice that:

$$\bar{f}_i = \bar{Z}_i(-\lambda_i^{-1}\mathcal{E}_{-i}'[Y_i - X\beta_i]) = \bar{Z}_i(-\lambda_i^{-1}\mathcal{E}_{-i}'\mathcal{E}_i) \quad (\text{a.4.11.83})$$

Further aspects of Bayesian vector autoregressions

derivations for equation (4.12.3)

Consider the general VAR model and reformulate it:

$$\begin{aligned}
 y_t &= Cz_t + A_1 y_{t-1} + \cdots + A_p y_{t-p} + \varepsilon_t \\
 \Leftrightarrow y_t - y_{t-1} &= Cz_t + A_1 y_{t-1} + \cdots + A_p y_{t-p} - y_{t-1} + \varepsilon_t \\
 \Leftrightarrow \Delta y_t &= Cz_t + \sum_{i=1}^p A_i y_{t-i} - y_{t-1} + \varepsilon_t \\
 \Leftrightarrow \Delta y_t &= Cz_t + \sum_{i=1}^p \left(A_i y_{t-i} + \sum_{j=i+1}^p A_j y_{t-i} - \sum_{j=i+1}^p A_j y_{t-i} \right) - y_{t-1} + \varepsilon_t \\
 \Leftrightarrow \Delta y_t &= Cz_t + \left(\sum_{i=1}^p A_i - I \right) y_{t-1} - \sum_{i=1}^{p-1} \left(\sum_{j=i+1}^p A_j \right) (y_{t-i} - y_{t-i-1}) + \varepsilon_t \\
 \Leftrightarrow \Delta y_t &= Cz_t + \left(\sum_{i=1}^p A_i - I \right) y_{t-1} + \sum_{i=1}^{p-1} B_i \Delta y_{t-i} + \varepsilon_t
 \end{aligned} \tag{a.4.12.1}$$

with:

$$B_i = - \sum_{j=i+1}^p A_j \tag{a.4.12.2}$$

derivations for equation (4.12.17)

Start from:

$$\begin{aligned}
 Y_{lrp} &= X_{lrp} \mathcal{B} + \mathcal{E}_{lrp} \\
 \Leftrightarrow Y_{lrp} &= (0_{n \times m} \quad \mathbf{1}'_p \otimes Y_{lrp}) \begin{pmatrix} C' \\ A'_1 \\ \vdots \\ A'_p \end{pmatrix} + \mathcal{E}_{lrp} \\
 \Leftrightarrow Y_{lrp} &= Y_{lrp} A'_1 + \cdots + Y_{lrp} A'_p + \mathcal{E}_{lrp} \\
 \Leftrightarrow Y'_{lrp} &= A_1 Y'_{lrp} + \cdots + A_p Y'_{lrp} + \mathcal{E}'_{lrp} \\
 \Leftrightarrow (A_1 + \cdots + A_p - I) Y'_{lrp} &= -\mathcal{E}'_{lrp} \\
 \Leftrightarrow (A_1 + \cdots + A_p - I) [diag(H \bar{y} / \pi_7) H^{-1}]' &= -\mathcal{E}'_{lrp} \\
 \Leftrightarrow (A_1 + \cdots + A_p - I) H^{-1} diag(H \bar{y} / \pi_7) &= -\mathcal{E}'_{lrp} \\
 \Leftrightarrow (A_1 + \cdots + A_p - I) H^{-1} &= -diag(\pi_7 / H \bar{y}) \mathcal{E}'_{lrp}
 \end{aligned} \tag{a.4.12.3}$$

derivations for equation (4.12.20)

Focus on the term within the integral and rearrange:

$$\begin{aligned}
 & (2\pi)^{-nT/2} |\bar{\Sigma}|^{-1/2} \exp \left(-\frac{1}{2} (y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) \right) \times (2\pi)^{-q/2} |V|^{-1/2} \exp \left(-\frac{1}{2} (\beta - b)' V^{-1} (\beta - b) \right) \\
 &= (2\pi)^{-nT/2} |\bar{\Sigma}|^{-1/2} (2\pi)^{-q/2} |V|^{-1/2} \exp \left(-\frac{1}{2} [(y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) + (\beta - b)' V^{-1} (\beta - b)] \right)
 \end{aligned} \tag{a.4.12.4}$$

Also:

$$|\bar{\Sigma}|^{-1/2} = |\Sigma \otimes I_T|^{-1/2} = |\Sigma|^{-T/2} |I_T|^{-n/2} = |\Sigma|^{-T/2} \tag{a.4.12.5}$$

And:

$$\begin{aligned}
 & (y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) + (\beta - b)' V^{-1} (\beta - b) \\
 &= y' \bar{\Sigma}^{-1} y + \beta' \bar{X}' \bar{\Sigma}^{-1} \bar{X} \beta - 2\beta' \bar{X}' \bar{\Sigma}^{-1} y + \beta' V^{-1} \beta + b' V^{-1} b - 2\beta' V^{-1} b \\
 &= \beta' (V^{-1} + \bar{X}' \bar{\Sigma}^{-1} \bar{X}) \beta - 2\beta' (V^{-1} b + \bar{X}' \bar{\Sigma}^{-1} y) + b' V^{-1} b + y' \bar{\Sigma}^{-1} y
 \end{aligned} \tag{a.4.12.6}$$

Complete the squares:

$$= \beta' (V^{-1} + \bar{X}' \bar{\Sigma}^{-1} \bar{X}) \beta - 2\beta' \bar{V}^{-1} \bar{V} (V^{-1} b + \bar{X}' \bar{\Sigma}^{-1} y) + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b} + b' V^{-1} b + y' \bar{\Sigma}^{-1} y \tag{a.4.12.7}$$

Define:

$$\bar{V} = (V^{-1} + \bar{X}' \bar{\Sigma}^{-1} \bar{X})^{-1} \quad \bar{b} = \bar{V} (V^{-1} b + \bar{X}' \bar{\Sigma}^{-1} y) \tag{a.4.12.8}$$

Then (a.4.12.7) rewrites:

$$\begin{aligned}
 &= \beta' \bar{V}^{-1} \beta - 2\beta' \bar{V}^{-1} \bar{b} + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b} + b' V^{-1} b + y' \bar{\Sigma}^{-1} y \\
 &= (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + (b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + y' \bar{\Sigma}^{-1} y)
 \end{aligned} \tag{a.4.12.9}$$

The terms in (a.4.12.8) simplify. Note first that:

$$\bar{X}' \bar{\Sigma}^{-1} \bar{X} = (I_n \otimes X)' (\Sigma \otimes I_T)^{-1} (I_n \otimes X) = (I_n \otimes X') (\Sigma^{-1} \otimes I_T) (I_n \otimes X) = (\Sigma^{-1} \otimes X') (I_n \otimes X) = \Sigma^{-1} \otimes X' X \tag{a.4.12.10}$$

Similarly:

$$\bar{X}' \bar{\Sigma}^{-1} y = (I_n \otimes X)' (\Sigma \otimes I_T)^{-1} \text{vec}(Y) = (I_n \otimes X') (\Sigma^{-1} \otimes I_T) \text{vec}(Y) = (\Sigma^{-1} \otimes X') \text{vec}(Y) = \text{vec}(X' Y \Sigma^{-1}) \tag{a.4.12.11}$$

Substituting (a.4.12.5) and (a.4.12.9) back in (a.4.12.4), the term within the integral rewrites:

$$\begin{aligned}
 & (2\pi)^{-nT/2} |\Sigma|^{-T/2} (2\pi)^{-q/2} |V|^{-1/2} \exp \left(-\frac{1}{2} [(\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + (b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + y' \bar{\Sigma}^{-1} y)] \right) \\
 &= (2\pi)^{-nT/2} |\Sigma|^{-T/2} (2\pi)^{-q/2} |V|^{-1/2} |\bar{V}|^{1/2} |\bar{V}|^{-1/2} \\
 &\times \exp \left(-\frac{1}{2} [(\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + (b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + y' \bar{\Sigma}^{-1} y)] \right) \\
 &= (2\pi)^{-nT/2} |\Sigma|^{-T/2} |V|^{-1/2} |\bar{V}|^{1/2} \exp \left(-\frac{1}{2} (b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + y' \bar{\Sigma}^{-1} y) \right) \\
 &\times (2\pi)^{-q/2} \bar{V}^{-1} \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right)
 \end{aligned} \tag{a.4.12.12}$$

The expression simplifies further. Note that:

$$\begin{aligned}
& |V|^{-1/2} |\bar{V}|^{1/2} \\
&= |V|^{-1/2} |(V^{-1} + \Sigma^{-1} \otimes X'X)^{-1}|^{1/2} \\
&= |V|^{-1/2} |(V^{-1} + \Sigma^{-1} \otimes X'X)|^{-1/2} \\
&= |V (V^{-1} + \Sigma^{-1} \otimes X'X)|^{-1/2} \\
&= |I + V(\Sigma^{-1} \otimes X'X)|^{-1/2}
\end{aligned} \tag{a.4.12.13}$$

Also:

$$\begin{aligned}
& y' \bar{\Sigma}^{-1} y \\
&= y' (\Sigma \otimes I_T)^{-1} y \\
&= y' (\Sigma^{-1} \otimes I_T) y \\
&= y' \text{vec}(Y \Sigma^{-1}) \\
&= \text{vec}(Y)' \text{vec}(Y \Sigma^{-1}) \\
&= \text{tr}(Y' Y \Sigma^{-1})
\end{aligned} \tag{a.4.12.14}$$

Thus, substituting again (a.4.12.13) and (a.4.12.14) back in (a.4.12.12), the term within the integral eventually rewrites as:

$$\begin{aligned}
& (2\pi)^{-nT/2} |\Sigma|^{-T/2} |I + V(\Sigma^{-1} \otimes X'X)|^{-1/2} \exp \left(-\frac{1}{2} [b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + \text{tr}(Y' Y \Sigma^{-1})] \right) \\
& \times (2\pi)^{-q/2} \bar{V}^{-1} \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right)
\end{aligned} \tag{a.4.12.15}$$

derivations for equation (4.12.23)

Focus on the term within the integral and rearrange:

$$\begin{aligned}
& (2\pi)^{-nT/2} |\Sigma|^{-T/2} \exp \left(-\frac{1}{2} (\beta - \hat{\beta})' (\Sigma \otimes (X'X)^{-1})^{-1} (\beta - \hat{\beta}) \right) \\
& \times \exp \left(-\frac{1}{2} \text{tr} [\Sigma^{-1} (Y - X\hat{B})' (Y - X\hat{B})] \right) \\
& \times (2\pi)^{-q/2} |\Sigma \otimes W|^{-1/2} \exp \left(-\frac{1}{2} (\beta - b)' (\Sigma \otimes W)^{-1} (\beta - b) \right) \\
& \times \frac{2^{-\alpha n/2}}{\Gamma_n(\frac{\alpha}{2})} |S|^{\alpha/2} |\Sigma|^{-(\alpha+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} S \} \right) \\
&= (2\pi)^{-nT/2} \frac{2^{-\alpha n/2}}{\Gamma_n(\frac{\alpha}{2})} |S|^{\alpha/2} \\
& \times (2\pi)^{-nk/2} |\Sigma \otimes W|^{-1/2} \exp \left(-\frac{1}{2} [(\beta - b)' (\Sigma \otimes W)^{-1} (\beta - b) + (\beta - \hat{\beta})' (\Sigma \otimes (X'X)^{-1})^{-1} (\beta - \hat{\beta})] \right) \\
& \times |\Sigma|^{-(\alpha+T+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} [S + (Y - X\hat{B})' (Y - X\hat{B})] \} \right)
\end{aligned} \tag{a.4.12.16}$$

Note first that:

$$|\Sigma \otimes W|^{-1/2} = |\Sigma|^{-k/2} |W|^{-n/2} \tag{a.4.12.17}$$

Then focus on the first term between square brackets to obtain:

$$\begin{aligned}
& (\beta - b)'(\Sigma \otimes W)^{-1}(\beta - b) + (\beta - \hat{\beta})'(\Sigma \otimes (X'X)^{-1})^{-1}(\beta - \hat{\beta}) \\
&= tr\{\Sigma^{-1}(\mathcal{B} - B)'W^{-1}(\mathcal{B} - B)\} + tr\{\Sigma^{-1}(\mathcal{B} - \hat{B})'(X'X)(\mathcal{B} - \hat{B})\} \\
&= tr\{\Sigma^{-1}[(\mathcal{B} - B)'W^{-1}(\mathcal{B} - B) + (\mathcal{B} - \hat{B})'(X'X)(\mathcal{B} - \hat{B})]\} \\
&= tr\{\Sigma^{-1}[(\mathcal{B} - B)'W^{-1}(\mathcal{B} - B) + (\mathcal{B} - \hat{B})'(X'X)(\mathcal{B} - \hat{B})]\} \\
&= tr\{\Sigma^{-1}[\mathcal{B}'W^{-1}\mathcal{B} + B'W^{-1}B - 2\mathcal{B}'W^{-1}B + \mathcal{B}'(X'X)\mathcal{B} + \hat{B}'(X'X)\hat{B} - 2\mathcal{B}'(X'X)\hat{B}]\} \\
&= tr\{\Sigma^{-1}[\mathcal{B}'(W^{-1} + X'X)\mathcal{B} - 2\mathcal{B}'(W^{-1}B + X'X\hat{B}) + B'W^{-1}B + \hat{B}'(X'X)\hat{B}]\} \tag{a.4.12.18}
\end{aligned}$$

Complete the squares:

$$= tr\{\Sigma^{-1}[\mathcal{B}'(W^{-1} + X'X)\mathcal{B} - 2\mathcal{B}'\bar{W}^{-1}\bar{W}(W^{-1}B + X'X\hat{B}) + \bar{B}'\bar{W}^{-1}\bar{B} - \bar{B}'\bar{W}^{-1}\bar{B} + B'W^{-1}B + \hat{B}'(X'X)\hat{B}]\} \tag{a.4.12.19}$$

Define:

$$\bar{W} = (W^{-1} + X'X)^{-1} \quad \bar{B} = \bar{W}(W^{-1}B + X'X\hat{B}) \tag{a.4.12.20}$$

then the expression becomes:

$$\begin{aligned}
&= tr\{\Sigma^{-1}[\mathcal{B}'\bar{W}^{-1}\mathcal{B} - 2\mathcal{B}'\bar{W}^{-1}\bar{B} + \bar{B}'\bar{W}^{-1}\bar{B} - \bar{B}'\bar{W}^{-1}\bar{B} + B'W^{-1}B + \hat{B}'(X'X)\hat{B}]\} \\
&= tr\{\Sigma^{-1}[(\mathcal{B}' - \bar{B})'\bar{W}^{-1}(\mathcal{B}' - \bar{B}) - \bar{B}'\bar{W}^{-1}\bar{B} + B'W^{-1}B + \hat{B}'(X'X)\hat{B}]\} \\
&= tr\{\Sigma^{-1}(\mathcal{B}' - \bar{B})'\bar{W}^{-1}(\mathcal{B}' - \bar{B})\} + tr\{\Sigma^{-1}[B'W^{-1}B + \hat{B}'(X'X)\hat{B} - \bar{B}'\bar{W}^{-1}\bar{B}]\} \tag{a.4.12.21}
\end{aligned}$$

Substituting (a.4.12.17) and (a.4.12.21) in (a.4.12.16) yields:

$$\begin{aligned}
& (2\pi)^{-nT/2} |W|^{-n/2} \frac{2^{-\alpha n/2}}{\Gamma_n(\frac{\alpha}{2})} |S|^{\alpha/2} \\
& \times (2\pi)^{-nk/2} |\Sigma|^{-k/2} \exp\left(-\frac{1}{2} tr\{\Sigma^{-1}(\mathcal{B}' - \bar{B})'\bar{W}^{-1}(\mathcal{B}' - \bar{B})\}\right) \\
& \times |\Sigma|^{-(\alpha + T + n + 1)/2} \exp\left(-\frac{1}{2} tr\{\Sigma^{-1}[S + (Y - X\hat{B})'(Y - X\hat{B}) + B'W^{-1}B + \hat{B}'(X'X)\hat{B} - \bar{B}'\bar{W}^{-1}\bar{B}]\}\right) \tag{a.4.12.22}
\end{aligned}$$

Define:

$$\bar{\alpha} = \alpha + T \quad \bar{S} = S + (Y - X\hat{B})'(Y - X\hat{B}) + B'W^{-1}B + \hat{B}'(X'X)\hat{B} - \bar{B}'\bar{W}^{-1}\bar{B} \tag{a.4.12.23}$$

Then the expression becomes:

$$\begin{aligned}
& (2\pi)^{-nT/2} |W|^{-n/2} \frac{2^{-\alpha n/2}}{\Gamma_n(\frac{\alpha}{2})} |S|^{\alpha/2} \\
& \times (2\pi)^{-nk/2} |\Sigma|^{-k/2} \exp\left(-\frac{1}{2} tr\{\Sigma^{-1}(\mathcal{B}' - \bar{B})'\bar{W}^{-1}(\mathcal{B}' - \bar{B})\}\right) \\
& \times |\Sigma|^{-(\bar{\alpha} + n + 1)/2} \exp\left(-\frac{1}{2} tr\{\Sigma^{-1}\bar{S}\}\right) \tag{a.4.12.24}
\end{aligned}$$

And finally, this rewrites:

$$\begin{aligned}
& (2\pi)^{-nT/2} |W|^{-n/2} |\bar{W}|^{n/2} \frac{2^{\bar{\alpha}n/2}}{2^{\alpha n/2}} \frac{\Gamma_n\left(\frac{\bar{\alpha}}{2}\right)}{\Gamma_n\left(\frac{\alpha}{2}\right)} |S|^{\alpha/2} |\bar{S}|^{-\bar{\alpha}/2} \\
& \times (2\pi)^{-nk/2} |\Sigma|^{-k/2} |\bar{W}|^{-n/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}(\mathcal{B}' - \bar{\mathcal{B}})' \bar{W}^{-1}(\mathcal{B}' - \bar{\mathcal{B}})\}\right) \\
& \times \frac{2^{-\bar{\alpha}n/2}}{\Gamma_n\left(\frac{\bar{\alpha}}{2}\right)} |\bar{S}|^{\bar{\alpha}/2} |\Sigma|^{-(\bar{\alpha}+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1} \bar{S}\}\right)
\end{aligned} \tag{a.4.12.25}$$

Go on simplifying. Note that:

$$(2\pi)^{-nT/2} \frac{2^{\bar{\alpha}n/2}}{2^{\alpha n/2}} = 2^{-nT/2} \pi^{-nT/2} \frac{2^{(\alpha+T)n/2}}{2^{\alpha n/2}} = \pi^{-nT/2} \tag{a.4.12.26}$$

Also:

$$\begin{aligned}
& |W|^{-n/2} |\bar{W}|^{n/2} \\
& = |W|^{-n/2} |(W^{-1} + X'X)^{-1}|^{n/2} \\
& = |W|^{-n/2} |(W^{-1} + X'X)|^{-n/2} \\
& = |W(W^{-1} + X'X)|^{-n/2} \\
& = |I + WX'X|^{-n/2}
\end{aligned} \tag{a.4.12.27}$$

Consider then:

$$\begin{aligned}
& (Y - X\hat{B})'(Y - X\hat{B}) + \hat{B}'X'X\hat{B} \\
& = Y'Y + \hat{B}'X'X\hat{B} - \hat{B}'X'Y - Y'X\hat{B} + \hat{B}'X'X\hat{B} \\
& = Y'Y + 2\hat{B}'X'X\hat{B} - \hat{B}'X'Y - Y'X\hat{B} \\
& = Y'Y + 2Y'X(X'X)^{-1}X'X(X'X)^{-1}X'Y - Y'X(X'X)^{-1}X'Y - Y'X(X'X)^{-1}X'Y \\
& = Y'Y + 2Y'X(X'X)^{-1}X'Y - Y'X(X'X)^{-1}X'Y - Y'X(X'X)^{-1}X'Y \\
& = Y'Y
\end{aligned} \tag{a.4.12.28}$$

Substitute back in (a.4.12.23):

$$\bar{S} = S + Y'Y + B'W^{-1}B - \bar{B}'\bar{W}^{-1}\bar{B} \tag{a.4.12.29}$$

Then:

$$\begin{aligned}
& = |S|^{\alpha/2} |\bar{S}|^{-\bar{\alpha}/2} \\
& = |S|^{\alpha/2} |S + Y'Y + B'W^{-1}B - \bar{B}'\bar{W}^{-1}\bar{B}|^{-\bar{\alpha}/2} \\
& = |S|^{-T/2} |S|^{(\alpha+T)/2} |S + Y'Y + B'W^{-1}B - \bar{B}'\bar{W}^{-1}\bar{B}|^{-\bar{\alpha}/2} \\
& = |S|^{-T/2} |S|^{\bar{\alpha}/2} |S + Y'Y + B'W^{-1}B - \bar{B}'\bar{W}^{-1}\bar{B}|^{-\bar{\alpha}/2} \\
& = |S|^{-T/2} |S^{-1}|^{-\bar{\alpha}/2} |S + Y'Y + B'W^{-1}B - \bar{B}'\bar{W}^{-1}\bar{B}|^{-\bar{\alpha}/2} \\
& = |S|^{-T/2} |S^{-1}(S + Y'Y + B'W^{-1}B - \bar{B}'\bar{W}^{-1}\bar{B})|^{-\bar{\alpha}/2} \\
& = |S|^{-T/2} |I + S^{-1}(Y'Y + B'W^{-1}B - \bar{B}'\bar{W}^{-1}\bar{B})|^{-\bar{\alpha}/2} \\
& = |S|^{-T/2} |I + S^{-1}(\bar{S} - S)|^{-\bar{\alpha}/2}
\end{aligned} \tag{a.4.12.30}$$

Substituting back (a.4.12.26), (a.4.12.27) and (a.4.12.30) in (a.4.12.25) finally yields:

$$\begin{aligned} & \pi^{-nT/2} |I + WX'X|^{-n/2} |S|^{-T/2} |I + S^{-1}(\bar{S} - S)|^{-\bar{\alpha}/2} \frac{\Gamma_n(\frac{\bar{\alpha}}{2})}{\Gamma_n(\frac{\alpha}{2})} \\ & \times (2\pi)^{-nk/2} |\Sigma|^{-k/2} |\bar{W}|^{-n/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}(\mathcal{B}' - \bar{B})' \bar{W}^{-1}(\mathcal{B}' - \bar{B})\}\right) \\ & \times \frac{2^{-\bar{\alpha}n/2}}{\Gamma_n(\frac{\bar{\alpha}}{2})} |\bar{S}|^{\bar{\alpha}/2} |\Sigma|^{-(\bar{\alpha}+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1} \bar{S}\}\right) \end{aligned} \quad (\text{a.4.12.31})$$

Therefore, substituting back in the integrals, we obtain:

$$\begin{aligned} f(y) &= \pi^{-nT/2} |I + WX'X|^{-n/2} |S|^{-T/2} |I + S^{-1}(\bar{S} - S)|^{-\bar{\alpha}/2} \frac{\Gamma_n(\frac{\bar{\alpha}}{2})}{\Gamma_n(\frac{\alpha}{2})} \\ & \times \int \int (2\pi)^{-nk/2} |\Sigma|^{-k/2} |\bar{W}|^{-n/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1}(\mathcal{B}' - \bar{B})' \bar{W}^{-1}(\mathcal{B}' - \bar{B})\}\right) d\beta \\ & \times \frac{2^{-\bar{\alpha}n/2}}{\Gamma_n(\frac{\bar{\alpha}}{2})} |\bar{S}|^{\bar{\alpha}/2} |\Sigma|^{-(\bar{\alpha}+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1} \bar{S}\}\right) d\sigma \end{aligned} \quad (\text{a.4.12.32})$$

derivations for equation (4.12.26)

$$\begin{aligned} & f(y) \\ & \approx \frac{f(y|\beta^*, \Sigma^*) \pi(\beta^*, \Sigma^*)}{\pi(\Sigma^*|y, \beta^*) \times \frac{1}{J} \sum_{j=1}^J \pi(\beta^*|\Sigma^{(j)}, y)} \\ & = (2\pi)^{-nT/2} |\bar{\Sigma}|^{-1/2} \exp\left(-\frac{1}{2} (y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta)\right) \\ & \times \frac{(2\pi)^{-q/2} |V|^{-1/2} \exp\left(-\frac{1}{2} (\beta - b)' V^{-1} (\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J (2\pi)^{-q/2} |\bar{V}|^{-1/2} \exp\left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b})\right)} \times \frac{\frac{2^{-\alpha n/2}}{\Gamma_n(\frac{\alpha}{2})} |S|^{\alpha/2} |\Sigma|^{-(\alpha+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1} S\}\right)}{\frac{2^{-\bar{\alpha}n/2}}{\Gamma_n(\frac{\bar{\alpha}}{2})} |\bar{S}|^{\bar{\alpha}/2} |\Sigma|^{-(\bar{\alpha}+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1} \bar{S}\}\right)} \end{aligned} \quad (\text{a.4.12.33})$$

Now, note that:

$$|\bar{\Sigma}|^{-1/2} = |\Sigma \otimes I_T|^{-1/2} = |\Sigma|^{-T/2} |I_T|^{-n/2} = |\Sigma|^{-T/2} \quad (\text{a.4.12.34})$$

Also:

$$\begin{aligned} & (y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) \\ & = (y - (I_n \otimes X)\beta)' (\Sigma \otimes I_T)^{-1} (y - (I_n \otimes X)\beta) \\ & = \text{tr}\{\Sigma^{-1} (Y - X\mathcal{B})' (Y - X\mathcal{B})\} \end{aligned} \quad (\text{a.4.12.35})$$

Substituting back (a.4.12.34) and (a.4.12.35) in (a.4.12.33):

$$\begin{aligned} & f(y) \\ & \approx (2\pi)^{-nT/2} |\Sigma|^{-T/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1} (Y - X\mathcal{B})' (Y - X\mathcal{B})\}\right) \\ & \times \frac{(2\pi)^{-q/2} |V|^{-1/2} \exp\left(-\frac{1}{2} (\beta - b)' V^{-1} (\beta - b)\right)}{\frac{1}{J} \sum_{j=1}^J (2\pi)^{-q/2} |\bar{V}|^{-1/2} \exp\left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b})\right)} \times \frac{\frac{2^{-\alpha n/2}}{\Gamma_n(\frac{\alpha}{2})} |S|^{\alpha/2} |\Sigma|^{-(\alpha+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1} S\}\right)}{\frac{2^{-\bar{\alpha}n/2}}{\Gamma_n(\frac{\bar{\alpha}}{2})} |\bar{S}|^{\bar{\alpha}/2} |\Sigma|^{-(\bar{\alpha}+n+1)/2} \exp\left(-\frac{1}{2} \text{tr}\{\Sigma^{-1} \bar{S}\}\right)} \end{aligned} \quad (\text{a.4.12.36})$$

Continue with the definitions of $\bar{\alpha}$ and $\bar{S}$:

$$\begin{aligned}
&= 2^{-nT/2} \pi^{-nT/2} |\Sigma|^{-T/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} (Y - X \mathcal{B})' (Y - X \mathcal{B}) \} \right) \\
&\times \frac{(2\pi)^{-q/2} |V|^{-1/2} \exp \left(-\frac{1}{2} (\beta - b)' V^{-1} (\beta - b) \right)}{\frac{1}{J} \sum_{j=1}^J (2\pi)^{-q/2} |\bar{V}|^{-1/2} \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right)} \\
&\times \frac{\frac{2^{-\alpha n/2}}{\Gamma_n(\frac{\alpha}{2})} |S|^{\alpha/2} |\Sigma|^{-(\alpha+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} S \} \right)}{\frac{2^{-(\alpha+T)n/2}}{\Gamma_n(\frac{\alpha}{2})} |\bar{S}|^{\bar{\alpha}/2} |\Sigma|^{-(\alpha+T+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} [S + (Y - X \mathcal{B})' (Y - X \mathcal{B})] \} \right)} \\
&= 2^{-nT/2} \pi^{-nT/2} |\Sigma|^{-T/2} \\
&\times \frac{(2\pi)^{-q/2} |V|^{-1/2} \exp \left(-\frac{1}{2} (\beta - b)' V^{-1} (\beta - b) \right)}{\frac{1}{J} \sum_{j=1}^J (2\pi)^{-q/2} |\bar{V}|^{-1/2} \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right)} \times \frac{\frac{2^{-\alpha n/2}}{\Gamma_n(\frac{\alpha}{2})} |S|^{\alpha/2} |\Sigma|^{-(\alpha+n+1)/2}}{\frac{2^{-(\alpha+T)n/2}}{\Gamma_n(\frac{\alpha}{2})} |\bar{S}|^{\bar{\alpha}/2} |\Sigma|^{-(\alpha+T+n+1)/2}} \\
&= 2^{-nT/2} \pi^{-nT/2} \times \frac{(2\pi)^{-q/2} |V|^{-1/2} \exp \left(-\frac{1}{2} (\beta - b)' V^{-1} (\beta - b) \right)}{\frac{1}{J} \sum_{j=1}^J (2\pi)^{-q/2} |\bar{V}|^{-1/2} \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right)} \times \frac{\frac{2^{-\alpha n/2}}{\Gamma_n(\frac{\alpha}{2})} |S|^{\alpha/2}}{\frac{2^{-(\alpha+T)n/2}}{\Gamma_n(\frac{\alpha}{2})} |\bar{S}|^{\bar{\alpha}/2}} \\
&= \pi^{-nT/2} \times \frac{(2\pi)^{-q/2} |V|^{-1/2} \exp \left(-\frac{1}{2} (\beta - b)' V^{-1} (\beta - b) \right)}{\frac{1}{J} \sum_{j=1}^J (2\pi)^{-q/2} |\bar{V}|^{-1/2} \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right)} \times \frac{\frac{1}{\Gamma_n(\frac{\alpha}{2})} |S|^{\alpha/2}}{\frac{1}{\Gamma_n(\frac{\alpha}{2})} |\bar{S}|^{\bar{\alpha}/2}} \\
&= \pi^{-nT/2} \frac{\exp \left(-\frac{1}{2} (\beta - b)' V^{-1} (\beta - b) \right)}{\frac{1}{J} \sum_{j=1}^J |V|^{1/2} |\bar{V}|^{-1/2} \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right)} \frac{\Gamma_n(\frac{\alpha}{2})}{\Gamma_n(\frac{\alpha}{2})} \frac{|S|^{\alpha/2}}{|\bar{S}|^{\bar{\alpha}/2}} \tag{a.4.12.37}
\end{aligned}$$

Note that:

$$\begin{aligned}
&= |S|^{\alpha/2} |\bar{S}|^{-\bar{\alpha}/2} \\
&= |S|^{\alpha/2} |S + (Y - X \mathcal{B})' (Y - X \mathcal{B})|^{-\bar{\alpha}/2} \\
&= |S|^{-T/2} |S|^{(\alpha+T)/2} |S + (Y - X \mathcal{B})' (Y - X \mathcal{B})|^{-\bar{\alpha}/2} \\
&= |S|^{-T/2} |S|^{\bar{\alpha}/2} |S + (Y - X \mathcal{B})' (Y - X \mathcal{B})|^{-\bar{\alpha}/2} \\
&= |S|^{-T/2} |S^{-1}|^{-\bar{\alpha}/2} |S + (Y - X \mathcal{B})' (Y - X \mathcal{B})|^{-\bar{\alpha}/2} \\
&= |S|^{-T/2} |S^{-1} (S + (Y - X \mathcal{B})' (Y - X \mathcal{B}))|^{-\bar{\alpha}/2} \\
&= |S|^{-T/2} |I + S^{-1} (Y - X \mathcal{B})' (Y - X \mathcal{B})|^{-\bar{\alpha}/2} \\
&= |S|^{-T/2} |I + S^{-1} (\bar{S} - S)|^{-\bar{\alpha}/2} \tag{a.4.12.38}
\end{aligned}$$

Also:

$$\begin{aligned}
&= |V|^{1/2} |\bar{V}|^{-1/2} \\
&= |V|^{1/2} |(V^{-1} + \Sigma^{-1} \otimes X'X)^{-1}|^{-1/2} \\
&= |V|^{1/2} |V^{-1} + \Sigma^{-1} \otimes X'X|^{1/2} \\
&= |I + V(\Sigma^{-1} \otimes X'X)|^{1/2} \tag{a.4.12.39}
\end{aligned}$$

Substituting (a.4.12.38) and (a.4.12.39) back in (a.4.12.37):

$$\begin{aligned}
&= \pi^{-nT/2} \frac{\Gamma_n(\frac{\alpha}{2})}{\Gamma_n(\frac{\alpha}{2})} |S|^{-T/2} |I + S^{-1} (\bar{S} - S)|^{-\bar{\alpha}/2} \\
&\times \frac{\exp \left(-\frac{1}{2} (\beta - b)' V^{-1} (\beta - b) \right)}{\frac{1}{J} \sum_{j=1}^J |I + V(\Sigma^{-1} \otimes X'X)|^{1/2} \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right)} \tag{a.4.12.40}
\end{aligned}$$

derivations for equation (4.12.29)

Use simple back recursion to obtain:

$$\begin{aligned}
\gamma_t &= \mu_t + F\gamma_{t-1} + \xi_t \\
\Leftrightarrow \gamma_t &= \mu_t + F(\mu_{t-1} + F\gamma_{t-2} + \xi_{t-1}) + \xi_t \\
\Leftrightarrow \gamma_t &= \mu_t + F\mu_{t-1} + F^2\gamma_{t-2} + \xi_t + F\xi_{t-1} \\
\Leftrightarrow \gamma_t &= \mu_t + F\mu_{t-1} + F^2(\mu_{t-2} + F\gamma_{t-3} + \xi_{t-2}) + \xi_t + F\xi_{t-1} \\
\Leftrightarrow \gamma_t &= \mu_t + F\mu_{t-1} + F^2\mu_{t-2} + F^3\gamma_{t-3} + \xi_t + F\xi_{t-1} + F^2\xi_{t-2}
\end{aligned} \tag{a.4.12.41}$$

Going on this way:

$$\gamma_t = \sum_{i=0}^j F^i \mu_{t-i} + F^j \gamma_{t-j} + \sum_{i=0}^j F^i \xi_{t-i} \tag{a.4.12.42}$$

derivations for equation (4.12.30)

Consider the general formulation of the VAR model:

$$y_t = Cz_t + A_1 y_{t-1} + \dots + A_p y_{t-p} + \varepsilon_t \tag{a.4.12.43}$$

Taking expectations on both sides, noting that $\mathbb{E}(y_t) = \mathbb{E}(y_{t-1}) = \dots = \mathbb{E}(y_{t-p}) = \mu$ by stationarity, and that $\mathbb{E}(z_t) = z_t$ and $\mathbb{E}(\varepsilon_t) = 0$, one obtains:

$$\mu = Cz_t + A_1 \mu + \dots + A_p \mu \tag{a.4.12.44}$$

Rearranging:

$$(I - A_1 - \dots - A_p)\mu = Cz_t \tag{a.4.12.45}$$

Which eventually yields:

$$\mu = (I - A_1 - \dots - A_p)^{-1} C z_t \tag{a.4.12.46}$$

derivations for equation (4.12.34)

Rearrange:

$$\begin{aligned}
&\exp\left(-\frac{1}{2}(\beta - \hat{\beta})'(\Sigma \otimes (X'X)^{-1})^{-1}(\beta - \hat{\beta})\right) \times \exp\left(-\frac{1}{2}(\beta - b)'(\Sigma \otimes W)^{-1}(\beta - b)\right) \\
&= \exp\left(-\frac{1}{2}\text{tr}\{\Sigma^{-1}(\mathcal{B} - \hat{\mathcal{B}})'(X'X)(\mathcal{B} - \hat{\mathcal{B}})\}\right) \times \exp\left(-\frac{1}{2}\text{tr}\{\Sigma^{-1}(\mathcal{B} - B)'W^{-1}(\mathcal{B} - B)\}\right) \\
&= \exp\left(-\frac{1}{2}\text{tr}\{\Sigma^{-1}[(\mathcal{B} - \hat{\mathcal{B}})'(X'X)(\mathcal{B} - \hat{\mathcal{B}}) + (\mathcal{B} - B)'W^{-1}(\mathcal{B} - B)]\}\right)
\end{aligned} \tag{a.4.12.47}$$

Consider the terms in square brackets and complete the squares:

$$\begin{aligned}
&(\mathcal{B} - \hat{\mathcal{B}})'(X'X)(\mathcal{B} - \hat{\mathcal{B}}) + (\mathcal{B} - B)'W^{-1}(\mathcal{B} - B) \\
&= B'(X'X)\mathcal{B} + \hat{B}'(X'X)\hat{B} - 2B'(X'X)\hat{B} + B'W^{-1}\mathcal{B} + B'W^{-1}B - 2\mathcal{B}W^{-1}B \\
&= B'(W^{-1} + X'X)\mathcal{B} - 2B'(W^{-1}B + X'X\hat{B}) + \hat{B}'(X'X)\hat{B} + B'W^{-1}B \\
&= B'(W^{-1} + X'X)\mathcal{B} - 2B'\bar{W}^{-1}\bar{W}(W^{-1}B + X'X\hat{B}) + \bar{B}'\bar{W}^{-1}\bar{B} - \bar{B}'\bar{W}^{-1}\bar{B} + \hat{B}'(X'X)\hat{B} + B'W^{-1}B
\end{aligned} \tag{a.4.12.48}$$

Define:

$$\bar{W} = (W^{-1} + X'X)^{-1} \quad \bar{B} = \bar{W}(W^{-1}B + X'X\hat{B}) \tag{a.4.12.49}$$

Then:

$$\begin{aligned}
&= \mathcal{B}' \bar{W}^{-1} \mathcal{B} - 2 \mathcal{B}' \bar{W}^{-1} \bar{B} + \bar{B}' \bar{W}^{-1} \bar{B} - \bar{B}' \bar{W}^{-1} \bar{B} + \hat{B}' (X'X) \hat{B} + B' W^{-1} B \\
&= (\mathcal{B} - \bar{B})' \bar{W}^{-1} (\mathcal{B} - \bar{B}) - \bar{B}' \bar{W}^{-1} \bar{B} + \hat{B}' (X'X) \hat{B} + B' W^{-1} B
\end{aligned} \tag{a.4.12.50}$$

Substitute back in (a.4.12.47):

$$\begin{aligned}
&= \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} [(\mathcal{B} - \bar{B})' \bar{W}^{-1} (\mathcal{B} - \bar{B}) - \bar{B}' \bar{W}^{-1} \bar{B} + \hat{B}' (X'X) \hat{B} + B' W^{-1} B] \} \right) \\
&= \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} (\mathcal{B} - \bar{B})' \bar{W}^{-1} (\mathcal{B} - \bar{B}) \} \right) \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} [-\bar{B}' \bar{W}^{-1} \bar{B} + \hat{B}' (X'X) \hat{B} + B' W^{-1} B] \} \right) \\
&\propto \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} (\mathcal{B} - \bar{B})' \bar{W}^{-1} (\mathcal{B} - \bar{B}) \} \right)
\end{aligned} \tag{a.4.12.51}$$

Finally, note again that:

$$\bar{B} = \bar{W} (W^{-1} B + X'X \hat{B}) = \bar{W} (W^{-1} B + X'X (X'X)^{-1} X'Y) = \bar{W} (W^{-1} B + X'Y) \tag{a.4.12.52}$$

derivations for equation (4.13.17)

The log-likelihood is given by:

$$\log(f(y|\beta, \Sigma)) = -\frac{nT}{2} \log(2\pi) - \frac{1}{2} \log(|\bar{\Sigma}|) - \frac{1}{2} (y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) \tag{a.4.12.53}$$

Then note that:

$$\log(|\bar{\Sigma}|) = \log(|\Sigma \otimes I_T|) = \log(|\Sigma|^T |I_T|^n) = \log(|\Sigma|^T) = T \log(|\Sigma|) \tag{a.4.12.54}$$

Also:

$$(y - \bar{X}\beta)' \bar{\Sigma}^{-1} (y - \bar{X}\beta) = (y - (I_n \otimes X)\beta)' (\Sigma \otimes I_T)^{-1} (y - (I_n \otimes X)\beta) = \text{tr} \{ \Sigma^{-1} (Y - X\mathcal{B})' (Y - X\mathcal{B}) \} \tag{a.4.12.55}$$

Substituting back (a.4.12.54) and (a.4.12.55) in (a.4.12.53):

$$\log(f(y|\beta, \Sigma)) = -\frac{nT}{2} \log(2\pi) - \frac{T}{2} \log(|\Sigma|) - \frac{1}{2} \text{tr} \{ \Sigma^{-1} (Y - X\mathcal{B})' (Y - X\mathcal{B}) \} \tag{a.4.12.56}$$

The function is estimated at the maximum likelihood values. Hence $\mathcal{B} = \hat{\mathcal{B}}$ and $\Sigma = \hat{\Sigma} = \mathcal{E}' \mathcal{E} / T$. Substituting in (a.4.12.56):

$$\begin{aligned}
&\log(f(y|\beta, \Sigma)) \\
&= -\frac{nT}{2} \log(2\pi) - \frac{T}{2} \log(|\hat{\Sigma}|) - \frac{1}{2} \text{tr} \{ \hat{\Sigma}^{-1} (Y - X\hat{\mathcal{B}})' (Y - X\hat{\mathcal{B}}) \} \\
&= -\frac{nT}{2} \log(2\pi) - \frac{T}{2} \log(|\hat{\Sigma}|) - \frac{1}{2} \text{tr} \{ \hat{\Sigma}^{-1} \hat{\Sigma} T \} \\
&= -\frac{nT}{2} \log(2\pi) - \frac{T}{2} \log(|\hat{\Sigma}|) - \frac{T}{2} \\
&= -\frac{nT}{2} \log(2\pi) - \frac{T}{2} (1 + \log(|\hat{\Sigma}|)) \\
&= -\frac{T}{2} (n \log(2\pi) + (1 + \log(|\hat{\Sigma}|)))
\end{aligned} \tag{a.4.12.57}$$

Following, the AIC obtains as:

$$\begin{aligned}
& AIC \\
&= 2q/T - 2\hat{L}/T \\
&= 2q/T - 2/T \left[-\frac{T}{2} (n \log(2\pi) + (1 + \log(|\hat{\Sigma}|))) \right] \\
&= 2q/T + n \log(2\pi) + 1 + \log(|\hat{\Sigma}|) \tag{a.4.12.58}
\end{aligned}$$

It follows immediately that the BIC is given by:

$$\begin{aligned}
& BIC \\
&= q \log(T)/T - 2\hat{L}/T \\
&= q \log(T)/T - 2/T \left[-\frac{T}{2} (n \log(2\pi) + (1 + \log(|\hat{\Sigma}|))) \right] \\
&= q \log(T)/T + n \log(2\pi) + 1 + \log(|\hat{\Sigma}|) \tag{a.4.12.59}
\end{aligned}$$

And the Hannan-Quinn criterion is given by:

$$\begin{aligned}
& HQ \\
&= 2q \log(\log(T))/T - 2\hat{L}/T \\
&= 2q \log(\log(T))/T - 2/T \left[-\frac{T}{2} (n \log(2\pi) + (1 + \log(|\hat{\Sigma}|))) \right] \\
&= 2q \log(\log(T))/T + n \log(2\pi) + 1 + \log(|\hat{\Sigma}|) \tag{a.4.12.60}
\end{aligned}$$

Removing the constants makes the value invariant to the number of endogenous variables:

$$AIC = 2q/T + \log(|\hat{\Sigma}|) \quad BIC = q \log(T)/T + \log(|\hat{\Sigma}|) \quad HQ = 2q \log(\log(T))/T + \log(|\hat{\Sigma}|) \tag{a.4.12.61}$$

Bayesian VAR: advanced applications

derivations for equation (4.14.18)

The distribution for the conditional forecasts is given by:

$$R \hat{y}_{T+1:T+h} \sim N(\bar{y}, \Omega) \quad (\text{a.4.14.1})$$

The distribution for the unconditional forecasts is given by:

$$\hat{y}_{T+1:T+h} \sim N(f_{T+1:T+h}, M(I_h \otimes \Gamma)M') \quad (\text{a.4.14.2})$$

Using the selection matrix R on the unconditional forecasts, we find that the distribution of the forecasts for the variables on which conditions apply is given by:

$$R \hat{y}_{T+1:T+h} \sim N(R f_{T+1:T+h}, D(I_h \otimes \Gamma)D') \quad (\text{a.4.14.3})$$

with D is a $k \times nh$ matrix such that $D = RM$. Now choose any $(nh - k) \times nh$ matrix $\hat{D}$ such that the rows of $\hat{D}$ form an orthonormal basis for the nullspace of D . This implies that:

$$\hat{D}\hat{D}' = I_{nh-k} \quad D\hat{D}' = 0_{k \times (nh-k)} \quad \hat{D}D' = 0_{(nh-k) \times k} \quad (\text{a.4.14.4})$$

Now define the random variable z as:

$$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} D \\ \hat{D} \end{pmatrix} M^{-1} \hat{y}_{T+1:T+h} = \begin{pmatrix} R \hat{y}_{T+1:T+h} \\ \hat{D} M^{-1} \hat{y}_{T+1:T+h} \end{pmatrix} \quad (\text{a.4.14.5})$$

Thus, the distribution of z is given by:

$$z \sim N \left[\begin{pmatrix} R f_{T+1:T+h} \\ \hat{D} M^{-1} f_{T+1:T+h} \end{pmatrix}, \begin{pmatrix} D(I_h \otimes \Gamma)D' & 0_{k \times (nh-k)} \\ 0_{(nh-k) \times k} & \hat{D}(I_h \otimes \Gamma)\hat{D}' \end{pmatrix} \right] \quad (\text{a.4.14.6})$$

The distribution of the conditions is given by (a.4.14.1). Because z_1 and z_2 are independent, we can simply substitute for (a.4.14.1) in (a.4.14.6) to obtain:

$$z \sim N \left[\begin{pmatrix} \bar{y} \\ \hat{D} M^{-1} f_{T+1:T+h} \end{pmatrix}, \begin{pmatrix} \Omega & 0_{k \times (nh-k)} \\ 0_{(nh-k) \times k} & \hat{D}(I_h \otimes \Gamma)\hat{D}' \end{pmatrix} \right] \quad (\text{a.4.14.7})$$

(a.4.14.5) permits to recover $\hat{y}_{T+1:T+h}$ from:

$$\hat{y}_{T+1:T+h} = M \begin{pmatrix} D \\ \hat{D} \end{pmatrix}^{-1} z \quad (\text{a.4.14.8})$$

Since D has full row rank, then its generalised inverse D^* is such that $DD^* = I_k$. Hence:

$$\begin{pmatrix} D \\ \hat{D} \end{pmatrix}^{-1} = (D^* \quad \hat{D}') \quad (\text{a.4.14.9})$$

which also implies that $\hat{D}D^* = 0$. Thus:

$$\hat{y}_{T+1:T+h} = M (D^* \quad \hat{D}') z \quad (\text{a.4.14.10})$$

Eventually combining (a.4.14.10) with (a.4.14.7), we obtain the restricted distribution of $\hat{y}_{T+1:T+h}$ as $\hat{y}_{T+1:T+h} \sim N(\hat{\mu}, \hat{\Omega})$, with:

$$\hat{\mu} = M(D^* \bar{y} + \hat{D}' \hat{D} M^{-1} f_{T+1:T+h}) \quad \hat{\Omega} = M(D^* \Omega D^{*'} + \hat{D}' \hat{D} (I_h \otimes \Gamma) \hat{D}' \hat{D}) M' \quad (\text{a.4.14.11})$$

Since $\hat{y}_{T+1:T+h} = f_{T+1:T+h} + M \xi_{T+1:T+h}$, it follows that $\xi_{T+1:T+h} = M^{-1}(\hat{y}_{T+1:T+h} - f_{T+1:T+h})$. Thus, from (a.4.14.11), the restricted distribution of the shocks $\xi_{T+1:T+h}$ is given by $\xi_{T+1:T+h} \sim N(\bar{\mu}, \bar{\Omega})$, with:

$$\bar{\mu} = D^* \bar{y} + \hat{D}' \hat{D} M^{-1} f_{T+1:T+h} - M^{-1} f_{T+1:T+h} \quad \bar{\Omega} = D^* \Omega D^{*'} + \hat{D}' \hat{D} (I_h \otimes \Gamma) \hat{D}' \hat{D} \quad (\text{a.4.14.12})$$

Also, it follows directly from post-multiplication of (a.4.14.9) that $D^* D + \hat{D}' \hat{D} = I_{nh}$, so that $\bar{\mu}$ in (a.4.14.12) rewrites:

$$\bar{\mu} = D^* \bar{y} - D^* D M^{-1} f_{T+1:T+h} = D^* \bar{y} - D^* R f_{T+1:T+h} = D^* (\bar{y} - R f_{T+1:T+h}) \quad (\text{a.4.14.13})$$

And $\bar{\Omega}$ in (a.4.14.12) rewrites:

$$\bar{\Omega} = D^* \Omega D^{*'} + (I_{nh} - D^* D)(I_h \otimes \Gamma)(I_{nh} - D^* D) \quad (\text{a.4.14.14})$$

derivations for equation (4.14.20)

Start from $\hat{y}_{T+1:T+h} = f_{T+1:T+h} + M \xi_{T+1:T+h}$. This implies that $\hat{y}_{T+1:T+h} \sim N(\hat{\mu}, \hat{\Omega})$, with:

$$\hat{\mu} = f_{T+1:T+h} + M \bar{\mu} \quad \hat{\Omega} = M \bar{\Omega} M' \quad (\text{a.4.14.15})$$

From (a.4.14.13), the first expression rewrites as:

$$\hat{\mu} = f_{T+1:T+h} + M \bar{\mu} = f_{T+1:T+h} + M D^* (\bar{y} - R f_{T+1:T+h}) \quad (\text{a.4.14.16})$$

And from (a.4.14.14), the second term obtains directly as:

$$\bar{\Omega} = M [D^* \Omega D^{*'} + (I_{nh} - D^* D)(I_h \otimes \Gamma)(I_{nh} - D^* D)] M' \quad (\text{a.4.14.17})$$

derivations for equation (4.14.45)

We show that $\bar{H}_0^{-1}$ is given by:

$$\bar{H}_0^{-1} = \begin{pmatrix} H_0^{-1} & 0_{n \times k} \\ -\Gamma_{0,2}^{-1} \Gamma_{0,1} H_0^{-1} & \Gamma_{0,2}^{-1} \end{pmatrix} \quad (\text{a.4.14.18})$$

Indeed:

$$\begin{aligned} & \bar{H}_0 \bar{H}_0^{-1} \\ &= \begin{pmatrix} H_0 & 0_{n \times k} \\ \Gamma_{0,1} & \Gamma_{0,2} \end{pmatrix} \begin{pmatrix} H_0^{-1} & 0_{n \times k} \\ -\Gamma_{0,2}^{-1} \Gamma_{0,1} H_0^{-1} & \Gamma_{0,2}^{-1} \end{pmatrix} \\ &= \begin{pmatrix} H_0 H_0^{-1} & 0_{n \times k} \\ \Gamma_{0,1} H_0^{-1} - \Gamma_{0,2} \Gamma_{0,2}^{-1} \Gamma_{0,1} H_0^{-1} & \Gamma_{0,2} \Gamma_{0,2}^{-1} \end{pmatrix} \\ &= \begin{pmatrix} I_n & 0_{n \times k} \\ 0_{k \times n} & I_k \end{pmatrix} \\ &= I_{\bar{n}} \end{aligned} \quad (\text{a.4.14.19})$$

derivations for equation (4.14.46)

Start from the stacked SVAR formulation and develop:

$$\begin{aligned}
 \bar{H}_0 \bar{y}_t &= \bar{H}_+ \bar{x}_t + \bar{\xi}_t \\
 \Leftrightarrow \bar{H}_0^{-1} \bar{H}_0 \bar{y}_t &= \bar{H}_0^{-1} \bar{H}_+ \bar{x}_t + \bar{H}_0^{-1} \bar{\xi}_t \\
 \Leftrightarrow \bar{y}_t &= \bar{H}_0^{-1} \bar{H}_+ \bar{x}_t + \bar{H}_0^{-1} \bar{\xi}_t \\
 \Leftrightarrow \begin{pmatrix} y_t \\ r_t \end{pmatrix} &= \bar{H}_0^{-1} \bar{H}_+ \bar{x}_t + \begin{pmatrix} H_0^{-1} & 0_{n \times k} \\ -\Gamma_{0,2}^{-1} \Gamma_{0,1} & H_0^{-1} \end{pmatrix} \begin{pmatrix} \xi_t \\ v_t \end{pmatrix}
 \end{aligned} \tag{a.4.14.20}$$

Consider the lower block for m_t :

$$r_t = \bar{H}_0^{-1} \bar{H}_+ \bar{x}_t - \Gamma_{0,2}^{-1} \Gamma_{0,1} H_0^{-1} \xi_t + \Gamma_{0,2}^{-1} v_t \tag{a.4.14.21}$$

It then follows that:

$$\begin{aligned}
 &\mathbb{E}(r_t \xi_t') \\
 &= \bar{H}_0^{-1} \bar{H}_+ \mathbb{E}(\bar{x}_t \xi_t') - \Gamma_{0,2}^{-1} \Gamma_{0,1} H_0^{-1} \mathbb{E}(\xi_t \xi_t') + \Gamma_{0,2}^{-1} \mathbb{E}(v_t \xi_t') \\
 &= -\Gamma_{0,2}^{-1} \Gamma_{0,1} H_0^{-1}
 \end{aligned} \tag{a.4.14.22}$$

where we have used a standard exogeneity assumption $Ex(\xi_t | \bar{x}_t) = 0$, and the fact that v_t and ξ_t are uncorrelated.

PART V

Vector autoregression extensions

Vector error correction

derivations for equation (5.15.3)

Start from the VAR model

$$y_t = Cz_t + A_1y_{t-1} + \cdots + A_py_{t-p} + \varepsilon_t \quad (\text{a.5.15.1})$$

Subtract y_{t-1} from both sides:

$$y_t = Cz_t + A_1y_{t-1} + \cdots + A_py_{t-p} + \varepsilon_t \quad (\text{a.5.15.2})$$

Rearrange:

$$\begin{aligned} \Leftrightarrow \Delta y_t &= Cz_t + \sum_{i=1}^p A_i y_{t-i} - y_{t-1} + \varepsilon_t \\ \Leftrightarrow \Delta y_t &= Cz_t + \sum_{i=1}^p \left(A_i y_{t-i} + \sum_{j=i+1}^p A_j y_{t-i} - \sum_{j=i+1}^p A_j y_{t-i} \right) - y_{t-1} + \varepsilon_t \\ \Leftrightarrow \Delta y_t &= Cz_t + \left(\sum_{i=1}^p A_i - I \right) y_{t-1} - \sum_{i=1}^{p-1} \left(\sum_{j=i+1}^p A_j \right) (y_{t-i} - y_{t-i-1}) + \varepsilon_t \\ \Leftrightarrow \Delta y_t &= Cz_t + \left(\sum_{i=1}^p A_i - I \right) y_{t-1} + \sum_{i=1}^{p-1} \left(- \sum_{j=i+1}^p A_j \right) (y_{t-i} - y_{t-i-1}) + \varepsilon_t \\ \Leftrightarrow \Delta y_t &= Cz_t + \Pi y_{t-1} + \sum_{i=1}^{p-1} F_i \Delta y_{t-i} + \varepsilon_t \quad (\text{a.5.15.3}) \end{aligned}$$

derivations for equation (5.15.13)

Start from:

$$\Delta Y = Y_{-1} K \Lambda' + Z \Phi + \mathcal{E} \quad (\text{a.5.15.4})$$

Use direct vectorization properties to obtain:

$$\text{vec}(\Delta Y) = (\Lambda \otimes Y_{-1}) \text{vec}(K) + (I_n \otimes Z) \text{vec}(\Phi) + \text{vec}(\mathcal{E}) \quad (\text{a.5.15.5})$$

Using previously introduced notations:

$$\Delta y = (\Lambda \otimes Y_{-1}) \kappa + \bar{Z} \phi + \varepsilon \quad (\text{a.5.15.6})$$

with $\kappa = \text{vec}(K)$. This immediately implies that the likelihood function is given by:

$$f(y|\kappa, \lambda, \phi, \Sigma) = (2\pi)^{-nT/2} |\bar{\Sigma}|^{-1/2} \exp \left(-\frac{1}{2} (\Delta y - (\Lambda \otimes Y_{-1}) \kappa - \bar{Z} \phi)' \bar{\Sigma}^{-1} (\Delta y - (\Lambda \otimes Y_{-1}) \kappa - \bar{Z} \phi) \right) \quad (\text{a.5.15.7})$$

derivations for equation (5.15.14)

Start from:

$$\Delta Y = Y_{-1}K\Lambda' + Z\Phi + \mathcal{E} \quad (\text{a.5.15.8})$$

Note that this rewrites as:

$$\Delta Y = (Y_{-1}K)\Lambda'I_n + Z\Phi + \mathcal{E} \quad (\text{a.5.15.9})$$

Use then direct vectorization properties to obtain:

$$\text{vec}(\Delta Y) = (I_n \otimes Y_{-1}K)\text{vec}(\Lambda') + (I_n \otimes Z)\text{vec}(\Phi) + \text{vec}(\mathcal{E}) \quad (\text{a.5.15.10})$$

Using previously introduced notations:

$$\Delta y = (I_n \otimes Y_{-1}K)\lambda + \bar{Z}\phi + \varepsilon \quad (\text{a.5.15.11})$$

with $\lambda = \text{vec}(\Lambda')$. This immediately implies that the likelihood function is given by:

$$f(y|\kappa, \lambda, \phi, \Sigma) = (2\pi)^{-nT/2} |\bar{\Sigma}|^{-1/2} \exp \left(-\frac{1}{2} (\Delta y - (I_n \otimes Y_{-1}K)\lambda - \bar{Z}\phi)' \bar{\Sigma}^{-1} (\Delta y - (I_n \otimes Y_{-1}K)\lambda - \bar{Z}\phi) \right) \quad (\text{a.5.15.12})$$

derivations for equation (5.15.21)

For convenience, define first:

$$\tilde{y} = \Delta y - \bar{Y}_{-1}\xi \quad (\text{a.5.15.13})$$

Then $\pi(\phi|y, \xi, \Sigma)$ rewrites:

$$\pi(\phi|y, \xi, \Sigma) \propto \exp \left(-\frac{1}{2} (\tilde{y} - \bar{Z}\phi)' \bar{\Sigma}^{-1} (\tilde{y} - \bar{Z}\phi) \right) \times \exp \left(-\frac{1}{2} (\phi - f)' Q^{-1} (\phi - f) \right) \quad (\text{a.5.15.14})$$

Group terms:

$$\begin{aligned} & \pi(\phi|y, \xi, \Sigma) \\ & \propto \exp \left(-\frac{1}{2} (\tilde{y} - \bar{Z}\phi)' \bar{\Sigma}^{-1} (\tilde{y} - \bar{Z}\phi) \right) \times \exp \left(-\frac{1}{2} (\phi - f)' Q^{-1} (\phi - f) \right) \\ & = \exp \left(-\frac{1}{2} [(\tilde{y} - \bar{Z}\phi)' \bar{\Sigma}^{-1} (\tilde{y} - \bar{Z}\phi) + (\phi - f)' Q^{-1} (\phi - f)] \right) \end{aligned} \quad (\text{a.5.15.15})$$

Consider the terms in square brackets:

$$\begin{aligned} & (\tilde{y} - \bar{Z}\phi)' \bar{\Sigma}^{-1} (\tilde{y} - \bar{Z}\phi) + (\phi - f)' Q^{-1} (\phi - f) \\ & = \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} + \phi' \bar{Z}' \bar{\Sigma}^{-1} \bar{Z} \phi - 2\phi' \bar{Z}' \bar{\Sigma}^{-1} \tilde{y} + \phi' Q^{-1} \phi + f' Q^{-1} f - 2\phi' Q^{-1} f \\ & = \phi' (Q^{-1} + \bar{Z}' \bar{\Sigma}^{-1} \bar{Z}) \phi - 2\phi' (Q^{-1} f + \bar{Z}' \bar{\Sigma}^{-1} \tilde{y}) + f' Q^{-1} f + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} \end{aligned} \quad (\text{a.5.15.16})$$

Complete the squares:

$$= \phi' (Q^{-1} + \bar{Z}' \bar{\Sigma}^{-1} \bar{Z}) \phi - 2\phi' \bar{Q}^{-1} \bar{Q} (Q^{-1} f + \bar{Z}' \bar{\Sigma}^{-1} \tilde{y}) + \bar{f}' \bar{Q}^{-1} \bar{f} - \bar{f}' \bar{Q}^{-1} \bar{f} + f' Q^{-1} f + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} \quad (\text{a.5.15.17})$$

Define:

$$\bar{Q} = (Q^{-1} + \bar{Z}' \bar{\Sigma}^{-1} \bar{Z})^{-1} \quad \bar{f} = \bar{Q} (Q^{-1} f + \bar{Z}' \bar{\Sigma}^{-1} \tilde{y}) \quad (\text{a.5.15.18})$$

Then (a.5.15.17) rewrites:

$$\begin{aligned}
&= \phi' \bar{Q}^{-1} \phi - 2\phi' \bar{Q}^{-1} \bar{f} + \bar{f}' \bar{Q}^{-1} \bar{f} - \bar{f}' \bar{Q}^{-1} \bar{f} + f' Q^{-1} f + \bar{y}' \bar{\Sigma}^{-1} \bar{y} \\
&= (\phi - \bar{f})' \bar{Q}^{-1} (\phi - \bar{f}) + (f' Q^{-1} f - \bar{f}' \bar{Q}^{-1} \bar{f} + \bar{y}' \bar{\Sigma}^{-1} \bar{y})
\end{aligned} \tag{a.5.15.19}$$

Substitute (a.5.15.19) back in (a.5.15.15):

$$\begin{aligned}
&\pi(\phi|y, \xi, \Sigma) \\
&= \exp \left(-\frac{1}{2} [(\phi - \bar{f})' \bar{Q}^{-1} (\phi - \bar{f}) + (f' Q^{-1} f - \bar{f}' \bar{Q}^{-1} \bar{f} + \bar{y}' \bar{\Sigma}^{-1} \bar{y})] \right) \\
&= \exp \left(-\frac{1}{2} (\phi - \bar{f})' \bar{Q}^{-1} (\phi - \bar{f}) \right) \exp \left(-\frac{1}{2} (f' Q^{-1} f - \bar{f}' \bar{Q}^{-1} \bar{f} + \bar{y}' \bar{\Sigma}^{-1} \bar{y}) \right) \\
&\propto \exp \left(-\frac{1}{2} (\phi - \bar{f})' \bar{Q}^{-1} (\phi - \bar{f}) \right)
\end{aligned} \tag{a.5.15.20}$$

Where the last line obtains by noting that the second term in row 2 does not involve ϕ and can hence be relegated to the normalization constant.

The terms in (a.5.15.18) simplify. Note first that:

$$\begin{aligned}
&\bar{Z}' \bar{\Sigma}^{-1} \bar{Z} \\
&= (I_n \otimes Z)' (\Sigma \otimes I_T)^{-1} (I_n \otimes Z) \\
&= (I_n \otimes Z') (\Sigma^{-1} \otimes I_T) (I_n \otimes Z) \\
&= (\Sigma^{-1} \otimes Z') (I_n \otimes Z) \\
&= \Sigma^{-1} \otimes Z' Z
\end{aligned} \tag{a.5.15.21}$$

Similarly:

$$\begin{aligned}
&\bar{Z}' \bar{\Sigma}^{-1} \bar{y} \\
&= \bar{Z}' \bar{\Sigma}^{-1} (\Delta y - \bar{Y}_{-1} \xi) \\
&= (I_n \otimes Z)' (\Sigma \otimes I_T)^{-1} \text{vec}(\Delta Y - Y_{-1} \Xi') \\
&= (I_n \otimes Z') (\Sigma^{-1} \otimes I_T) \text{vec}(\Delta Y - Y_{-1} \Xi') \\
&= (\Sigma^{-1} \otimes Z') \text{vec}(\Delta Y - Y_{-1} \Xi') \\
&= \text{vec}(Z' [\Delta Y - Y_{-1} \Xi'] \Sigma^{-1})
\end{aligned} \tag{a.5.15.22}$$

Then (a.5.15.18) rewrites:

$$\bar{Q} = (Q^{-1} + \Sigma^{-1} \otimes Z' Z)^{-1} \quad \bar{f} = \bar{Q} (Q^{-1} f + \text{vec}(Z' [\Delta Y - Y_{-1} \Xi'] \Sigma^{-1})) \tag{a.5.15.23}$$

derivations for equation (5.15.24)

Note that:

$$|\bar{\Sigma}|^{-1/2} = |\Sigma \otimes I_T|^{-1/2} = |\Sigma|^{-T/2} |I_T|^{-n/2} = |\Sigma|^{-T/2} \tag{a.5.15.24}$$

Also:

$$\begin{aligned}
&(\Delta y - \bar{Y}_{-1} \xi - \bar{Z} \phi)' \bar{\Sigma}^{-1} (\Delta y - \bar{Y}_{-1} \xi - \bar{Z} \phi) \\
&= (\Delta y - (I_n \otimes Y_{-1}) \xi - (I_n \otimes Z) \phi)' (\Sigma \otimes I_T)^{-1} (\Delta y - (I_n \otimes Y_{-1}) \xi - (I_n \otimes Z) \phi) \\
&= \text{tr} \{ \Sigma^{-1} (\Delta Y - Y_{-1} \Xi' - Z \Phi)' (\Delta Y - Y_{-1} \Xi' - Z \Phi) \}
\end{aligned} \tag{a.5.15.25}$$

Then substituting in the original expression:

$$\begin{aligned}
& \pi(\Sigma|y, \phi, \xi) \\
& \propto |\bar{\Sigma}|^{-1/2} \exp\left(-\frac{1}{2}(\Delta y - \bar{Y}_{-1}\xi - \bar{Z}\phi)' \bar{\Sigma}^{-1}(\Delta y - \bar{Y}_{-1}\xi - \bar{Z}\phi)\right) \\
& \times |\Sigma|^{-(\alpha+n+1)/2} \exp\left(-\frac{1}{2}tr\{\Sigma^{-1}S\}\right) \\
& = |\Sigma|^{-T/2} \exp\left(-\frac{1}{2}tr\{\Sigma^{-1}(\Delta Y - Y_{-1}\Xi' - Z\Phi)'(\Delta Y - Y_{-1}\Xi' - Z\Phi)\}\right) \times |\Sigma|^{-(\alpha+n+1)/2} \exp\left(-\frac{1}{2}tr\{\Sigma^{-1}S\}\right) \\
& = |\Sigma|^{-(\alpha+T+n+1)/2} \exp\left(-\frac{1}{2}tr\{\Sigma^{-1}[S + (\Delta Y - Y_{-1}\Xi' - Z\Phi)'(\Delta Y - Y_{-1}\Xi' - Z\Phi)]\}\right) \\
& = |\Sigma|^{-(\bar{\alpha}+n+1)/2} \exp\left(-\frac{1}{2}tr\{\Sigma^{-1}\bar{S}\}\right) \tag{a.5.15.26}
\end{aligned}$$

with:

$$\bar{\alpha} = \alpha + T \quad \bar{S} = S + (\Delta Y - Y_{-1}\Xi' - Z\Phi)'(\Delta Y - Y_{-1}\Xi' - Z\Phi) \tag{a.5.15.27}$$

derivations for equation (5.15.28)

For convenience, define first:

$$\tilde{y} = \Delta y - \bar{Z}\phi \tag{a.5.15.28}$$

Then $\pi(\xi|y, \phi, \Sigma)$ rewrites:

$$\pi(\xi|y, \phi, \Sigma) \propto \exp\left(-\frac{1}{2}(\tilde{y} - \bar{Y}_{-1}\xi)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{Y}_{-1}\xi)\right) \times \exp\left(-\frac{1}{2}\xi' U^{-1} \xi\right) \tag{a.5.15.29}$$

Group terms:

$$\begin{aligned}
& \pi(\xi|y, \phi, \Sigma) \\
& \propto \exp\left(-\frac{1}{2}(\tilde{y} - \bar{Y}_{-1}\xi)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{Y}_{-1}\xi)\right) \times \exp\left(-\frac{1}{2}\xi' U^{-1} \xi\right) \\
& = \exp\left(-\frac{1}{2}[(\tilde{y} - \bar{Y}_{-1}\xi)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{Y}_{-1}\xi) + \xi' U^{-1} \xi]\right) \tag{a.5.15.30}
\end{aligned}$$

Consider the terms in square brackets:

$$\begin{aligned}
& (\tilde{y} - \bar{Y}_{-1}\xi)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{Y}_{-1}\xi) + \xi' U^{-1} \xi \\
& = \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} + \xi' \bar{Y}_{-1}' \bar{\Sigma}^{-1} \bar{Y}_{-1} \xi - 2\xi' \bar{Y}_{-1}' \bar{\Sigma}^{-1} \tilde{y} + \xi' U^{-1} \xi \\
& = \xi'(U^{-1} + \bar{Y}_{-1}' \bar{\Sigma}^{-1} \bar{Y}_{-1}) \xi - 2\xi' (\bar{Y}_{-1}' \bar{\Sigma}^{-1} \tilde{y}) + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} \tag{a.5.15.31}
\end{aligned}$$

Complete the squares:

$$= \xi'(U^{-1} + \bar{Y}_{-1}' \bar{\Sigma}^{-1} \bar{Y}_{-1}) \xi - 2\xi' \bar{U}^{-1} \bar{U} (\bar{Y}_{-1}' \bar{\Sigma}^{-1} \tilde{y}) + \bar{d}' \bar{U}^{-1} \bar{d} - \bar{d}' \bar{U}^{-1} \bar{d} + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} \tag{a.5.15.32}$$

Define:

$$\bar{U} = (U^{-1} + \bar{Y}_{-1}' \bar{\Sigma}^{-1} \bar{Y}_{-1})^{-1} \quad \bar{d} = \bar{U} (\bar{Y}_{-1}' \bar{\Sigma}^{-1} \tilde{y}) \tag{a.5.15.33}$$

Then (a.5.15.32) rewrites:

$$\begin{aligned}
&= \xi' \bar{U}^{-1} \xi - 2\xi' \bar{U}^{-1} \bar{d} + \bar{d}' \bar{U}^{-1} \bar{d} - \bar{d}' \bar{U}^{-1} \bar{d} + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} \\
&= (\xi - \bar{d})' \bar{U}^{-1} (\xi - \bar{d}) + (\tilde{y}' \bar{\Sigma}^{-1} \tilde{y} - \bar{d}' \bar{U}^{-1} \bar{d})
\end{aligned} \tag{a.5.15.34}$$

Substitute (a.5.15.34) back in (a.5.15.30):

$$\begin{aligned}
&\pi(\xi|y, \phi, \Sigma) \\
&= \exp\left(-\frac{1}{2} [(\xi - \bar{d})' \bar{U}^{-1} (\xi - \bar{d}) + (\tilde{y}' \bar{\Sigma}^{-1} \tilde{y} - \bar{d}' \bar{U}^{-1} \bar{d})]\right) \\
&= \exp\left(-\frac{1}{2} (\xi - \bar{d})' \bar{U}^{-1} (\xi - \bar{d})\right) \exp\left(-\frac{1}{2} (\tilde{y}' \bar{\Sigma}^{-1} \tilde{y} - \bar{d}' \bar{U}^{-1} \bar{d})\right) \\
&\propto \exp\left(-\frac{1}{2} (\xi - \bar{d})' \bar{U}^{-1} (\xi - \bar{d})\right)
\end{aligned} \tag{a.5.15.35}$$

Where the last line obtains by noting that the second term in row 2 does not involve ξ and can hence be relegated to the normalization constant.

The terms in (a.5.15.33) simplify. Note first that:

$$\begin{aligned}
&\bar{Y}'_{-1} \bar{\Sigma}^{-1} \bar{Y}_{-1} \\
&= (I_n \otimes Y_{-1})' (\Sigma \otimes I_T)^{-1} (I_n \otimes Y_{-1}) \\
&= (I_n \otimes Y'_{-1}) (\Sigma^{-1} \otimes I_T) (I_n \otimes Y_{-1}) \\
&= (\Sigma^{-1} \otimes Y'_{-1}) (I_n \otimes Y_{-1}) \\
&= \Sigma^{-1} \otimes Y'_{-1} Y_{-1}
\end{aligned} \tag{a.5.15.36}$$

Similarly:

$$\begin{aligned}
&\bar{Y}'_{-1} \bar{\Sigma}^{-1} \tilde{y} \\
&= \bar{Y}'_{-1} \bar{\Sigma}^{-1} (\Delta y - \bar{Z} \phi) \\
&= (I_n \otimes Y_{-1})' (\Sigma \otimes I_T)^{-1} (\Delta y - \bar{Z} \phi) \\
&= (I_n \otimes Y'_{-1}) (\Sigma^{-1} \otimes I_T) (\Delta y - \bar{Z} \phi) \\
&= (\Sigma^{-1} \otimes Y'_{-1}) (\Delta y - \bar{Z} \phi) \\
&= (\Sigma^{-1} \otimes Y'_{-1}) \text{vec}(\Delta Y - Z \Phi) \\
&= \text{vec}(Y'_{-1} [\Delta Y - Z \Phi] \Sigma^{-1})
\end{aligned} \tag{a.5.15.37}$$

Then (a.5.15.33) rewrites:

$$\bar{U} = (U^{-1} + \Sigma^{-1} \otimes Y'_{-1} Y_{-1})^{-1} \quad \bar{d} = \bar{U} \text{vec}(Y'_{-1} [\Delta Y - Z \Phi] \Sigma^{-1}) \tag{a.5.15.38}$$

derivations for equation (5.15.34)

For convenience, define first:

$$\tilde{y} = \Delta y - \bar{Z} \phi \quad \tilde{X} = \Lambda \otimes Y_{-1} \tag{a.5.15.39}$$

Then $\pi(\kappa|y, \lambda, \phi, \Sigma)$ rewrites:

$$\pi(\kappa|y, \lambda, \phi, \Sigma) \propto \exp\left(-\frac{1}{2} (\tilde{y} - \tilde{X} \kappa)' \bar{\Sigma}^{-1} (\tilde{y} - \tilde{X} \kappa)\right) \times \exp\left(-\frac{1}{2} \kappa' R^{-1} \kappa\right) \tag{a.5.15.40}$$

Group terms:

$$\begin{aligned}
& \pi(\kappa|y, \lambda, \phi, \Sigma) \\
& \propto \exp\left(-\frac{1}{2}(\tilde{y} - \tilde{X}\kappa)' \tilde{\Sigma}^{-1}(\tilde{y} - \tilde{X}\kappa)\right) \times \exp\left(-\frac{1}{2}\kappa' R^{-1}\kappa\right) \\
& = \exp\left(-\frac{1}{2}[(\tilde{y} - \tilde{X}\kappa)' \tilde{\Sigma}^{-1}(\tilde{y} - \tilde{X}\kappa) + \kappa' R^{-1}\kappa]\right) \tag{a.5.15.41}
\end{aligned}$$

Consider the terms in square brackets:

$$\begin{aligned}
& (\tilde{y} - \tilde{X}\kappa)' \tilde{\Sigma}^{-1}(\tilde{y} - \tilde{X}\kappa) + \kappa' R^{-1}\kappa \\
& = \tilde{y}' \tilde{\Sigma}^{-1} \tilde{y} + \kappa' \tilde{X}' \tilde{\Sigma}^{-1} \tilde{X} \kappa - 2\kappa' \tilde{X}' \tilde{\Sigma}^{-1} \tilde{y} + \kappa' R^{-1} \kappa \\
& = \kappa' (R^{-1} + \tilde{X}' \tilde{\Sigma}^{-1} \tilde{X}) \kappa - 2\kappa' (\tilde{X}' \tilde{\Sigma}^{-1} \tilde{y}) + \tilde{y}' \tilde{\Sigma}^{-1} \tilde{y} \tag{a.5.15.42}
\end{aligned}$$

Complete the squares:

$$= \kappa' (R^{-1} + \tilde{X}' \tilde{\Sigma}^{-1} \tilde{X}) \kappa - 2\kappa' \bar{R}^{-1} \bar{R} (\tilde{X}' \tilde{\Sigma}^{-1} \tilde{y}) + \bar{g}' \bar{R}^{-1} \bar{g} - \bar{g}' \bar{R}^{-1} \bar{g} + \tilde{y}' \tilde{\Sigma}^{-1} \tilde{y} \tag{a.5.15.43}$$

Define:

$$\bar{R} = (R^{-1} + \tilde{X}' \tilde{\Sigma}^{-1} \tilde{X})^{-1} \quad \bar{g} = \bar{R} (\tilde{X}' \tilde{\Sigma}^{-1} \tilde{y}) \tag{a.5.15.44}$$

Then (a.5.15.43) rewrites:

$$\begin{aligned}
& = \kappa' \bar{R}^{-1} \kappa - 2\kappa' \bar{R}^{-1} \bar{g} + \bar{g}' \bar{R}^{-1} \bar{g} - \bar{g}' \bar{R}^{-1} \bar{g} + \tilde{y}' \tilde{\Sigma}^{-1} \tilde{y} \\
& = (\kappa - \bar{g})' \bar{R}^{-1} (\kappa - \bar{g}) + (\tilde{y}' \tilde{\Sigma}^{-1} \tilde{y} - \bar{g}' \bar{R}^{-1} \bar{g}) \tag{a.5.15.45}
\end{aligned}$$

Substitute (a.5.15.45) back in (a.5.15.41):

$$\begin{aligned}
& \pi(\kappa|y, \lambda, \phi, \Sigma) \\
& = \exp\left(-\frac{1}{2}[(\kappa - \bar{g})' \bar{R}^{-1} (\kappa - \bar{g}) + (\tilde{y}' \tilde{\Sigma}^{-1} \tilde{y} - \bar{g}' \bar{R}^{-1} \bar{g})]\right) \\
& = \exp\left(-\frac{1}{2}(\kappa - \bar{g})' \bar{R}^{-1} (\kappa - \bar{g})\right) \exp\left(-\frac{1}{2}(\tilde{y}' \tilde{\Sigma}^{-1} \tilde{y} - \bar{g}' \bar{R}^{-1} \bar{g})\right) \\
& \propto \exp\left(-\frac{1}{2}(\kappa - \bar{g})' \bar{R}^{-1} (\kappa - \bar{g})\right) \tag{a.5.15.46}
\end{aligned}$$

Where the last line obtains by noting that the second term in row 2 does not involve κ and can hence be relegated to the normalization constant.

The terms in (a.5.15.44) simplify. Note first that:

$$\begin{aligned}
& \tilde{X}' \tilde{\Sigma}^{-1} \tilde{X} \\
& = (\Lambda \otimes Y_{-1})' (\Sigma \otimes I_T)^{-1} (\Lambda \otimes Y_{-1}) \\
& = (\Lambda' \otimes Y'_{-1}) (\Sigma^{-1} \otimes I_T) (\Lambda \otimes Y_{-1}) \\
& = (\Lambda' \Sigma^{-1} \otimes Y'_{-1}) (\Lambda \otimes Y_{-1}) \\
& = (\Lambda' \Sigma^{-1} \Lambda) \otimes (Y'_{-1} Y_{-1}) \tag{a.5.15.47}
\end{aligned}$$

Similarly:

$$\begin{aligned}
& \tilde{X}'\tilde{\Sigma}^{-1}\tilde{y} \\
&= \tilde{X}'\tilde{\Sigma}^{-1}(\Delta y - \bar{Z}\phi) \\
&= (\Lambda \otimes Y_{-1})'(\Sigma \otimes I_T)^{-1}(\Delta y - \bar{Z}\phi) \\
&= (\Lambda' \otimes Y'_{-1})(\Sigma^{-1} \otimes I_T)(\Delta y - \bar{Z}\phi) \\
&= (\Lambda'\Sigma^{-1} \otimes Y'_{-1})(\Delta y - \bar{Z}\phi) \\
&= (\Lambda'\Sigma^{-1} \otimes Y'_{-1}) \text{vec}(\Delta Y - Z\Phi) \\
&= \text{vec}(Y'_{-1}[\Delta Y - Z\Phi]\Sigma^{-1}\Lambda)
\end{aligned} \tag{a.5.15.48}$$

Then (a.5.15.44) rewrites:

$$\bar{R} = (R^{-1} + \Lambda'\Sigma^{-1}\Lambda \otimes Y'_{-1}Y_{-1})^{-1} \quad \bar{g} = \bar{R} \text{vec}(Y'_{-1}[\Delta Y - Z\Phi]\Sigma^{-1}\Lambda) \tag{a.5.15.49}$$

derivations for equation (5.15.37)

For convenience, define first:

$$\tilde{y} = \Delta y - \bar{Z}\phi \quad \tilde{X} = I_n \otimes Y_{-1}K \tag{a.5.15.50}$$

Then $\pi(\lambda|y, \kappa, \phi, \Sigma)$ rewrites:

$$\pi(\lambda|y, \kappa, \phi, \Sigma) \propto \exp\left(-\frac{1}{2}(\tilde{y} - \tilde{X}\lambda)'\tilde{\Sigma}^{-1}(\tilde{y} - \tilde{X}\lambda)\right) \times \exp\left(-\frac{1}{2}\lambda'P^{-1}\lambda\right) \tag{a.5.15.51}$$

Group terms:

$$\begin{aligned}
& \pi(\lambda|y, \kappa, \phi, \Sigma) \\
& \propto \exp\left(-\frac{1}{2}(\tilde{y} - \tilde{X}\lambda)'\tilde{\Sigma}^{-1}(\tilde{y} - \tilde{X}\lambda)\right) \times \exp\left(-\frac{1}{2}\lambda'P^{-1}\lambda\right) \\
& = \exp\left(-\frac{1}{2}\left[(\tilde{y} - \tilde{X}\lambda)'\tilde{\Sigma}^{-1}(\tilde{y} - \tilde{X}\lambda) + \lambda'P^{-1}\lambda\right]\right)
\end{aligned} \tag{a.5.15.52}$$

Consider the terms in square brackets:

$$\begin{aligned}
& (\tilde{y} - \tilde{X}\lambda)'\tilde{\Sigma}^{-1}(\tilde{y} - \tilde{X}\lambda) + \lambda'P^{-1}\lambda \\
&= \tilde{y}'\tilde{\Sigma}^{-1}\tilde{y} + \lambda'\tilde{X}'\tilde{\Sigma}^{-1}\tilde{X}\lambda - 2\lambda'\tilde{X}'\tilde{\Sigma}^{-1}\tilde{y} + \lambda'P^{-1}\lambda \\
&= \lambda'(P^{-1} + \tilde{X}'\tilde{\Sigma}^{-1}\tilde{X})\lambda - 2\lambda'(\tilde{X}'\tilde{\Sigma}^{-1}\tilde{y}) + \tilde{y}'\tilde{\Sigma}^{-1}\tilde{y}
\end{aligned} \tag{a.5.15.53}$$

Complete the squares:

$$= \lambda'(P^{-1} + \tilde{X}'\tilde{\Sigma}^{-1}\tilde{X})\lambda - 2\lambda'\bar{P}^{-1}\bar{P}(\tilde{X}'\tilde{\Sigma}^{-1}\tilde{y}) + \bar{h}'\bar{P}^{-1}\bar{h} - \bar{h}'\bar{P}^{-1}\bar{h} + \tilde{y}'\tilde{\Sigma}^{-1}\tilde{y} \tag{a.5.15.54}$$

Define:

$$\bar{P} = (P^{-1} + \tilde{X}'\tilde{\Sigma}^{-1}\tilde{X})^{-1} \quad \bar{h} = \bar{P}(\tilde{X}'\tilde{\Sigma}^{-1}\tilde{y}) \tag{a.5.15.55}$$

Then (a.5.15.54) rewrites:

$$\begin{aligned}
&= \lambda'\bar{P}^{-1}\lambda - 2\lambda'\bar{P}^{-1}\bar{h} + \bar{h}'\bar{P}^{-1}\bar{h} - \bar{h}'\bar{P}^{-1}\bar{h} + \tilde{y}'\tilde{\Sigma}^{-1}\tilde{y} \\
&= (\lambda - \bar{h})'\bar{P}^{-1}(\lambda - \bar{h}) + (\tilde{y}'\tilde{\Sigma}^{-1}\tilde{y} - \bar{h}'\bar{P}^{-1}\bar{h})
\end{aligned} \tag{a.5.15.56}$$

Substitute (a.5.15.56) back in (a.5.15.52):

$$\begin{aligned}
& \pi(\lambda | y, \kappa, \phi, \Sigma) \\
&= \exp \left(-\frac{1}{2} [(\lambda - \bar{h})' \bar{P}^{-1} (\lambda - \bar{h}) + (\tilde{y}' \bar{\Sigma}^{-1} \tilde{y} - \bar{h}' \bar{P}^{-1} \bar{h})] \right) \\
&= \exp \left(-\frac{1}{2} (\lambda - \bar{h})' \bar{P}^{-1} (\lambda - \bar{h}) \right) \exp \left(-\frac{1}{2} (\tilde{y}' \bar{\Sigma}^{-1} \tilde{y} - \bar{h}' \bar{P}^{-1} \bar{h}) \right) \\
&\propto \exp \left(-\frac{1}{2} (\lambda - \bar{h})' \bar{P}^{-1} (\lambda - \bar{h}) \right) \tag{a.5.15.57}
\end{aligned}$$

Where the last line obtains by noting that the second term in row 2 does not involve λ and can hence be relegated to the normalization constant.

The terms in (a.5.15.55) simplify. Note first that:

$$\begin{aligned}
& \tilde{X}' \bar{\Sigma}^{-1} \tilde{X} \\
&= (I_n \otimes Y_{-1} K)' (\Sigma \otimes I_T)^{-1} (I_n \otimes Y_{-1} K) \\
&= (I_n \otimes K' Y_{-1}') (\Sigma^{-1} \otimes I_T) (I_n \otimes Y_{-1} K) \\
&= (\Sigma^{-1} \otimes K' Y_{-1}') (I_n \otimes Y_{-1} K) \\
&= \Sigma^{-1} \otimes (K' Y_{-1}' Y_{-1} K) \tag{a.5.15.58}
\end{aligned}$$

Similarly:

$$\begin{aligned}
& \tilde{X}' \bar{\Sigma}^{-1} \tilde{y} \\
&= \tilde{X}' \bar{\Sigma}^{-1} (\Delta y - \bar{Z} \phi) \\
&= (I_n \otimes Y_{-1} K)' (\Sigma \otimes I_T)^{-1} (\Delta y - \bar{Z} \phi) \\
&= (I_n \otimes K' Y_{-1}') (\Sigma^{-1} \otimes I_T) (\Delta y - \bar{Z} \phi) \\
&= (\Sigma^{-1} \otimes K' Y_{-1}') (\Delta y - \bar{Z} \phi) \\
&= (\Sigma^{-1} \otimes K' Y_{-1}') \text{vec}(\Delta Y - Z \Phi) \\
&= \text{vec}(K' Y_{-1}' [\Delta Y - Z \Phi] \Sigma^{-1}) \tag{a.5.15.59}
\end{aligned}$$

Then (a.5.15.55) rewrites:

$$\bar{P} = (P^{-1} + \Sigma^{-1} \otimes K' Y_{-1}' Y_{-1} K)^{-1} \quad \bar{h} = \bar{P} \text{vec}(K' Y_{-1}' [\Delta Y - Z \Phi] \Sigma^{-1}) \tag{a.5.15.60}$$

derivations for equation (5.15.44)

Bayes rule in its raw form is given by:

$$\pi(\xi, \tau^2, v, \psi^2, \eta, \phi, \Sigma | y) \propto f(y | \xi, \phi, \Sigma) \pi(\xi, \tau^2, v, \psi^2, \eta, \phi, \Sigma) \tag{a.5.15.61}$$

where the parameters τ^2, v, ψ^2 and η are omitted from the likelihood function as they don't intervene in the likelihood once ξ is determined. Consider first the rightmost term, which is the joint prior. Using a standard independence assumption between ϕ, Σ and the other parameters, it rewrites as:

$$\pi(\xi, \tau^2, v, \psi^2, \eta, \phi, \Sigma) = \pi(\xi, \tau^2, v, \psi^2, \eta) \pi(\phi) \pi(\Sigma) \tag{a.5.15.62}$$

The first term on the right-hand-side of (a.5.15.62) rewrites as:

$$\begin{aligned}
& \pi(\xi, \tau^2, \nu, \psi^2, \eta) \\
&= \frac{\pi(\xi, \tau^2, \nu, \psi^2, \eta)}{\pi(\tau^2, \nu)} \frac{\pi(\tau^2, \nu)}{\pi(\nu)} \pi(\nu) \\
&= \frac{\pi(\xi, \tau^2, \nu, \psi^2, \eta)}{\pi(\tau^2, \nu)} \pi(\tau^2 | \nu) \pi(\nu) \\
&= \frac{\pi(\xi, \tau^2, \nu, \psi^2, \eta)}{\pi(\tau^2, \nu, \psi^2, \eta)} \frac{\pi(\tau^2, \nu, \psi^2, \eta)}{\pi(\tau^2, \nu)} \pi(\tau^2 | \nu) \pi(\nu) \\
&= \pi(\xi | \tau^2, \nu, \psi^2, \eta) \frac{\pi(\tau^2, \nu, \psi^2, \eta)}{\pi(\tau^2, \nu)} \pi(\tau^2 | \nu) \pi(\nu)
\end{aligned} \tag{a.5.15.63}$$

Assuming independence between τ^2, ν and ψ^2, η :

$$\begin{aligned}
&= \pi(\xi | \tau^2, \nu, \psi^2, \eta) \frac{\pi(\tau^2, \nu) \pi(\psi^2, \eta)}{\pi(\tau^2, \nu)} \pi(\tau^2 | \nu) \pi(\nu) \\
&= \pi(\xi | \tau^2, \nu, \psi^2, \eta) \pi(\psi^2, \eta) \pi(\tau^2 | \nu) \pi(\nu)
\end{aligned} \tag{a.5.15.64}$$

Assuming independence between the different ξ_i, τ_i and ν_i :

$$\begin{aligned}
&= \left(\prod_{i=1}^{n^2} \pi(\xi_i | \tau^2, \nu, \psi_i^2, \eta_i) \pi(\psi_i^2, \eta_i) \right) \pi(\tau^2 | \nu) \pi(\nu) \\
&= \left(\prod_{i=1}^{n^2} \pi(\xi_i | \tau^2, \nu, \psi_i^2, \eta_i) \frac{\pi(\psi_i^2, \eta_i)}{\pi(\eta_i)} \pi(\eta_i) \right) \pi(\tau^2 | \nu) \pi(\nu) \\
&= \left(\prod_{i=1}^{n^2} \pi(\xi_i | \tau^2, \nu, \psi_i^2, \eta_i) \pi(\psi_i^2 | \eta_i) \pi(\eta_i) \right) \pi(\tau^2 | \nu) \pi(\nu)
\end{aligned} \tag{a.5.15.65}$$

Noting that conditioning on ν and η_i is irrelevant once τ and ψ_i^2 are known, this becomes:

$$= \left(\prod_{i=1}^{n^2} \pi(\xi_i | \tau^2, \psi_i^2) \pi(\psi_i^2 | \eta_i) \pi(\eta_i) \right) \pi(\tau^2 | \nu) \pi(\nu) \tag{a.5.15.66}$$

Replacing (a.5.15.66) in (a.5.15.61):

$$\pi(\xi, \tau^2, \nu, \psi^2, \eta, \phi, \Sigma | y) \propto f(y | \xi, \phi, \Sigma) \left(\prod_{i=1}^{n^2} \pi(\xi_i | \tau^2, \psi_i^2) \pi(\psi_i^2 | \eta_i) \pi(\eta_i) \right) \pi(\tau^2 | \nu) \pi(\nu) \pi(\phi) \pi(\Sigma) \tag{a.5.15.67}$$

derivations for equation (5.15.46)

Rearrange:

$$\begin{aligned}
& \pi(\tau^2 | y, \xi, \nu, \psi^2, \eta, \phi, \Sigma) \\
& \propto \left(\prod_{i=1}^{n^2} (\tau^2)^{-1/2} \exp \left(-\frac{1}{2} \frac{\xi_i^2}{\tau^2 \psi_i^2} \right) \right) (\tau^2)^{-1/2-1} \exp \left(-\frac{1}{\nu \tau^2} \right) \\
&= (\tau^2)^{-n^2/2} \exp \left(-\frac{1}{2} \sum_{i=1}^{n^2} \frac{\xi_i^2}{\tau^2 \psi_i^2} \right) (\tau^2)^{-1/2-1} \exp \left(-\frac{1}{\nu \tau^2} \right) \\
&= (\tau^2)^{-(n^2+1)/2-1} \exp \left(-\frac{1/2 \sum_{i=1}^{n^2} \xi_i^2 / \psi_i^2 + 1/\nu}{\tau^2} \right)
\end{aligned} \tag{a.5.15.68}$$

Define:

$$a_\tau = \frac{n^2 + 1}{2} \quad b_\tau = \frac{1}{2} \sum_{i=1}^{n^2} \frac{\xi_i^2}{\psi_i^2} + \frac{1}{v} \quad (\text{a.5.15.69})$$

Then this rewrites:

$$\pi(\tau^2|y, \xi, v, \psi^2, \eta, \phi, \Sigma) \propto (\tau^2)^{-a_\tau-1} \exp\left(-\frac{b_\tau}{\tau^2}\right) \quad (\text{a.5.15.70})$$

derivations for equation (5.15.49)

Rearrange:

$$\begin{aligned} & \pi(v|y, \xi, \tau^2, \psi^2, \eta, \phi, \Sigma) \\ & \propto v^{-1/2} \exp\left(-\frac{1}{v\tau^2}\right) v^{-1/2-1} \exp\left(-\frac{1}{v}\right) \\ & = v^{-1-1} \exp\left(-\frac{1/\tau^2 + 1}{v}\right) \end{aligned} \quad (\text{a.5.15.71})$$

Define:

$$a_v = 1 \quad b_v = \frac{1}{\tau^2} + 1 \quad (\text{a.5.15.72})$$

Then this rewrites:

$$\pi(v|y, \xi, \tau^2, \psi^2, \eta, \phi, \Sigma) \propto v^{-a_v-1} \exp\left(-\frac{b_v}{v}\right) \quad (\text{a.5.15.73})$$

derivations for equation (5.15.52)

Rearrange:

$$\begin{aligned} & \pi(\psi_i^2|y, \xi, \tau^2, v, \eta, \phi, \Sigma) \\ & \propto (\psi_i^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau^2 \psi_i^2}\right) (\psi_i^2)^{-1/2-1} \exp\left(-\frac{1}{\eta_i \psi_i^2}\right) \\ & = (\psi_i^2)^{-1-1} \exp\left(-\frac{\xi_i^2/2\tau^2 + 1/\eta_i}{\psi_i^2}\right) \end{aligned} \quad (\text{a.5.15.74})$$

Define:

$$a_\psi = 1 \quad b_\psi = \frac{\xi_i^2}{2\tau^2} + \frac{1}{\eta_i} \quad (\text{a.5.15.75})$$

Then this rewrites:

$$\pi(\psi_i^2|y, \xi, \tau^2, v, \eta, \phi, \Sigma) \propto (\psi_i^2)^{-a_\psi-1} \exp\left(-\frac{b_\psi}{\psi_i^2}\right) \quad (\text{a.5.15.76})$$

derivations for equation (5.15.55)

Rearrange:

$$\begin{aligned}
 & \pi(\eta_i|y, \xi, \tau^2, v, \psi^2, \phi, \Sigma) \\
 & \propto \eta_i^{-1/2} \exp\left(-\frac{1}{\eta_i \psi_i^2}\right) \eta_i^{-1/2-1} \exp\left(-\frac{1}{\eta_i}\right) \\
 & = \eta_i^{-1-1} \exp\left(-\frac{1/\psi_i^2 + 1}{\eta_i}\right)
 \end{aligned} \tag{a.5.15.77}$$

Define:

$$a_\eta = 1 \quad b_\eta = \frac{1}{\psi_i^2} + 1 \tag{a.5.15.78}$$

Then this rewrites:

$$\pi(\psi_i^2|y, \xi, \tau^2, v, \eta, \phi, \Sigma) \propto \eta_i^{-a_\eta-1} \exp\left(-\frac{b_\eta}{\eta_i}\right) \tag{a.5.15.79}$$

derivations for equation (5.15.65)

Start from the joint version of Bayes rule:

$$\pi(\kappa, \lambda, \tau^2, v, \psi^2, \eta, \zeta^2, \omega, \phi, \Sigma|y) \propto f(y|\kappa, \lambda, \phi, \Sigma) \pi(\kappa, \lambda, \tau^2, v, \psi^2, \eta, \zeta^2, \omega, \phi, \Sigma) \tag{a.5.15.80}$$

where the parameters $\tau^2, v, \psi^2, \eta, \zeta^2$ and ω are omitted from the likelihood function as they don't intervene in the likelihood once κ and λ are determined. Consider first the rightmost term, which is the joint prior. Using a standard independence assumption between ϕ, Σ and the other parameters, it rewrites as:

$$\pi(\kappa, \lambda, \tau^2, v, \psi^2, \eta, \zeta^2, \omega, \phi, \Sigma) = \pi(\kappa, \lambda, \tau^2, v, \psi^2, \eta, \zeta^2, \omega) \pi(\phi) \pi(\Sigma) \tag{a.5.15.81}$$

The first term on the right-hand-side of (a.5.15.81) rewrites as:

$$\begin{aligned}
 & \pi(\kappa, \lambda, \tau^2, v, \psi^2, \eta, \zeta^2, \omega) \\
 & = \frac{\pi(\kappa, \lambda, \tau^2, v, \psi^2, \eta, \zeta^2, \omega)}{\pi(\tau^2, v, \psi^2, \eta, \zeta^2, \omega)} \pi(\tau^2, v, \psi^2, \eta, \zeta^2, \omega) \\
 & = \pi(\kappa, \lambda|\tau^2, v, \psi^2, \eta, \zeta^2, \omega) \pi(\tau^2, v, \psi^2, \eta, \zeta^2, \omega)
 \end{aligned} \tag{a.5.15.82}$$

Assume independence between κ and λ , and also between ψ^2, η and ζ^2, ω and τ^2, v :

$$= \pi(\kappa|\tau^2, v, \psi^2, \eta, \zeta^2, \omega) \pi(\lambda|\tau^2, v, \psi^2, \eta, \zeta^2, \omega) \pi(\tau^2, v) \pi(\psi^2, \eta) \pi(\zeta^2, \omega) \tag{a.5.15.83}$$

For the first term, the conditioning on v, η, ζ^2 and ω can be dropped as only τ^2 and ψ^2 are relevant to determine κ . Similarly, for the second term, the conditioning on v, ψ^2, η and ω can be dropped as only τ^2 and η^2 are relevant to determine λ . Hence:

$$= \pi(\kappa|\tau^2, \psi^2) \pi(\lambda|\tau^2, \zeta^2) \pi(\tau^2, v) \pi(\psi^2, \eta) \pi(\zeta^2, \omega) \tag{a.5.15.84}$$

Continue reformulating:

$$\begin{aligned}
 & = \pi(\kappa|\tau^2, \psi^2) \pi(\lambda|\tau^2, \zeta^2) \frac{\pi(\tau^2, v)}{\pi(v)} \pi(v) \frac{\pi(\psi^2, \eta)}{\pi(\eta)} \pi(\eta) \frac{\pi(\zeta^2, \omega)}{\pi(\omega)} \pi(\omega) \\
 & = \pi(\kappa|\tau^2, \psi^2) \pi(\lambda|\tau^2, \zeta^2) \pi(\tau^2|v) \pi(v) \pi(\psi^2|\eta) \pi(\eta) \pi(\zeta^2|\omega) \pi(\omega)
 \end{aligned} \tag{a.5.15.85}$$

Assuming finally independence across the row parameters ($i = 1, \dots, n$), and also independence between the column parameters ($j = 1, \dots, r$), one obtains:

$$= \left(\prod_{i=1}^n \prod_{j=1}^r \pi(\kappa_{ij} | \psi_i^2, \tau_j^2) \pi(\lambda_{ij} | \zeta_i^2, \tau_j^2) \right) \left(\prod_{i=1}^n \pi(\psi_i^2 | \eta_i) \pi(\eta_i) \pi(\zeta_i^2 | \omega_i) \pi(\omega_i) \right) \left(\prod_{j=1}^r \pi(\tau_j^2 | \nu_j) \pi(\nu_j) \right) \quad (\text{a.5.15.86})$$

Substitute back in (a.5.15.81):

$$\pi(\kappa, \lambda, \tau^2, \nu, \psi^2, \eta, \zeta^2, \omega, \phi, \Sigma) = \pi(\phi) \pi(\Sigma) \left(\prod_{i=1}^n \prod_{j=1}^r \pi(\kappa_{ij} | \psi_i^2, \tau_j^2) \pi(\lambda_{ij} | \zeta_i^2, \tau_j^2) \right) \left(\prod_{i=1}^n \pi(\psi_i^2 | \eta_i) \pi(\eta_i) \pi(\zeta_i^2 | \omega_i) \pi(\omega_i) \right) \left(\prod_{j=1}^r \pi(\tau_j^2 | \nu_j) \pi(\nu_j) \right) \quad (\text{a.5.15.87})$$

Substitute back (a.5.15.87) in (a.5.15.80):

$$\pi(\kappa, \lambda, \tau^2, \nu, \psi^2, \eta, \zeta^2, \omega, \phi, \Sigma | y) \propto f(y | \kappa, \lambda, \phi, \Sigma) \pi(\phi) \pi(\Sigma) \left(\prod_{i=1}^n \prod_{j=1}^r \pi(\kappa_{ij} | \psi_i^2, \tau_j^2) \pi(\lambda_{ij} | \zeta_i^2, \tau_j^2) \right) \left(\prod_{i=1}^n \pi(\psi_i^2 | \eta_i) \pi(\eta_i) \pi(\zeta_i^2 | \omega_i) \pi(\omega_i) \right) \left(\prod_{j=1}^r \pi(\tau_j^2 | \nu_j) \pi(\nu_j) \right) \quad (\text{a.5.15.88})$$

derivations for equation (5.15.67)

Rearrange:

$$\begin{aligned} & \pi(\tau_j^2 | y, \kappa, \lambda, \nu, \psi^2, \eta, \zeta^2, \omega, \phi, \Sigma) \\ & \propto \left(\prod_{i=1}^n (\tau_j^2)^{-1/2} \exp \left(-\frac{1}{2} \frac{\kappa_{ij}^2}{\psi_i^2 \tau_j^2} \right) (\tau_j^2)^{-1/2} \exp \left(-\frac{1}{2} \frac{\lambda_{ij}^2}{\zeta_i^2 \tau_j^2} \right) \right) (\tau_j^2)^{-1/2-1} \exp \left(-\frac{1}{\nu_j \tau_j^2} \right) \\ & = (\tau_j^2)^{-n/2} \exp \left(-\frac{1}{2} \sum_{i=1}^n \frac{\kappa_{ij}^2}{\psi_i^2 \tau_j^2} \right) (\tau_j^2)^{-n/2} \exp \left(-\frac{1}{2} \sum_{i=1}^n \frac{\lambda_{ij}^2}{\zeta_i^2 \tau_j^2} \right) (\tau_j^2)^{-1/2-1} \exp \left(-\frac{1}{\nu_j \tau_j^2} \right) \\ & = (\tau_j^2)^{-n} \exp \left(-\frac{1}{2} \sum_{i=1}^n \left[\frac{\kappa_{ij}^2}{\psi_i^2 \tau_j^2} + \frac{\lambda_{ij}^2}{\zeta_i^2 \tau_j^2} \right] \right) (\tau_j^2)^{-1/2-1} \exp \left(-\frac{1}{\nu_j \tau_j^2} \right) \\ & = (\tau_j^2)^{-(n+1/2)-1} \exp \left(-\frac{1/2 \sum_{i=1}^n (\kappa_{ij}^2 / \psi_i^2 + \lambda_{ij}^2 / \zeta_i^2) + 1/\nu_j}{\tau_j^2} \right) \end{aligned} \quad (\text{a.5.15.89})$$

Define:

$$a_\tau = n + \frac{1}{2} \quad b_\tau = \frac{1}{2} \sum_{i=1}^n \left(\frac{\kappa_{ij}^2}{\psi_i^2} + \frac{\lambda_{ij}^2}{\zeta_i^2} \right) + \frac{1}{\nu_j} \quad (\text{a.5.15.90})$$

Then this rewrites:

$$\pi(\tau_j^2 | y, \kappa, \lambda, \nu, \psi^2, \eta, \zeta^2, \omega, \phi, \Sigma) \propto (\tau_j^2)^{-a_\tau-1} \exp \left(-\frac{b_\tau}{\tau_j^2} \right) \quad (\text{a.5.15.91})$$

derivations for equation (5.15.70)

Rearrange:

$$\begin{aligned}
& \pi(v_j|y, \kappa, \lambda, \tau^2, \psi^2, \eta, \zeta^2, \omega, \phi, \Sigma) \\
& \propto v_j^{-1/2} \exp\left(-\frac{1}{v_j \tau_j^2}\right) v_j^{-1/2-1} \exp\left(-\frac{1}{v_j}\right) \\
& = v_j^{-1-1} \exp\left(-\frac{1/\tau_j^2 + 1}{v_j}\right)
\end{aligned} \tag{a.5.15.92}$$

Define:

$$a_v = 1 \quad b_v = \frac{1}{\tau_j^2} + 1 \tag{a.5.15.93}$$

Then this rewrites:

$$\pi(v_j|y, \kappa, \lambda, \tau^2, \psi^2, \eta, \zeta^2, \omega, \phi, \Sigma) \propto v_j^{-a_v-1} \exp\left(-\frac{b_v}{v_j}\right) \tag{a.5.15.94}$$

derivations for equation (5.15.73)

Rearrange:

$$\begin{aligned}
& \pi(\psi_i^2|y, \kappa, \lambda, \tau^2, v, \eta, \zeta^2, \omega, \phi, \Sigma) \\
& \propto \left(\prod_{j=1}^r (\psi_i^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\kappa_{ij}^2}{\psi_i^2 \tau_j^2}\right) \right) (\psi_i^2)^{-1/2-1} \exp\left(-\frac{1}{\eta_i \psi_i^2}\right) \\
& = (\psi_i^2)^{-r/2} \exp\left(-\frac{1}{2} \sum_{j=1}^r \frac{\kappa_{ij}^2}{\psi_i^2 \tau_j^2}\right) (\psi_i^2)^{-1/2-1} \exp\left(-\frac{1}{\eta_i \psi_i^2}\right) \\
& = (\psi_i^2)^{-(r+1)/2-1} \exp\left(-\frac{1/2 \sum_{j=1}^r \kappa_{ij}^2 / \tau_j^2 + 1/\eta_i}{\psi_i^2}\right)
\end{aligned} \tag{a.5.15.95}$$

Define:

$$a_\psi = \frac{r+1}{2} \quad b_\psi = \frac{1}{2} \sum_{j=1}^r \frac{\kappa_{ij}^2}{\tau_j^2} + \frac{1}{\eta_i} \tag{a.5.15.96}$$

Then this rewrites:

$$\pi(\psi_i^2|y, \kappa, \lambda, \tau^2, v, \eta, \zeta^2, \omega, \phi, \Sigma) \propto (\psi_i^2)^{-a_\psi-1} \exp\left(-\frac{b_\psi}{\psi_i^2}\right) \tag{a.5.15.97}$$

derivations for equation (5.15.76)

Rearrange:

$$\begin{aligned}
& \pi(\eta_i|y, \kappa, \lambda, \tau^2, v, \psi^2, \zeta^2, \omega, \phi, \Sigma) \\
& \propto \eta_i^{-1/2} \exp\left(-\frac{1}{\eta_i \psi_i^2}\right) \eta_i^{-1/2-1} \exp\left(-\frac{1}{\eta_i}\right) \\
& = \eta_i^{-1-1} \exp\left(-\frac{1/\psi_i^2 + 1}{\eta_i}\right)
\end{aligned} \tag{a.5.15.98}$$

Define:

$$a_\eta = 1 \quad b_\eta = \frac{1}{\psi_i^2} + 1 \quad (\text{a.5.15.99})$$

Then this rewrites:

$$\pi(\eta_i|y, \kappa, \lambda, \tau^2, \nu, \psi^2, \zeta^2, \omega, \phi, \Sigma) \propto \eta_i^{-a_\eta-1} \exp\left(-\frac{b_\eta}{\eta_i}\right) \quad (\text{a.5.15.100})$$

derivations for equation (5.15.79)

Rearrange:

$$\begin{aligned} & \pi(\zeta_i^2|y, \kappa, \lambda, \tau^2, \nu, \psi^2, \eta, \omega, \phi, \Sigma) \\ & \propto \left(\prod_{j=1}^r (\zeta_i^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\lambda_{ij}^2}{\zeta_i^2 \tau_j^2}\right) \right) (\zeta_i^2)^{-1/2-1} \exp\left(-\frac{1}{\omega_i \zeta_i^2}\right) \\ & = (\zeta_i^2)^{-r/2} \exp\left(-\frac{1}{2} \sum_{j=1}^r \frac{\lambda_{ij}^2}{\zeta_i^2 \tau_j^2}\right) (\zeta_i^2)^{-1/2-1} \exp\left(-\frac{1}{\omega_i \zeta_i^2}\right) \\ & = (\zeta_i^2)^{-(r+1)/2-1} \exp\left(-\frac{1/2 \sum_{j=1}^r \lambda_{ij}^2 / \tau_j^2 + 1/\omega_i}{\zeta_i^2}\right) \end{aligned} \quad (\text{a.5.15.101})$$

Define:

$$a_\zeta = \frac{r+1}{2} \quad b_\zeta = \frac{1}{2} \sum_{j=1}^r \frac{\lambda_{ij}^2}{\tau_j^2} + \frac{1}{\omega_i} \quad (\text{a.5.15.102})$$

Then this rewrites:

$$\pi(\zeta_i^2|y, \kappa, \lambda, \tau^2, \nu, \psi^2, \eta, \omega, \phi, \Sigma) \propto (\zeta_i^2)^{-a_\zeta-1} \exp\left(-\frac{b_\zeta}{\zeta_i^2}\right) \quad (\text{a.5.15.103})$$

derivations for equation (5.15.82)

Rearrange:

$$\begin{aligned} & \pi(\omega_i|y, \kappa, \lambda, \tau^2, \nu, \psi^2, \eta, \zeta^2, \phi, \Sigma) \\ & \propto \omega_i^{-1/2} \exp\left(-\frac{1}{\omega_i \zeta_i^2}\right) \omega_i^{-1/2-1} \exp\left(-\frac{1}{\omega_i}\right) \\ & = \omega_i^{-1-1} \exp\left(-\frac{1/\zeta_i^2 + 1}{\omega_i}\right) \end{aligned} \quad (\text{a.5.15.104})$$

Define:

$$a_\omega = 1 \quad b_\omega = \frac{1}{\zeta_i^2} + 1 \quad (\text{a.5.15.105})$$

Then this rewrites:

$$\pi(\omega_i|y, \kappa, \lambda, \tau^2, \nu, \psi^2, \eta, \zeta^2, \phi, \Sigma) \propto \omega_i^{-a_\omega-1} \exp\left(-\frac{b_\omega}{\omega_i}\right) \quad (\text{a.5.15.106})$$

derivations for equation (5.15.86)

Bayes rule in its raw form is given by:

$$\pi(\xi, \delta, \phi, \Sigma|y) \propto f(y|\xi, \phi, \Sigma) \pi(\xi, \delta, \phi, \Sigma) \quad (\text{a.5.15.107})$$

where the parameters δ are omitted from the likelihood function as they don't intervene in the likelihood once ξ is determined. Consider first the rightmost term, which is the joint prior. Using a standard independence assumption between ϕ , Σ and the other parameters, it rewrites as:

$$\pi(\xi, \delta, \phi, \Sigma) = \pi(\xi, \delta) \pi(\phi) \pi(\Sigma) \quad (\text{a.5.15.108})$$

The first term on the right-hand-side of (a.5.15.62) rewrites as:

$$\pi(\xi, \delta) = \frac{\pi(\xi, \delta)}{\pi(\delta)} \pi(\delta) = \pi(\xi|\delta) \pi(\delta) \quad (\text{a.5.15.109})$$

Assuming independence between the different ξ_i and δ_i :

$$\pi(\xi, \delta) = \pi(\xi|\delta) \pi(\delta) = \left(\prod_{i=1}^{n^2} \pi(\xi_i|\delta_i) \pi(\delta_i) \right) \quad (\text{a.5.15.110})$$

Replacing (a.5.15.110) back in (a.5.15.108):

$$\pi(\xi, \delta, \phi, \Sigma) = \left(\prod_{i=1}^{n^2} \pi(\xi_i|\delta_i) \pi(\delta_i) \right) \pi(\phi) \pi(\Sigma) \quad (\text{a.5.15.111})$$

Eventually replacing (a.5.15.111) in (a.5.15.107):

$$\pi(\xi, \delta, \phi, \Sigma|y) \propto f(y|\xi, \phi, \Sigma) \left(\prod_{i=1}^{n^2} \pi(\xi_i|\delta_i) \pi(\delta_i) \right) \pi(\phi) \pi(\Sigma) \quad (\text{a.5.15.112})$$

derivations for equation (5.15.88)

The joint conditional posterior is given by:

$$\pi(\delta_i|y, \xi, \phi, \Sigma) \propto \left[(2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_1^2}\right) \right]^{\delta_i} \left[(2\pi\tau_2^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_2^2}\right) \right]^{(1-\delta_i)} \mu^{\delta_i} (1-\mu)^{(1-\delta_i)} \quad (\text{a.5.15.113})$$

Since δ_i is a binary variable, it can only take values 0 or 1. Consider the case $\delta_i = 1$. In this case, the posterior becomes:

$$\pi(\delta_i|y, \xi, \phi, \Sigma) \propto (2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_1^2}\right) \mu \quad (\text{a.5.15.114})$$

In the case $\delta_i = 0$, the posterior becomes:

$$\pi(\delta_i|y, \xi, \phi, \Sigma) \propto (2\pi\tau_2^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_2^2}\right) (1-\mu) \quad (\text{a.5.15.115})$$

Because δ_i is binary, its posterior is Bernoulli as well. The sum of the probabilities for $\delta_i = 0$ and $\delta_i = 1$ must therefore be 1. This obtains by normalizing the kernels by the sum of their values. Thus for $\delta_i = 1$:

$$\begin{aligned}
& \pi(\delta_i|y, \xi, \phi, \Sigma) \\
&= \frac{(2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_1^2}\right) \mu}{(2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_1^2}\right) \mu + (2\pi\tau_2^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_2^2}\right) (1-\mu)} \\
&= \frac{\tau_1^{-1} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_1^2}\right) \mu}{\tau_1^{-1} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_1^2}\right) \mu + \tau_2^{-1} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_2^2}\right) (1-\mu)} \\
&= \frac{\frac{\mu}{\tau_1} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_1^2}\right)}{\frac{\mu}{\tau_1} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_1^2}\right) + \frac{1-\mu}{\tau_2} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_2^2}\right)} \tag{a.5.15.116}
\end{aligned}$$

Define:

$$\mu_\xi = \frac{a_1}{a_1 + a_2} \quad a_1 = \frac{\mu}{\tau_1} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_1^2}\right) \quad a_2 = \frac{1-\mu}{\tau_2} \exp\left(-\frac{1}{2} \frac{\xi_i^2}{\tau_2^2}\right) \tag{a.5.15.117}$$

Hence the posterior is Bernoulli with a probability of success μ_ξ , which writes as:

$$\pi(\delta_i|y, \xi, \phi, \Sigma) = \mu_\xi^{\delta_i} (1 - \mu_\xi)^{(1-\delta_i)} \tag{a.5.15.118}$$

derivations for equation (5.15.94)

Bayes rule in raw form is given by:

$$\pi(\kappa, \chi, \lambda, \gamma, \phi, \Sigma|y) \propto f(y|\kappa, \chi, \phi, \Sigma) \pi(\kappa, \chi, \lambda, \gamma, \phi, \Sigma) \tag{a.5.15.119}$$

A standard independence assumption between ϕ, Σ and the groups made of κ and χ , then λ and γ yields:

$$\pi(\kappa, \chi, \lambda, \gamma, \phi, \Sigma|y) \propto f(y|\kappa, \chi, \phi, \Sigma) \pi(\kappa, \chi) \pi(\lambda, \gamma) \pi(\phi) \pi(\Sigma) \tag{a.5.15.120}$$

Rearranging:

$$\begin{aligned}
& \pi(\kappa, \chi, \lambda, \gamma, \phi, \Sigma|y) \\
& \propto f(y|\kappa, \chi, \phi, \Sigma) \pi(\kappa, \chi) \pi(\lambda, \gamma) \pi(\phi) \pi(\Sigma) \\
& = f(y|\kappa, \chi, \phi, \Sigma) \frac{\pi(\kappa, \chi)}{\pi(\chi)} \pi(\chi) \frac{\pi(\lambda, \gamma)}{\pi(\gamma)} \pi(\gamma) \pi(\phi) \pi(\Sigma) \\
& = f(y|\kappa, \chi, \phi, \Sigma) \pi(\kappa|\chi) \pi(\chi) \pi(\lambda|\gamma) \pi(\gamma) \pi(\phi) \pi(\Sigma) \tag{a.5.15.121}
\end{aligned}$$

Finally, assume independence between the different $\kappa_i, \chi_i, \lambda_i$ and γ_i to obtain:

$$\pi(\kappa, \chi, \lambda, \gamma, \phi, \Sigma|y) \propto f(y|\kappa, \chi, \phi, \Sigma) \left(\prod_{i=1}^{nr} \pi(\kappa_i|\chi_i) \pi(\chi_i) \pi(\lambda_i|\gamma_i) \pi(\gamma_i) \right) \pi(\phi) \pi(\Sigma) \tag{a.5.15.122}$$

derivations for equation (5.15.96)

The joint conditional posterior is given by:

$$\pi(\chi_i|y, \kappa, \lambda, \gamma, \phi, \Sigma) \propto \left[(2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_1^2}\right) \right]^{\chi_i} \left[(2\pi\tau_2^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_2^2}\right) \right]^{(1-\chi_i)} \mu^{\chi_i} (1-\mu)^{(1-\chi_i)} \tag{a.5.15.123}$$

Since χ_i is a binary variable, it can only take values 0 or 1. Consider the case $\chi_i = 1$. In this case, the posterior becomes:

$$\pi(\chi_i|y, \kappa, \lambda, \gamma, \phi, \Sigma) \propto (2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_1^2}\right) \mu \quad (\text{a.5.15.124})$$

In the case $\chi_i = 0$, the posterior becomes:

$$\pi(\chi_i|y, \kappa, \lambda, \gamma, \phi, \Sigma) \propto (2\pi\tau_2^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_2^2}\right) (1 - \mu) \quad (\text{a.5.15.125})$$

Because χ_i is binary, its posterior is Bernoulli as well. The sum of the probabilities for $\chi_i = 0$ and $\chi_i = 1$ must therefore be 1. This obtains by normalizing the kernels by the sum of their values. Thus for $\chi_i = 1$:

$$\begin{aligned} & \pi(\chi_i|y, \kappa, \lambda, \gamma, \phi, \Sigma) \\ &= \frac{(2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_1^2}\right) \mu}{(2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_1^2}\right) \mu + (2\pi\tau_2^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_2^2}\right) (1 - \mu)} \\ &= \frac{\tau_1^{-1} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_1^2}\right) \mu}{\tau_1^{-1} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_1^2}\right) \mu + \tau_2^{-1} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_2^2}\right) (1 - \mu)} \\ &= \frac{\frac{\mu}{\tau_1} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_1^2}\right)}{\frac{\mu}{\tau_1} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_1^2}\right) + \frac{1-\mu}{\tau_2} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_2^2}\right)} \end{aligned} \quad (\text{a.5.15.126})$$

Define:

$$\mu_\kappa = \frac{a_1}{a_1 + a_2} \quad a_1 = \frac{\mu}{\tau_1} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_1^2}\right) \quad a_2 = \frac{1-\mu}{\tau_2} \exp\left(-\frac{1}{2} \frac{\kappa_i^2}{\tau_2^2}\right) \quad (\text{a.5.15.127})$$

Hence the posterior is Bernoulli with a probability of success μ_κ , which writes as:

$$\pi(\chi_i|y, \kappa, \lambda, \gamma, \phi, \Sigma) = \mu_\kappa^{\chi_i} (1 - \mu_\kappa)^{(1-\chi_i)} \quad (\text{a.5.15.128})$$

derivations for equation (5.15.99)

The joint conditional posterior is given by:

$$\pi(\gamma|y, \kappa, \chi, \lambda, \phi, \Sigma) \propto \left[(2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_1^2}\right) \right]^{\gamma_i} \left[(2\pi\tau_2^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_2^2}\right) \right]^{(1-\gamma_i)} \mu^{\gamma_i} (1 - \mu)^{(1-\gamma_i)} \quad (\text{a.5.15.129})$$

Since γ_i is a binary variable, it can only take values 0 or 1. Consider the case $\gamma_i = 1$. In this case, the posterior becomes:

$$\pi(\gamma_i|y, \kappa, \chi, \lambda, \phi, \Sigma) \propto (2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_1^2}\right) \mu \quad (\text{a.5.15.130})$$

In the case $\gamma_i = 0$, the posterior becomes:

$$\pi(\gamma_i|y, \kappa, \chi, \lambda, \phi, \Sigma) \propto (2\pi\tau_2^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_2^2}\right) (1 - \mu) \quad (\text{a.5.15.131})$$

Because γ_i is binary, its posterior is Bernoulli as well. The sum of the probabilities for $\gamma_i = 0$ and $\gamma_i = 1$ must therefore be 1. This obtains by normalizing the kernels by the sum of their values. Thus for $\gamma_i = 1$:

$$\begin{aligned}
& \pi(\gamma_i|y, \kappa, \chi, \lambda, \phi, \Sigma) \\
&= \frac{(2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_1^2}\right) \mu}{(2\pi\tau_1^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_1^2}\right) \mu + (2\pi\tau_2^2)^{-1/2} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_2^2}\right) (1-\mu)} \\
&= \frac{\tau_1^{-1} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_1^2}\right) \mu}{\tau_1^{-1} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_1^2}\right) \mu + \tau_2^{-1} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_2^2}\right) (1-\mu)} \\
&= \frac{\frac{\mu}{\tau_1} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_1^2}\right)}{\frac{\mu}{\tau_1} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_1^2}\right) + \frac{1-\mu}{\tau_2} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_2^2}\right)} \tag{a.5.15.132}
\end{aligned}$$

Define:

$$\mu_\lambda = \frac{a_1}{a_1 + a_2} \quad a_1 = \frac{\mu}{\tau_1} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_1^2}\right) \quad a_2 = \frac{1-\mu}{\tau_2} \exp\left(-\frac{1}{2} \frac{\lambda_i^2}{\tau_2^2}\right) \tag{a.5.15.133}$$

Hence the posterior is Bernoulli with a probability of success μ_κ , which writes as:

$$\pi(\gamma_i|y, \kappa, \chi, \lambda, \phi, \Sigma) = \mu_\lambda^{\gamma_i} (1 - \mu_\lambda)^{(1-\gamma_i)} \tag{a.5.15.134}$$

Vector autoregressive moving average

derivations for equation (5.16.16)

For convenience, define first:

$$\tilde{y} = y - \bar{Z}\kappa \quad (\text{a.5.16.1})$$

Then $\pi(\beta|y, Z, \kappa, \Sigma)$ rewrites:

$$\pi(\beta|y, \kappa, Z, \Sigma) \propto \exp\left(-\frac{1}{2}(\tilde{y} - \bar{X}\beta)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{X}\beta)\right) \times \exp\left(-\frac{1}{2}(\beta - b)' V^{-1}(\beta - b)\right) \quad (\text{a.5.16.2})$$

Group terms:

$$\begin{aligned} & \pi(\beta|y, \kappa, Z, \Sigma) \\ & \propto \exp\left(-\frac{1}{2}(\tilde{y} - \bar{X}\beta)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{X}\beta)\right) \times \exp\left(-\frac{1}{2}(\beta - b)' V^{-1}(\beta - b)\right) \\ & = \exp\left(-\frac{1}{2}[(\tilde{y} - \bar{X}\beta)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{X}\beta) + (\beta - b)' V^{-1}(\beta - b)]\right) \end{aligned} \quad (\text{a.5.16.3})$$

Consider the terms in square brackets:

$$\begin{aligned} & (\tilde{y} - \bar{X}\beta)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{X}\beta) + (\beta - b)' V^{-1}(\beta - b) \\ & = \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} + \beta' \bar{X}' \bar{\Sigma}^{-1} \bar{X} \beta - 2\beta' \bar{X}' \bar{\Sigma}^{-1} \tilde{y} + \beta' V^{-1} \beta + b' V^{-1} b - 2\beta' V^{-1} b \\ & = \beta' (V^{-1} + \bar{X}' \bar{\Sigma}^{-1} \bar{X}) \beta - 2\beta' (V^{-1} b + \bar{X}' \bar{\Sigma}^{-1} \tilde{y}) + b' V^{-1} b + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} \end{aligned} \quad (\text{a.5.16.4})$$

Complete the squares:

$$= \beta' (V^{-1} + \bar{X}' \bar{\Sigma}^{-1} \bar{X}) \beta - 2\beta' \bar{V}^{-1} \bar{V} (V^{-1} b + \bar{X}' \bar{\Sigma}^{-1} \tilde{y}) + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b} + b' V^{-1} b + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} \quad (\text{a.5.16.5})$$

Define:

$$\bar{V} = (V^{-1} + \bar{X}' \bar{\Sigma}^{-1} \bar{X})^{-1} \quad \bar{b} = \bar{V} (V^{-1} b + \bar{X}' \bar{\Sigma}^{-1} \tilde{y}) \quad (\text{a.5.16.6})$$

Then (a.5.16.5) rewrites:

$$\begin{aligned} &= \beta' \bar{V}^{-1} \beta - 2\beta' \bar{V}^{-1} \bar{b} + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b} + b' V^{-1} b + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} \\ &= (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + (b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y}) \end{aligned} \quad (\text{a.5.16.7})$$

Substitute (a.5.16.7) back in (a.5.16.3):

$$\begin{aligned} &\pi(\beta|y, \kappa, Z, \Sigma) \\ &= \exp \left(-\frac{1}{2} [(\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) + (b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y})] \right) \\ &= \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right) \exp \left(-\frac{1}{2} (b' V^{-1} b - \bar{b}' \bar{V}^{-1} \bar{b} + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y}) \right) \\ &\propto \exp \left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) \right) \end{aligned} \quad (\text{a.5.16.8})$$

Where the last line obtains by noting that the second term in row 2 does not involve β and can hence be relegated to the normalization constant.

The terms in (a.5.16.6) simplify. Note first that:

$$\begin{aligned} &\bar{X}' \bar{\Sigma}^{-1} \bar{X} \\ &= (I_n \otimes X)' (\Sigma \otimes I_T)^{-1} (I_n \otimes X) \\ &= (I_n \otimes X') (\Sigma^{-1} \otimes I_T) (I_n \otimes X) \\ &= (\Sigma^{-1} \otimes X') (I_n \otimes X) \\ &= \Sigma^{-1} \otimes X' X \end{aligned} \quad (\text{a.5.16.9})$$

Similarly:

$$\begin{aligned} &\bar{X}' \bar{\Sigma}^{-1} \tilde{y} \\ &= \bar{X}' \bar{\Sigma}^{-1} (y - \bar{Z} \kappa) \\ &= (I_n \otimes X)' (\Sigma \otimes I_T)^{-1} \text{vec}(Y - ZK) \\ &= (I_n \otimes X') (\Sigma^{-1} \otimes I_T) \text{vec}(Y - ZK) \\ &= (\Sigma^{-1} \otimes X') \text{vec}(Y - ZK) \\ &= \text{vec}(X' [Y - ZK] \Sigma^{-1}) \end{aligned} \quad (\text{a.5.16.10})$$

Then (a.5.16.6) rewrites:

$$\bar{V} = (V^{-1} + \Sigma^{-1} \otimes X' X)^{-1} \quad \bar{b} = \bar{V} (V^{-1} b + \text{vec}(X' [Y - ZK] \Sigma^{-1})) \quad (\text{a.5.16.11})$$

derivations for equation (5.16.19)

For convenience, define first:

$$\tilde{y} = y - \bar{X} \beta \quad (\text{a.5.16.12})$$

Then $\pi(\kappa|y, \beta, Z, \Sigma)$ rewrites:

$$\pi(\kappa|y, \beta, Z, \Sigma) \propto \exp \left(-\frac{1}{2} (\tilde{y} - \bar{Z} \kappa)' \bar{\Sigma}^{-1} (\tilde{y} - \bar{Z} \kappa) \right) \exp \left(-\frac{1}{2} (\kappa - g)' W^{-1} (\kappa - g) \right) \quad (\text{a.5.16.13})$$

Group terms:

$$\begin{aligned}
& \pi(\kappa|y, \beta, Z, \Sigma) \\
& \propto \exp\left(-\frac{1}{2}(\tilde{y} - \bar{Z}\kappa)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{Z}\kappa)\right) \times \exp\left(-\frac{1}{2}(\kappa - g)' W^{-1}(\kappa - g)\right) \\
& = \exp\left(-\frac{1}{2}[(\tilde{y} - \bar{Z}\kappa)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{Z}\kappa) + (\kappa - g)' W^{-1}(\kappa - g)]\right)
\end{aligned} \tag{a.5.16.14}$$

Consider the terms in square brackets:

$$\begin{aligned}
& (\tilde{y} - \bar{Z}\kappa)' \bar{\Sigma}^{-1}(\tilde{y} - \bar{Z}\kappa) + (\kappa - g)' W^{-1}(\kappa - g) \\
& = \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} + \kappa' \bar{Z}' \bar{\Sigma}^{-1} \bar{Z} \kappa - 2\kappa' \bar{Z}' \bar{\Sigma}^{-1} \tilde{y} + \kappa' W^{-1} \kappa + g' W^{-1} g - 2\kappa' W^{-1} g \\
& = \kappa'(W^{-1} + \bar{Z}' \bar{\Sigma}^{-1} \bar{Z}) \kappa - 2\kappa'(W^{-1} g + \bar{Z}' \bar{\Sigma}^{-1} \tilde{y}) + g' W^{-1} g + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y}
\end{aligned} \tag{a.5.16.15}$$

Complete the squares:

$$= \kappa'(W^{-1} + \bar{Z}' \bar{\Sigma}^{-1} \bar{Z}) \kappa - 2\kappa' \bar{W}^{-1} \bar{W} (W^{-1} g + \bar{Z}' \bar{\Sigma}^{-1} \tilde{y}) + \bar{g}' \bar{W}^{-1} \bar{g} - \bar{g}' \bar{W}^{-1} \bar{g} + g' W^{-1} g + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} \tag{a.5.16.16}$$

Define:

$$\bar{W} = (W^{-1} + \bar{Z}' \bar{\Sigma}^{-1} \bar{Z})^{-1} \quad \bar{g} = \bar{W} (W^{-1} g + \bar{Z}' \bar{\Sigma}^{-1} \tilde{y}) \tag{a.5.16.17}$$

Then (a.5.16.16) rewrites:

$$\begin{aligned}
& = \kappa' \bar{W}^{-1} \kappa - 2\kappa' \bar{W}^{-1} \bar{g} + \bar{g}' \bar{W}^{-1} \bar{g} - \bar{g}' \bar{W}^{-1} \bar{g} + g' W^{-1} g + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y} \\
& = (\kappa - \bar{g})' \bar{W}^{-1} (\kappa - \bar{g}) + (g' W^{-1} g - \bar{g}' \bar{W}^{-1} \bar{g} + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y})
\end{aligned} \tag{a.5.16.18}$$

Substitute (a.5.16.18) back in (a.5.16.14):

$$\begin{aligned}
& \pi(\kappa|y, \beta, Z, \Sigma) \\
& = \exp\left(-\frac{1}{2}[(\kappa - \bar{g})' \bar{W}^{-1} (\kappa - \bar{g}) + (g' W^{-1} g - \bar{g}' \bar{W}^{-1} \bar{g} + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y})]\right) \\
& = \exp\left(-\frac{1}{2}(\kappa - \bar{g})' \bar{W}^{-1} (\kappa - \bar{g})\right) \exp\left(-\frac{1}{2}(g' W^{-1} g - \bar{g}' \bar{W}^{-1} \bar{g} + \tilde{y}' \bar{\Sigma}^{-1} \tilde{y})\right) \\
& \propto \exp\left(-\frac{1}{2}(\kappa - \bar{g})' \bar{W}^{-1} (\kappa - \bar{g})\right)
\end{aligned} \tag{a.5.16.19}$$

Where the last line obtains by noting that the second term in row 2 does not involve κ and can hence be relegated to the normalization constant.

The terms in (a.5.16.17) simplify. Note first that:

$$\begin{aligned}
& \bar{Z}' \bar{\Sigma}^{-1} \bar{Z} \\
& = (I_n \otimes Z)' (\Sigma \otimes I_T)^{-1} (I_n \otimes Z) \\
& = (I_n \otimes Z') (\Sigma^{-1} \otimes I_T) (I_n \otimes Z) \\
& = (\Sigma^{-1} \otimes Z') (I_n \otimes Z) \\
& = \Sigma^{-1} \otimes Z' Z
\end{aligned} \tag{a.5.16.20}$$

Similarly:

$$\begin{aligned}
& \bar{Z}'\bar{\Sigma}^{-1}\bar{y} \\
&= \bar{Z}'\bar{\Sigma}^{-1}(y - \bar{X}\beta) \\
&= (I_n \otimes Z)'(\Sigma \otimes I_T)^{-1} \text{vec}(Y - XB) \\
&= (I_n \otimes Z')(\Sigma^{-1} \otimes I_T) \text{vec}(Y - XB) \\
&= (\Sigma^{-1} \otimes Z') \text{vec}(Y - XB) \\
&= \text{vec}(Z' [Y - XB] \Sigma^{-1})
\end{aligned} \tag{a.5.16.21}$$

Then (a.5.16.17) rewrites:

$$\bar{W} = (W^{-1} + \Sigma^{-1} \otimes Z'Z)^{-1} \quad \bar{g} = \bar{W}(W^{-1}g + \text{vec}(Z' [Y - XB] \Sigma^{-1})) \tag{a.5.16.22}$$

derivations for equation (5.16.22)

Note that:

$$|\bar{\Sigma}|^{-1/2} = |\Sigma \otimes I_T|^{-1/2} = |\Sigma|^{-T/2} |I_T|^{-n/2} = |\Sigma|^{-T/2} \tag{a.5.16.23}$$

Also:

$$\begin{aligned}
& (y - \bar{X}\beta - \bar{Z}\kappa)' \bar{\Sigma}^{-1} (y - \bar{X}\beta - \bar{Z}\kappa) \\
&= (y - (I_n \otimes X)\beta - (I_n \otimes Z)\kappa)' (\Sigma \otimes I_T)^{-1} (y - (I_n \otimes X)\beta - (I_n \otimes Z)\kappa) \\
&= \text{tr} \{ \Sigma^{-1} (Y - XB - ZK)' (Y - XB - ZK) \}
\end{aligned} \tag{a.5.16.24}$$

Then substituting in the original expression:

$$\begin{aligned}
& \pi(\Sigma|y, \beta, \kappa, Z) \\
&\propto |\bar{\Sigma}|^{-1/2} \exp \left(-\frac{1}{2} (y - \bar{X}\beta - \bar{Z}\kappa)' \bar{\Sigma}^{-1} (y - \bar{X}\beta - \bar{Z}\kappa) \right) \\
&\times |\Sigma|^{-(\alpha+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} S \} \right) \\
&= |\Sigma|^{-T/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} (Y - XB - ZK)' (Y - XB - ZK) \} \right) \times |\Sigma|^{-(\alpha+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} S \} \right) \\
&= |\Sigma|^{-(\alpha+T+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} [S + (Y - XB - ZK)' (Y - XB - ZK)] \} \right) \\
&= |\Sigma|^{-(\bar{\alpha}+n+1)/2} \exp \left(-\frac{1}{2} \text{tr} \{ \Sigma^{-1} \bar{S} \} \right)
\end{aligned} \tag{a.5.16.25}$$

with:

$$\bar{\alpha} = \alpha + T \quad \bar{S} = S + (Y - XB - ZK)' (Y - XB - ZK) \tag{a.5.16.26}$$

PART VI

Nowcasting

Mixed frequency vector autoregression

Dynamic factor model

derivations for equation (6.18.18)

Start from Bayes rule and use an independence assumption between λ, f , and ε :

$$\begin{aligned}
 & \pi(\lambda, f, \beta, \varepsilon, \gamma|x) \\
 & \propto f(x|\lambda, f, \varepsilon) \pi(\lambda, f, \beta, \varepsilon, \gamma) \\
 & = f(x|\lambda, f, \varepsilon) \pi(\lambda) \pi(f, \beta) \pi(\varepsilon, \gamma) \\
 & = f(x|\lambda, f, \varepsilon) \pi(\lambda) \frac{\pi(f, \beta)}{\pi(\beta)} \pi(\beta) \frac{\pi(\varepsilon, \gamma)}{\pi(\gamma)} \pi(\gamma) \\
 & = f(x|\lambda, f, \varepsilon) \pi(\lambda) \pi(f|\beta) \pi(\beta) \pi(\varepsilon|\gamma) \pi(\gamma)
 \end{aligned} \tag{a.6.18.1}$$

Assume independence between the $\lambda_i, \beta_j, \varepsilon_i$:

$$\pi(\lambda, f, \beta, \varepsilon, \gamma|x) \propto f(x|\lambda, f, \varepsilon) \left(\prod_{i=1}^n \pi(\lambda_i) \right) \left(\prod_{j=1}^m \pi(f_j|\beta_j) \right) \left(\prod_{j=1}^m \pi(\beta_j) \right) \left(\prod_{i=1}^n \pi(\varepsilon_i|\gamma_i) \right) \left(\prod_{i=1}^n \pi(\gamma_i) \right) \tag{a.6.18.2}$$

derivations for equation (6.18.19)

Combine the two expressions to obtain:

$$x_i = W_i \lambda_i + E_i \gamma_i + \varepsilon_i \quad \varepsilon_i \sim N(0, \sigma_i^2 I_{T_i}) \tag{a.6.18.3}$$

Following, the likelihood for x_i is given by:

$$f(x_i|\lambda, f, \varepsilon) = (2\pi)^{-T_i/2} \exp \left(-\frac{1}{2} (x_i - W_i \lambda_i - E_i \gamma_i)' (x_i - W_i \lambda_i - E_i \gamma_i) / \sigma_i^2 \right) \tag{a.6.18.4}$$

Then, using an independence assumption between the different x_i , the full likelihood rewrites as:

$$f(x|\lambda, f, \varepsilon) = \left[\prod_{i=1}^n (2\pi)^{-T_i/2} \right] \left[\prod_{i=1}^n \exp \left(-\frac{1}{2} (x_i - W_i \lambda_i - E_i \gamma_i)' (x_i - W_i \lambda_i - E_i \gamma_i) / \sigma_i^2 \right) \right] \tag{a.6.18.5}$$

derivations for equation (6.18.29)

Start from the expression, and define for convenience $\hat{x}_i = x_i - E_i \gamma_i$:

$$\begin{aligned}
 & \pi(\lambda_i | y, \lambda_i) \\
 & = \exp \left(-\frac{1}{2} (\hat{x}_i - W_i \lambda_i)' (\hat{x}_i - W_i \lambda_i) / \sigma_i^2 \right) \times \exp \left(-\frac{1}{2} (\lambda_i - h_i)' U_i^{-1} (\lambda_i - h_i) \right) \\
 & = \exp \left(-\frac{1}{2} [(\hat{x}_i - W_i \lambda_i)' (\hat{x}_i - W_i \lambda_i) / \sigma_i^2 + (\lambda_i - h_i)' U_i^{-1} (\lambda_i - h_i)] \right)
 \end{aligned} \tag{a.6.18.6}$$

Consider the quadratic term, develop and complete the squares:

$$\begin{aligned}
& (\hat{x}_i - W_i \lambda_i)' (\hat{x}_i - W_i \lambda_i) / \sigma_i + (\lambda_i - h_i)' U_i^{-1} (\lambda_i - h_i) \\
&= \hat{x}_i' \hat{x}_i / \sigma_i + \lambda_i' (W_i' W_i / \sigma_i) \lambda_i - 2 \lambda_i' (W_i' \hat{x}_i / \sigma_i) + \lambda_i' U_i^{-1} \lambda_i + h_i' U_i^{-1} h_i - 2 \lambda_i' U_i^{-1} h_i \\
&= \lambda_i' (U_i^{-1} + W_i' W_i / \sigma_i) \lambda_i - 2 \lambda_i' (U_i^{-1} h_i + W_i' \hat{x}_i / \sigma_i) + \hat{x}_i' \hat{x}_i / \sigma_i + h_i' U_i^{-1} h_i \\
&= \lambda_i' (U_i^{-1} + W_i' W_i / \sigma_i) \lambda_i - 2 \lambda_i' \bar{U}_i^{-1} \bar{U}_i (U_i^{-1} h_i + W_i' \hat{x}_i / \sigma_i) + \hat{x}_i' \hat{x}_i / \sigma_i + h_i' U_i^{-1} h_i + \bar{g}_i' \bar{U}_i^{-1} \bar{g}_i - \bar{g}_i' \bar{U}_i^{-1} \bar{g}_i
\end{aligned} \tag{a.6.18.7}$$

Define:

$$\bar{U}_i = (U_i^{-1} + W_i' W_i / \sigma_i)^{-1} \quad \bar{h}_i = \bar{U}_i (U_i^{-1} h_i + W_i' \hat{x}_i / \sigma_i) \tag{a.6.18.8}$$

Then (a.6.18.7) rewrites:

$$\begin{aligned}
& (\hat{x}_i - W_i \lambda_i)' (\hat{x}_i - W_i \lambda_i) / \sigma_i + (\lambda_i - h_i)' U_i^{-1} (\lambda_i - h_i) \\
&= \lambda_i' \bar{U}_i^{-1} \lambda_i - 2 \lambda_i' \bar{U}_i^{-1} \bar{h}_i + \bar{h}_i' \bar{U}_i^{-1} \bar{h}_i + \hat{x}_i' \hat{x}_i / \sigma_i + h_i' U_i^{-1} h_i - \bar{h}_i' \bar{U}_i^{-1} \bar{h}_i \\
&= (\lambda_i - \bar{h}_i)' \bar{U}_i^{-1} (\lambda_i - \bar{h}_i) + \hat{x}_i' \hat{x}_i / \sigma_i + h_i' U_i^{-1} h_i - \bar{h}_i' \bar{U}_i^{-1} \bar{h}_i
\end{aligned} \tag{a.6.18.9}$$

Substitute back in (a.6.18.6):

$$\begin{aligned}
& \pi(\lambda_i | y, \lambda_i) \\
&= \exp \left(-\frac{1}{2} [(\lambda_i - \bar{h}_i)' \bar{U}_i^{-1} (\lambda_i - \bar{h}_i) + \hat{x}_i' \hat{x}_i + h_i' U_i^{-1} h_i - \bar{h}_i' \bar{U}_i^{-1} \bar{h}_i] \right) \\
&= \exp \left(-\frac{1}{2} (\lambda_i - \bar{h}_i)' \bar{U}_i^{-1} (\lambda_i - \bar{h}_i) \right) \exp \left(-\frac{1}{2} [\hat{x}_i' \hat{x}_i + h_i' U_i^{-1} h_i - \bar{h}_i' \bar{U}_i^{-1} \bar{h}_i] \right) \\
&\propto \exp \left(-\frac{1}{2} (\lambda_i - \bar{h}_i)' \bar{U}_i^{-1} (\lambda_i - \bar{h}_i) \right)
\end{aligned} \tag{a.6.18.10}$$

Reformulate the posterior parameters:

$$\bar{h}_i = \bar{U}_i (U_i^{-1} h_i + W_i' \hat{x}_i / \sigma_i) = \bar{U}_i (U_i^{-1} h_i + W_i' [x_i - E_i \gamma] / \sigma_i) \tag{a.6.18.11}$$

derivations for equation (6.18.32)

Start from:

$$\begin{aligned}
& \pi(\beta_j | y, \beta_j) \\
&= \exp \left(-\frac{1}{2} (f_j - Z \beta_j)' (f_j - Z \beta_j) / \omega_j \right) \times \exp \left(-\frac{1}{2} (\beta_j - b_j)' V_j^{-1} (\beta_j - b_j) \right) \\
&= \exp \left(-\frac{1}{2} [(f_j - Z \beta_j)' (f_j - Z \beta_j) / \omega_j + (\beta_j - b_j)' V_j^{-1} (\beta_j - b_j)] \right)
\end{aligned} \tag{a.6.18.12}$$

Consider the quadratic term, develop and complete the squares:

$$\begin{aligned}
& (f_j - Z \beta_j)' (f_j - Z \beta_j) / \omega_j + (\beta_j - b_j)' V_j^{-1} (\beta_j - b_j) \\
&= f_j' f_j / \omega_j + \beta_j' (Z' Z / \omega_j) \beta_j - 2 \beta_j' Z' f_j / \omega_j + \beta_j' V_j^{-1} \beta_j + b_j' V_j^{-1} b_j - 2 \beta_j' V_j^{-1} b_j \\
&= \beta_j' (V_j^{-1} + Z' Z / \omega_j) \beta_j - 2 \beta_j' (V_j^{-1} b_j + Z' f_j / \omega_j) + f_j' f_j / \omega_j + b_j' V_j^{-1} b_j \\
&= \beta_j' (V_j^{-1} + Z' Z / \omega_j) \beta_j - 2 \beta_j' \bar{V}_j^{-1} \bar{V}_j (V_j^{-1} b_j + Z' f_j / \omega_j) + f_j' f_j / \omega_j + b_j' V_j^{-1} b_j + \bar{b}_j' \bar{V}_j^{-1} \bar{b}_j - \bar{b}_j' \bar{V}_j^{-1} \bar{b}_j
\end{aligned} \tag{a.6.18.13}$$

Define:

$$\bar{V}_j = (V_j^{-1} + Z' Z / \omega_j)^{-1} \quad \bar{b}_j = \bar{V}_j (V_j^{-1} b_j + Z' f_j / \omega_j) \tag{a.6.18.14}$$

Then (a.6.18.13) rewrites:

$$\begin{aligned} & \beta_j' \bar{V}_j^{-1} \beta_j - 2\beta_j' \bar{V}_j^{-1} \bar{b}_j + \bar{b}_j' \bar{V}_j^{-1} \bar{b}_j + f_j' f_j / \omega_j + b_j' V_j^{-1} b_j - \bar{b}_j' \bar{V}_j^{-1} \bar{b}_j \\ = & (\beta_j - \bar{b}_j)' \bar{V}_j^{-1} (\beta_j - \bar{b}_j) + f_j' f_j / \omega_j + b_j' V_j^{-1} b_j - \bar{b}_j' \bar{V}_j^{-1} \bar{b}_j \end{aligned} \quad (\text{a.6.18.15})$$

Substitute back in (a.6.18.12):

$$\begin{aligned} & \pi(\beta_j | y, \setminus \beta_j) \\ = & \exp \left(-\frac{1}{2} \left[(\beta_j - \bar{b}_j)' \bar{V}_j^{-1} (\beta_j - \bar{b}_j) + f_j' f_j / \omega_j + b_j' V_j^{-1} b_j - \bar{b}_j' \bar{V}_j^{-1} \bar{b}_j \right] \right) \\ = & \exp \left(-\frac{1}{2} (\beta_j - \bar{b}_j)' \bar{V}_j^{-1} (\beta_j - \bar{b}_j) \right) \exp \left(-\frac{1}{2} \left[f_j' f_j / \omega_j + b_j' V_j^{-1} b_j - \bar{b}_j' \bar{V}_j^{-1} \bar{b}_j \right] \right) \\ \propto & \exp \left(-\frac{1}{2} (\beta_j - \bar{b}_j)' \bar{V}_j^{-1} (\beta_j - \bar{b}_j) \right) \end{aligned} \quad (\text{a.6.18.16})$$

derivations for equation (6.18.35)

Start from the expression:

$$\begin{aligned} & \pi(\gamma_i | y, \setminus \gamma_i) \\ = & \exp \left(-\frac{1}{2} (\varepsilon_i - E_i \gamma_i)' (\varepsilon_i - E_i \gamma_i) / \sigma_i \right) \times \exp \left(-\frac{1}{2} (\gamma_i - g_i)' Q_i^{-1} (\gamma_i - g_i) \right) \\ = & \exp \left(-\frac{1}{2} \left[(\varepsilon_i - E_i \gamma_i)' (\varepsilon_i - E_i \gamma_i) / \sigma_i + (\gamma_i - g_i)' Q_i^{-1} (\gamma_i - g_i) \right] \right) \end{aligned} \quad (\text{a.6.18.17})$$

Consider the quadratic term, develop and complete the squares:

$$\begin{aligned} & (\varepsilon_i - E_i \gamma_i)' (\varepsilon_i - E_i \gamma_i) / \sigma_i + (\gamma_i - g_i)' Q_i^{-1} (\gamma_i - g_i) \\ = & \varepsilon_i' \varepsilon_i / \sigma_i + \gamma_i' (E_i' E_i / \sigma_i) \gamma_i - 2\gamma_i' E_i' \varepsilon_i / \sigma_i + \gamma_i' Q_i^{-1} \gamma_i + g_i' Q_i^{-1} g_i - 2\gamma_i' Q_i^{-1} g_i \\ = & \gamma_i' (Q_i^{-1} + E_i' E_i / \sigma_i) \gamma_i - 2\gamma_i' (Q_i^{-1} g_i + E_i' \varepsilon_i / \sigma_i) + \varepsilon_i' \varepsilon_i / \sigma_i + g_i' Q_i^{-1} g_i \\ = & \gamma_i' (Q_i^{-1} + E_i' E_i / \sigma_i) \gamma_i - 2\gamma_i' \bar{Q}_i^{-1} \bar{Q}_i (Q_i^{-1} g_i + E_i' \varepsilon_i / \sigma_i) + \varepsilon_i' \varepsilon_i / \sigma_i + g_i' Q_i^{-1} g_i + \bar{g}_i' \bar{Q}_i^{-1} \bar{g}_i - \bar{g}_i' \bar{Q}_i^{-1} \bar{g}_i \end{aligned} \quad (\text{a.6.18.18})$$

Define:

$$\bar{Q}_i = (Q_i^{-1} + E_i' E_i / \sigma_i)^{-1} \quad \bar{g}_i = \bar{Q}_i (Q_i^{-1} g_i + E_i' \varepsilon_i / \sigma_i) \quad (\text{a.6.18.19})$$

Then (a.6.18.18) rewrites:

$$\begin{aligned} & \gamma_i' \bar{Q}_i^{-1} \gamma_i - 2\gamma_i' \bar{Q}_i^{-1} \bar{g}_i + \bar{g}_i' \bar{Q}_i^{-1} \bar{g}_i + \varepsilon_i' \varepsilon_i / \sigma_i + g_i' Q_i^{-1} g_i - \bar{g}_i' \bar{Q}_i^{-1} \bar{g}_i \\ = & (\gamma_i - \bar{g}_i)' \bar{Q}_i^{-1} (\gamma_i - \bar{g}_i) + \varepsilon_i' \varepsilon_i / \sigma_i + g_i' Q_i^{-1} g_i - \bar{g}_i' \bar{Q}_i^{-1} \bar{g}_i \end{aligned} \quad (\text{a.6.18.20})$$

Substitute back in (a.6.18.17):

$$\begin{aligned} & \pi(\gamma_i | y, \setminus \gamma_i) \\ = & \exp \left(-\frac{1}{2} \left[(\gamma_i - \bar{g}_i)' \bar{Q}_i^{-1} (\gamma_i - \bar{g}_i) + \varepsilon_i' \varepsilon_i / \sigma_i + g_i' Q_i^{-1} g_i - \bar{g}_i' \bar{Q}_i^{-1} \bar{g}_i \right] \right) \\ = & \exp \left(-\frac{1}{2} (\gamma_i - \bar{g}_i)' \bar{Q}_i^{-1} (\gamma_i - \bar{g}_i) \right) \exp \left(-\frac{1}{2} \left[\varepsilon_i' \varepsilon_i / \sigma_i + g_i' Q_i^{-1} g_i - \bar{g}_i' \bar{Q}_i^{-1} \bar{g}_i \right] \right) \\ \propto & \exp \left(-\frac{1}{2} (\gamma_i - \bar{g}_i)' \bar{Q}_i^{-1} (\gamma_i - \bar{g}_i) \right) \end{aligned} \quad (\text{a.6.18.21})$$

derivations for equation (6.18.46)

The Wold representation for f_t is given by:

$$f_t = \Psi_0 \xi_t + \Psi_1 \xi_{t-1} + \Psi_2 \xi_{t-2} \dots \quad (\text{a.6.18.22})$$

This can rewrite for any period $t - h$ as:

$$f_{t-h} = \Psi_0 \xi_{t-h} + \Psi_1 \xi_{t-h-1} + \Psi_2 \xi_{t-h-2} \dots \quad (\text{a.6.18.23})$$

Following, one obtains:

$$\begin{aligned} y_t &= \Lambda_0 f_t + \Lambda_1 f_{t-1} + \dots + \Lambda_q f_{t-q} + \varepsilon_t \\ &= \Lambda_0 (\Psi_0 \xi_t + \Psi_1 \xi_{t-1} + \Psi_2 \xi_{t-2} \dots) \\ &\quad + \Lambda_1 (\Psi_0 \xi_{t-1} + \Psi_1 \xi_{t-2} + \Psi_2 \xi_{t-3} \dots) \\ &\quad + \dots \\ &\quad + \Lambda_q (\Psi_0 \xi_{t-q} + \Psi_1 \xi_{t-q-1} + \Psi_2 \xi_{t-q-2} \dots) \\ &= (\Lambda_0 \Psi_0) \xi_t \\ &\quad + (\Lambda_0 \Psi_1 + \Lambda_1 \Psi_0) \xi_{t-1} \\ &\quad + (\Lambda_0 \Psi_2 + \Lambda_1 \Psi_1 + \Lambda_2 \Psi_0) \xi_{t-2} \\ &\quad + \dots \\ &\quad + \varepsilon_t \end{aligned} \quad (\text{a.6.18.24})$$

For $q = 2$, this yields:

$$\begin{aligned} y_t &= (\Lambda_0 \Psi_0) \xi_t \\ &\quad + (\Lambda_0 \Psi_1 + \Lambda_1 \Psi_0) \xi_{t-1} \\ &\quad + (\Lambda_0 \Psi_2 + \Lambda_1 \Psi_1 + \Lambda_2 \Psi_0) \xi_{t-2} \\ &\quad + (\Lambda_0 \Psi_3 + \Lambda_1 \Psi_2 + \Lambda_2 \Psi_1) \xi_{t-3} \\ &\quad + (\Lambda_0 \Psi_4 + \Lambda_1 \Psi_3 + \Lambda_2 \Psi_2) \xi_{t-4} \\ &\quad + \dots + \varepsilon_t \\ &= \bar{\Psi}_0 \xi_t + \bar{\Psi}_1 \xi_{t-1} + \bar{\Psi}_2 \xi_{t-2} + \bar{\Psi}_3 \xi_{t-3} + \bar{\Psi}_4 \xi_{t-4} + \dots + \varepsilon_t \end{aligned} \quad (\text{a.6.18.25})$$

In general:

$$\bar{\Psi}_i = \sum_{j=0}^i \Lambda_j \Psi_{i-j} \quad i < q \quad \text{and} \quad \bar{\Psi}_i = \sum_{j=0}^q \Lambda_j \Psi_{i-j} \quad i \geq q \quad (\text{a.6.18.26})$$

Substituting for the Wold representation of the residuals:

$$y_t = \bar{\Psi}_0 \xi_t + \bar{\Psi}_1 \xi_{t-1} + \bar{\Psi}_2 \xi_{t-2} + \dots + \Phi_0 e_t + \Phi_1 e_{t-1} + \Phi_2 e_{t-2} + \dots \quad (\text{a.6.18.27})$$

MIDAS regression

derivations for equation (6.19.16)

Start from :

$$\begin{aligned}
 & \pi(\beta|y, \sigma) \\
 & \propto \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \exp\left(-\frac{1}{2} \beta' V^{-1} \beta\right) \\
 & = \exp\left(-\frac{1}{2} [(y - X\beta)'(y - X\beta)/\sigma + \beta' V^{-1} \beta]\right) \quad (\text{a.6.19.1})
 \end{aligned}$$

Consider the term in square brackets:

$$\begin{aligned}
 & (y - X\beta)'(y - X\beta)/\sigma + \beta' V^{-1} \beta \\
 & = y'y/\sigma + \beta' X' X \beta / \sigma - 2\beta' X' y / \sigma + \beta' V^{-1} \beta \\
 & = \beta' (V^{-1} + X' X / \sigma) \beta - 2\beta' X' y / \sigma + y'y/\sigma \quad (\text{a.6.19.2})
 \end{aligned}$$

Complete the squares:

$$= \beta' (V^{-1} + X' X / \sigma) \beta - 2\beta' \bar{V}^{-1} \bar{V} X' y / \sigma + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b} + y'y/\sigma \quad (\text{a.6.19.3})$$

Define:

$$\bar{V} = (V^{-1} + X' X / \sigma)^{-1} \quad \bar{b} = \bar{V} X' y / \sigma \quad (\text{a.6.19.4})$$

Then (a.6.19.3) rewrites:

$$\begin{aligned}
 & = \beta' \bar{V}^{-1} \beta - 2\beta' \bar{V}^{-1} \bar{b} + \bar{b}' \bar{V}^{-1} \bar{b} - \bar{b}' \bar{V}^{-1} \bar{b} + y'y/\sigma \\
 & = (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) - \bar{b}' \bar{V}^{-1} \bar{b} + y'y/\sigma \quad (\text{a.6.19.5})
 \end{aligned}$$

Substitute (a.6.19.5) back in (a.6.19.1):

$$\begin{aligned}
 & \pi(\beta|y, \sigma) \\
 & \propto \exp\left(-\frac{1}{2} [(\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b}) - \bar{b}' \bar{V}^{-1} \bar{b} + y'y/\sigma]\right) \\
 & = \exp\left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b})\right) \exp\left(-\frac{1}{2} [-\bar{b}' \bar{V}^{-1} \bar{b} + y'y/\sigma]\right) \\
 & \propto \exp\left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b})\right) \quad (\text{a.6.19.6})
 \end{aligned}$$

derivations for equation (6.19.19)

Start from:

$$\begin{aligned} \pi(\sigma|y, \beta) &\propto \sigma^{-T/2} \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \\ &\quad \times \sigma^{-\kappa/2-1} \exp\left(-\frac{\lambda}{2\sigma}\right) \end{aligned} \quad (\text{a.6.19.7})$$

Rearrange:

$$\pi(\sigma|y, \beta) \propto \sigma^{-(T+\kappa)/2-1} \times \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta) + \lambda}{\sigma}\right) \quad (\text{a.6.19.8})$$

define:

$$\bar{\kappa} = (T + \kappa)/2 \quad \bar{\lambda} = \frac{1}{2} ((y - X\beta)'(y - X\beta) + \lambda) \quad (\text{a.6.19.9})$$

Then (a.6.19.8) rewrites:

$$\pi(\sigma|y, \beta) \propto \sigma^{-\bar{\kappa}-1} \exp\left(-\frac{\bar{\lambda}}{\sigma}\right) \quad (\text{a.6.19.10})$$

derivations for equation (6.19.28)

Start from the joint formulation:

$$\begin{aligned} &\pi(\beta, \tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma|y) \\ &\propto f(y|\beta, \tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma) \\ &\quad \times \pi(\beta, \tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma) \\ &= f(y|\beta, \sigma) \pi(\beta_1, \dots, \beta_l, \tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma) \\ &= f(y|\beta, \sigma) \\ &\quad \times \frac{\pi(\beta, \tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma)}{\pi(\tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma)} \pi(\tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma) \\ &= f(y|\beta, \sigma) \\ &\quad \times \pi(\beta|\tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma) \pi(\tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma) \\ &= f(y|\beta, \sigma) \pi(\beta_1, \dots, \beta_l|\tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma) \\ &\quad \times \pi(\tau^2, v) \pi(\psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l) \pi(\sigma) \\ &= f(y|\beta, \sigma) \pi(\beta|\tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma) \\ &\quad \times \frac{\pi(\tau^2, v)}{\pi(v)} \pi(v) \frac{\pi(\psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l)}{\pi(\eta_1, \dots, \eta_l)} \pi(\eta_1, \dots, \eta_l) \pi(\sigma) \\ &= f(y|\beta, \sigma) \pi(\beta|\tau^2, v, \psi_1^2, \dots, \psi_l^2, \eta_1, \dots, \eta_l, \sigma) \\ &\quad \times \pi(\tau^2|v) \pi(v) \pi(\psi_1^2, \dots, \psi_l^2|\eta_1, \dots, \eta_l) \pi(\eta_1, \dots, \eta_l) \pi(\sigma) \end{aligned} \quad (\text{a.6.19.11})$$

Eventually assuming independence between the l coefficients in β :

$$= f(y|\beta, \sigma) \pi(\beta|\tau^2, \psi_1^2, \dots, \psi_l^2, \sigma) \pi(\tau^2|v) \pi(v) \left(\prod_{h=1}^l \pi(\psi_h^2|\eta_h) \pi(\eta_h) \right) \pi(\sigma) \quad (\text{a.6.19.12})$$

derivations for equation (6.19.30)

Start from :

$$\pi(\beta|y, \sigma) \propto \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \times \exp\left(-\frac{1}{2} \beta' V^{-1} \beta\right) \quad (\text{a.6.19.13})$$

This is similar to (a.6.19.1). Thus a similar reasoning leads to:

$$\pi(\beta|y, \sigma) \propto \exp\left(-\frac{1}{2} (\beta - \bar{b})' \bar{V}^{-1} (\beta - \bar{b})\right) \quad (\text{a.6.19.14})$$

with:

$$\bar{V} = (V^{-1} + X'X/\sigma)^{-1} \quad \bar{b} = \bar{V}X'y/\sigma \quad (\text{a.6.19.15})$$

Simplify the posterior terms:

$$\begin{aligned} & \bar{V} \\ &= (V^{-1} + X'X/\sigma)^{-1} \\ &= ([\sigma \tau^2 \text{diag}(\psi_1^2, \psi_2^2, \dots, \psi_l^2)]^{-1} + X'X/\sigma)^{-1} \\ &= (\text{diag}(1/\psi_1^2, 1/\psi_2^2, \dots, 1/\psi_l^2)/(\tau^2 \sigma) + X'X/\sigma)^{-1} \\ &= \sigma(\text{diag}(1/\psi_1^2, 1/\psi_2^2, \dots, 1/\psi_l^2)/\tau^2 + X'X)^{-1} \\ &= \sigma(\Psi^{-1}/\tau^2 + X'X)^{-1} \\ &= \sigma V^* \end{aligned} \quad (\text{a.6.19.16})$$

with:

$$V^* = (\Psi^{-1}/\tau^2 + X'X)^{-1} \quad \Psi = \text{diag}(\psi_1^2, \psi_2^2, \dots, \psi_l^2) \quad (\text{a.6.19.17})$$

Also:

$$\bar{b} = \bar{V}X'y/\sigma = \sigma V^*X'y/\sigma = V^*X'y \quad (\text{a.6.19.18})$$

derivations for equation (6.19.33)

Start from:

$$\pi(\tau^2|y, \tau^2) \propto |V|^{-1/2} \exp\left(-\frac{1}{2} \beta' V^{-1} \beta\right) \times (\tau^2)^{-1/2-1} \exp\left(-\frac{1}{\tau^2}\right) \quad (\text{a.6.19.19})$$

$$V = \sigma \tau^2 \text{diag}(\psi_1^2, \psi_2^2, \dots, \psi_l^2) \quad (\text{a.6.19.20})$$

Thus:

$$\begin{aligned} & |V|^{-1/2} \\ &= |\sigma \tau^2 \text{diag}(\psi_1^2, \psi_2^2, \dots, \psi_l^2)|^{-1/2} \\ &= [\sigma^l (\tau^2)^l |\text{diag}(\psi_1^2, \psi_2^2, \dots, \psi_l^2)|]^{-1/2} \\ &= \sigma^{-l/2} (\tau^2)^{-l/2} \left(\prod_{h=1}^l \psi_h^2\right)^{-1/2} \\ &= \sigma^{-l/2} (\tau^2)^{-l/2} \prod_{h=1}^l (\psi_h^2)^{-1/2} \end{aligned} \quad (\text{a.6.19.21})$$

Also:

$$\begin{aligned}
& \beta' V^{-1} \beta \\
&= (\beta_1 \ \beta_2 \ \cdots \ \beta_l) \left[\sigma \tau^2 \begin{pmatrix} \psi_1^2 & 0 & \cdots & 0 \\ 0 & \psi_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \psi_l^2 \end{pmatrix} \right]^{-1} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_l \end{pmatrix} \\
&= \sigma^{-1} (\tau^2)^{-1} (\beta_1 \ \beta_2 \ \cdots \ \beta_l) \begin{pmatrix} 1/\psi_1^2 & 0 & \cdots & 0 \\ 0 & 1/\psi_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\psi_l^2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_l \end{pmatrix} \\
&= \sigma^{-1} (\tau^2)^{-1} \left[\sum_{h=1}^l \beta_h^2 / \psi_h^2 \right] \tag{a.6.19.22}
\end{aligned}$$

Therefore:

$$\exp \left(-\frac{1}{2} \beta' V^{-1} \beta \right) = \exp \left(-\frac{1}{2} \sigma^{-1} (\tau^2)^{-1} \left[\sum_{h=1}^l \beta_h^2 / \psi_h^2 \right] \right) = \prod_{h=1}^l \exp \left(-\frac{1}{2} \sigma^{-1} (\tau^2)^{-1} \beta_h^2 / \psi_h^2 \right) \tag{a.6.19.23}$$

Following:

$$|V|^{-1/2} \exp \left(-\frac{1}{2} \beta' V^{-1} \beta \right) = \sigma^{-l/2} (\tau^2)^{-l/2} \prod_{h=1}^l (\psi_h^2)^{-1/2} \left[\prod_{h=1}^l \exp \left(-\frac{1}{2} \sigma^{-1} (\tau^2)^{-1} \beta_h^2 / \psi_h^2 \right) \right] \tag{a.6.19.24}$$

Substitute back in (a.6.19.19) and relegate to the normalization constant any term not involving τ :

$$\begin{aligned}
& \pi(\tau^2 | y, \setminus \tau^2) \\
& \propto \sigma^{-l/2} (\tau^2)^{-l/2} \prod_{h=1}^l (\psi_h^2)^{-1/2} \\
& \times \left[\prod_{h=1}^l \exp \left(-\frac{1}{2} \sigma^{-1} (\tau^2)^{-1} \beta_h^2 / \psi_h^2 \right) \right] \\
& \times (\tau^2)^{-1/2-1} \exp \left(-\frac{1}{v \tau^2} \right) \\
& \propto (\tau^2)^{-l/2} \left[\prod_{h=1}^l \exp \left(-\frac{1}{2} \sigma^{-1} (\tau^2)^{-1} \beta_h^2 / \psi_h^2 \right) \right] (\tau^2)^{-1/2-1} \exp \left(-\frac{1}{v \tau^2} \right) \\
& \propto (\tau^2)^{-l/2} \exp \left(-\frac{1}{2} \sigma^{-1} (\tau^2)^{-1} \sum_{h=1}^l \beta_h^2 / \psi_h^2 \right) (\tau^2)^{-1/2-1} \exp \left(-\frac{1}{v \tau^2} \right) \\
& \propto (\tau^2)^{-(l+1)/2-1} \exp \left(-\frac{1/v + 1/2 \sum_{h=1}^l \beta_h^2 / (\psi_h^2 \sigma)}{\tau^2} \right) \tag{a.6.19.25}
\end{aligned}$$

Define:

$$\gamma_\tau = \frac{l+1}{2} \quad \phi_\tau = \frac{1}{v} + \frac{1}{2\sigma} \sum_{h=1}^l \frac{\beta_h^2}{\psi_h^2} \tag{a.6.19.26}$$

Then (a.6.19.25) rewrites:

$$\pi(\tau^2|y, \backslash \tau) \propto (\tau^2)^{-\gamma_\tau-1} \exp\left(-\frac{\phi_\tau}{\tau^2}\right) \quad (\text{a.6.19.27})$$

derivations for equation (6.19.39)

Start from:

$$\pi(\psi_h^2|y, \backslash \psi_h) \propto |V|^{-1/2} \exp\left(-\frac{1}{2}\beta'V^{-1}\beta\right) \times (\psi_h^2)^{-1/2-1} \exp\left(-\frac{1}{\eta_h \psi_h^2}\right) \quad (\text{a.6.19.28})$$

From (a.6.19.24), this rewrites:

$$\begin{aligned} & \pi(\psi_h^2|y, \backslash \psi_h) \\ & \propto \sigma^{-l/2} (\tau^2)^{-l/2} \\ & \times \prod_{h=1}^l (\psi_h^2)^{-1/2} \left[\prod_{h=1}^l \exp\left(-\frac{1}{2}\sigma^{-1} (\tau^2)^{-1} \beta_h^2 / \psi_h^2\right) \right] \\ & \times (\psi_h^2)^{-1/2-1} \exp\left(-\frac{1}{\eta_h \psi_h^2}\right) \end{aligned} \quad (\text{a.6.19.29})$$

Relegate to the normalization constant any term that does not involve ψ_{ij}^2 :

$$\begin{aligned} & \pi(\psi_h^2|y, \backslash \psi_h) \\ & \propto (\psi_h^2)^{-1/2} \exp\left(-\frac{1}{2}\sigma^{-1} (\tau^2)^{-1} \beta_h^2 / \psi_h^2\right) (\psi_h^2)^{-1/2-1} \exp\left(-\frac{1}{\eta_h \psi_h^2}\right) \\ & = (\psi_h^2)^{-2} \exp\left(-\left[\frac{1}{2}\sigma^{-1} (\tau^2)^{-1} \beta_h^2 / \psi_h^2 + \frac{1}{\eta_h \psi_h^2}\right]\right) \end{aligned} \quad (\text{a.6.19.30})$$

Define:

$$\gamma_\psi = 1 \quad \phi_\psi = \frac{1}{\eta_h} + \frac{\beta_h^2}{2\sigma\tau^2} \quad (\text{a.6.19.31})$$

Then (a.6.19.30) rewrites:

$$\pi(\psi_h^2|y, \backslash \psi_h^2) \propto (\psi_h^2)^{-\gamma_\psi-1} \exp\left(-\frac{\phi_\psi}{\psi_h^2}\right) \quad (\text{a.6.19.32})$$

derivations for equation (6.19.45)

Start from:

$$\pi(\sigma|y, \backslash \sigma) \propto \sigma^{-T/2} \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \times |V|^{-1/2} \exp\left(-\frac{1}{2}\beta'V^{-1}\beta\right) \times \sigma^{-\kappa/2-1} \exp\left(-\frac{\lambda}{2\sigma}\right) \quad (\text{a.6.19.33})$$

Using (a.6.19.24), this rewrites:

$$\begin{aligned}
& \pi(\sigma|y, \backslash \sigma) \\
& \propto \sigma^{-T/2} \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \\
& \times \sigma^{-l/2} \left[\prod_{h=1}^l \exp\left(-\frac{1}{2} \sigma^{-1} (\tau^2)^{-1} \beta_h^2 / \psi_h^2\right) \right] \\
& \times \sigma^{-\kappa/2-1} \exp\left(-\frac{\lambda}{2\sigma}\right)
\end{aligned} \tag{a.6.19.34}$$

Rearrange:

$$\pi(\sigma|y, \backslash \sigma) \propto \sigma^{-(T+l+\kappa)/2-1} \times \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta) + \sum_{h=1}^l \beta_h^2 / \tau^2 \psi_h^2 + \lambda}{\sigma}\right) \tag{a.6.19.35}$$

define:

$$\bar{\kappa} = \frac{(T+l+\kappa)}{2} \quad \bar{\lambda} = \frac{1}{2} \left((y - X\beta)'(y - X\beta) + \sum_{h=1}^l \beta_h^2 / \tau^2 \psi_h^2 + \lambda \right) \tag{a.6.19.36}$$

Then (a.6.19.35) rewrites:

$$\pi(\sigma|y, \backslash \sigma) \propto \sigma^{-\bar{\kappa}-1} \exp\left(-\frac{\bar{\lambda}}{\sigma}\right) \tag{a.6.19.37}$$

derivations for equation (6.19.52)

Start from:

$$\pi(\beta, \xi, \mu, \sigma|y) \propto f(y|\beta, \sigma) \pi(\beta, \xi, \mu, \sigma) \tag{a.6.19.38}$$

Assume independence between the different β_h and σ :

$$\begin{aligned}
& \pi(\beta, \xi, \mu, \sigma|y) \\
& \propto f(y|\beta, \sigma) \pi(\beta, \xi, \mu, \sigma) \\
& = f(y|\beta, \sigma) \frac{\pi(\beta, \xi, \mu, \sigma)}{\pi(\xi, \mu, \sigma)} \pi(\xi, \mu, \sigma) \\
& = f(y|\beta, \sigma) \pi(\beta|\xi, \mu, \sigma) \pi(\xi, \mu) \pi(\sigma) \\
& = f(y|\beta, \sigma) \pi(\beta|\xi, \mu, \sigma) \frac{\pi(\xi, \mu)}{\pi(\mu)} \pi(\mu) \pi(\sigma) \\
& = f(y|\beta, \sigma) \pi(\beta|\xi, \mu, \sigma) \pi(\xi|\mu) \pi(\mu) \pi(\sigma) \\
& = f(y|\beta, \sigma) \pi(\beta|\xi_1, \dots, \xi_l, \sigma) \left(\prod_{h=1}^l \pi(\xi_h|\mu) \right) \pi(\mu) \pi(\sigma)
\end{aligned} \tag{a.6.19.39}$$

derivations for equation (6.19.54)

Start from:

$$\pi(\beta|y, \backslash \beta) \propto \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \left(\prod_{h=1}^l \mathbb{1}(|\beta_h| < \sqrt{\sigma} \xi_h) \right) \tag{a.6.19.40}$$

Rearrange:

$$\begin{aligned}
& \pi(\beta|y, \backslash\beta) \\
& \propto \exp\left(-\frac{1}{2} \frac{(y - X\beta)'(y - X\beta)}{\sigma}\right) \left(\prod_{h=1}^l \mathbb{1}(|\beta_h| < \sqrt{\sigma}\xi_h)\right) \\
& = \exp\left(-\frac{1}{2} [y'y/\sigma + \beta'(X'X/\sigma)\beta - 2\beta'X'y/\sigma]\right) \prod_{h=1}^l \mathbb{1}(|\beta_h| < \sqrt{\sigma}\xi_h) \\
& = \exp\left(-\frac{1}{2} [y'y/\sigma + \beta'(X'X/\sigma)\beta - 2\beta'\bar{V}^{-1}\bar{V}X'y/\sigma + \bar{b}'\bar{V}^{-1}\bar{b} - \bar{b}'\bar{V}^{-1}\bar{b}]\right) \prod_{h=1}^l \mathbb{1}(|\beta_h| < \sqrt{\sigma}\xi_h)
\end{aligned} \tag{a.6.19.41}$$

Define:

$$\bar{V} = (X'X/\sigma)^{-1} = \sigma(X'X)^{-1} \quad \bar{b} = \bar{V}X'y/\sigma = \sigma(X'X)^{-1}X'y/\sigma = (X'X)^{-1}(X'y) \tag{a.6.19.42}$$

Then (a.6.19.41) rewrites:

$$\begin{aligned}
& \pi(\beta|y, \backslash\beta) \\
& \propto \exp\left(-\frac{1}{2} [y'y/\sigma + \beta'\bar{V}^{-1}\beta - 2\beta'\bar{V}^{-1}\bar{b} + \bar{b}'\bar{V}^{-1}\bar{b} - \bar{b}'\bar{V}^{-1}\bar{b}]\right) \prod_{h=1}^l \mathbb{1}(|\beta_h| < \sqrt{\sigma}\xi_h) \\
& = \exp\left(-\frac{1}{2} [y'y/\sigma + (\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b}) - \bar{b}'\bar{V}^{-1}\bar{b}]\right) \prod_{h=1}^l \mathbb{1}(|\beta_h| < \sqrt{\sigma}\xi_h) \\
& = \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right) \exp\left(-\frac{1}{2} [y'y/\sigma - \bar{b}'\bar{V}^{-1}\bar{b}]\right) \prod_{h=1}^l \mathbb{1}(|\beta_h| < \sqrt{\sigma}\xi_h) \\
& \propto \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right) \prod_{h=1}^l \mathbb{1}(|\beta_h| < \sqrt{\sigma}\xi_h) \\
& = \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right) \mathbb{1}(|\beta| < \sqrt{\sigma}\xi)
\end{aligned} \tag{a.6.19.43}$$

define:

$$\chi = \sqrt{\sigma}\xi \tag{a.6.19.44}$$

Then (a.6.19.43) rewrites:

$$\pi(\beta|y, \backslash\beta) \propto \exp\left(-\frac{1}{2}(\beta - \bar{b})'\bar{V}^{-1}(\beta - \bar{b})\right) \mathbb{1}(|\beta| < \chi) \tag{a.6.19.45}$$

derivations for equation (6.19.63)

Rearrange:

$$\begin{aligned}
& \pi(\sigma|y, \backslash \sigma) \\
& \propto \sigma^{-T/2} \exp\left(-\frac{1}{2} \frac{(y-X\beta)'(y-X\beta)}{\sigma}\right) \times \prod_{h=1}^l [(\sqrt{\sigma})^{-1} \mathbb{1}(|\beta_h| < \sqrt{\sigma}\xi_h)] \times \sigma^{-\kappa/2-1} \exp\left(-\frac{\lambda}{2\sigma}\right) \\
& = \sigma^{-T/2} \exp\left(-\frac{1}{2} \frac{(y-X\beta)'(y-X\beta)}{\sigma}\right) \times \sigma^{-l/2} \prod_{h=1}^l [\mathbb{1}(|\beta_h| < \sqrt{\sigma}\xi_h)] \times \sigma^{-\kappa/2-1} \exp\left(-\frac{\lambda}{2\sigma}\right) \\
& = \sigma^{-(T+l+\kappa)/2-1} \exp\left(-\frac{1}{2} \frac{(y-X\beta)'(y-X\beta) + \lambda}{\sigma}\right) \prod_{h=1}^l [\mathbb{1}(|\beta_h| < \sqrt{\sigma}\xi_h)] \\
& = \sigma^{-(T+l+\kappa)/2-1} \exp\left(-\frac{1}{2} \frac{(y-X\beta)'(y-X\beta) + \lambda}{\sigma}\right) \prod_{h=1}^l \mathbb{1}\left(\sigma > \frac{\beta_h^2}{\xi_h^2}\right) \\
& = \sigma^{-(T+l+\kappa)/2-1} \exp\left(-\frac{1}{2} \frac{(y-X\beta)'(y-X\beta) + \lambda}{\sigma}\right) \mathbb{1}(\sigma > \text{Max}_{(h)} [\beta_h^2 / \xi_h^2]) \tag{a.6.19.46}
\end{aligned}$$

Define:

$$\bar{\kappa} = (\kappa + T + l)/2 \quad \bar{\lambda} = [\lambda + (y-X\beta)'(y-X\beta)]/2 \quad \bar{l} = \text{Max}_{(h)} [\beta_h^2 / \xi_h^2] \tag{a.6.19.47}$$

Then (a.6.19.46) rewrites:

$$\pi(\sigma|y, \backslash \sigma) \propto \sigma^{-\bar{\kappa}-1} \exp\left(-\frac{\bar{\lambda}}{\sigma}\right) \mathbb{1}(\sigma > \bar{l}) \tag{a.6.19.48}$$

Bibliography

